

VERIFICATION REPORT

of

BANTARGEBAK LANDFILL GAS MANAGEMENT & POWER GENERATION, INDONESIA



Document Prepared By Germanischer Lloyd Certification GmbH

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Summary:

Germanischer Lloyd Certification GmbH (GLC) has performed the pre-CDM VCS combined validation + verification of the project “Bantargebang Landfill Gas Management & Power Generation” in Indonesia (West Java Province) with regard to the relevant requirements for VCS project activities.

The project was designed with the objective to capture, flare and utilise landfill gas (LFG; which is rich in methane / CH₄) generated at the Bantargebang Landfill in order to avoid emissions of methane to the atmosphere and generate renewable electricity to be exported through the grid. Additionally, the Project Activity aims to improve the management of the landfill and its infrastructure.

This is the combined validation + 1st verification of the project under VCS (Verified Carbon Standard) program which covers the period from 2011-02-26 to 2012-07-31 (both days are included). The project is verified against the VCS Version 3^{/2/}. The verification is based on the monitoring plan of the drafted VCS Project Description document (VCS PD)^{/1/}, the Project Design Document^{/15/} (PDD) as per Clean Development Mechanism, the requirements of VCS (VCS Standard and VCS Program Guide of VCS Version 3)^{/2/}, CDM Validation and Verification Standard^{/3/}, and VCS Validation and Verification Manual^{/61/}. The project was validated and registered with reference no. 6486 under the CDM (Clean Development Mechanism), on 2012-07-05.

The project activity has already been validated by TÜV NORD CERT GmbH and registered as a CDM project on 2012-07-05. GLC conducted the additional validation of the cover page and sections 1.2, 1.3, 1.5, 1.6, 1.7, 1.9, 1.10, 1.12.1, 1.12.2, 1.12.3, 1.12.4 and 1.13 set out in the VCS Project Description for project validation under Verified Carbon Standard. The starting date of the crediting period of the project activity is 2011-02-26. The emission reductions are claimed as Verified Carbon Units (VCUs) under the Verified Carbon Standard Program in its version 3.

The methodology applied for the project to estimate the GHG emission reduction is ACM0001 (version 11) “Consolidated baseline and monitoring methodology for landfill gas project activities”. Monitoring report (MR)^{/4/}, Emission reduction calculation spread sheets (ER sheets)^{/5/}, Project Description document (VCS PD)^{/1/} and other supporting documents have also been made available to the verification team by project participants.

The combined validation + verification scope is defined as an independent review and ex post determination by the validation/verification body (VVB) of the monitored reductions in GHG emissions during the verification period from 2011-02-26 to 2012-07-31, and consisted of the following three phases:

- desk review of project design, baseline and monitoring plan;
- follow-up on-site visit and interviews with project stakeholders;
- resolution of outstanding issues and the issuance of the final verification report and opinion.

The overall verification, from Contract Review to Verification Report & Opinion, was conducted using

Germanischer Lloyd Certification GmbH internal procedures.

A risk based approach has been followed in this verification. In the course of verification (as a result of both a desk review and an on-site visit), 38 Corrective Action Requests (CAR), 18 Clarification Requests (CL) and 1 Forward Action Request (FAR) have been raised. All the CARs and CLs have been closed out successfully¹.

As a result of the verification, the GLC's Assessment Team confirms that the GHG emission reduction in the verification period from 2011-02-26 to 2012-07-31 (including both days), is 281,951 tCO₂e.

GLC does not assume any responsibility towards the issuance and utilization of VCUs hereby verified and certified. Request for issuance of VCUs shall be made by the project proponent to an approved VCS Program Registry based on the requirements set out under the most recent version of the VCS Program Guidelines clause on VCS Registration. The verification of reported emission reductions is based on the information made available to GLC and the engagement conditions detailed in this report. Hence, GLC cannot be held liable by any party for decisions made or not made based on this report.

¹ It is noteworthy that a part of the findings are based on desk review by 2012-09-28 (prior to on-site verification) without auditors' full access to all necessary documents. These documents have been made available during the on-site visit.

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1 INTRODUCTION

PT Navigat Organic Energy Indonesia has commissioned Germanischer Lloyd Certification GmbH (GLC) to carry out the combined validation + 1st verification of the emission reductions reported for the project activity “Bantargebang Landfill Gas Management & Power Generation” (yet to be registered under the VCS scheme) with regard to the relevant requirements for VCS program as per the VCS version 3^{/2/}.

This combined validation + verification report summarizes:

- (i) The findings from the VCS combined validation + verification for the current monitoring period from 2011-02-26 to 2012-07-31; and
- (ii) VCU certification statement for the emission reductions.

For validation part, refer to chapter 2.

1.1 Objective

VCS Verification is the independent review and ex post determination by a Verification Entity or Designated Operational Entity (DOE) (which is approved/registered as a verifier by VCSA) of the monitored reductions in GHG emissions that have occurred as a result of the registered VCS project activity during the defined verification period.

VCS Certification is the written assurance by a Certification Entity (which is approved/ registered as a certifier by VCSA) that, during a specific period in time, a project activity achieved the emission reductions as verified.

The objective of this verification was to verify and certify emission reductions reported for the “Bantargebang Landfill Gas Management & Power Generation” project for the monitoring period covering from 2011-02-26 to 2012-07-31 (both days included).

One of the objectives is also to confirm that the monitoring system is implemented and fully functional to generate Verified Carbon Units (VCUs) without any double counting.

1.2 Scope and Criteria

The scope of the verification is:

- To verify that the project activity has been implemented and operated as per the latest version of the VCS PD and that all physical features (technology, project equipment, and monitoring and metering equipment) of the project are in place.
- To verify that actual monitoring systems and procedures are in compliance with the monitoring systems and procedures described in the monitoring plan as per the latest version of the VCS PD.
- To evaluate the GHG emission reduction data and express a conclusion with a reasonable level of assurance about whether the reported GHG emission reduction data is free from significant misstatement.

- To verify that the reported GHG emission data is sufficiently supported by evidence.

The verification shall ensure that reported emission reductions are complete and accurate in order to be certified.

The project design, its eligibility as a CDM project activity and the correct application of the Approved Consolidated Methodology ACM0001 “Consolidated baseline and monitoring methodology for landfill gas project activities”(version 11) ^{/7/} have all been already validated by the DOE TÜV NORD CERT GmbH (Report No.: 8000374274-09/272, dated 2012-06-22) ^{/6/}.

As stated in the Validation Report for the project “Bantargebang Landfill Gas Management & Power Generation”, the validation opinion of TÜV NORD is that the project as described in the registered CDM-PDD (version 2 dated 2012-06-21 ^{/15/}) meets all relevant UNFCCC requirements for the CDM, all relevant host country criteria and correctly applies the Approved Consolidated Methodology ACM0001 “Consolidated baseline and monitoring methodology for landfill gas project activities”(version 11) ^{/7/}.

While the project “Bantargebang Landfill Gas Management & Power Generation” was already previously validated as a CDM project activity, VCSA recognizes the CDM as a GHG Program that meets its VCU Verification Criteria. This verification report, as required by applicable guidance of the VCS Program Guide ^{/2/}, includes the validation of additional VCS specific and unique requirements in terms of project design, applicability to the adopted methodology and additionality demonstration that were not previously assessed/validated in the Validation Report ^{/6/} as per CDM rules and requirements.

According to VCS 3, the Project registered under CDM program is eligible to claim Voluntary Carbon Units (VCUs) generated prior to the registration date under CDM program. Based on VCS 3, GLC has conducted additional validation that covers validation of the following items of the VCS Project Description document ^{/1/}:

Cover page and sections 1.2, 1.3, 1.5, 1.6, 1.7, 1.9, 1.10, 1.12.1, 1.12.2, 1.12.3, 1.12.4 and 1.13

The validation opinion of GLC is that the Project Activity “Bantargebang Landfill Gas Management & Power Generation” as described in the CDM PDD ^{/15/} and VCS PD of 2012-12-03 ^{/1/} meets all VCS Program requirements as set out in the VCS Rules.

1.3 Level of assurance

As the VCS Version 3 only recognizes verified emission reductions, GLC has focused on providing a reasonable level of assurance that the emission reduction calculation methodology used is appropriate and correctly applied, and that emission reductions have been accurately monitored.

In accordance with the recommendations in the Validation and Verification Standard ^{/3/} and VCS Validation and Verification Manual ^{/61/}, GLC may discount verified emission reductions or requests a discount of these by using conservative assumptions for uncertainties in emission reduction calculations that cannot be fully quantified or that cannot give a desired level of assurance. For verifying/certifying VCUs, the desired level of assurance was based on the combined quantitative assessment of the accuracy of monitoring project performance and the identification of material risks. For this verification, GLC is able to give a reasonable level of assurance.

1.4 Summary Description of the Project

The project activity encompasses the collection of LFG generated at the Bantargebang landfill, its destruction (by combustion in a high temperature enclosed flare) and its utilization as gaseous fuel for electricity generation in an electricity generation facility.

The main purpose of the Project Activity is the utilisation of LFG as gaseous fuel for electricity generation in the electricity generation facility. All net electricity generated at the electricity generation facility is exported through the electricity grid, thus displacing equivalent amount of electricity that would otherwise be generated by fossil-fuel sources (power plants) and new additions in the absence of the project activity (baseline scenario). For periods when the electricity generation facility is not (fully) operational or for periods when more LFG is available than the electricity generation facility can convert into power, one enclosed high temperature flare is available for safe combustion of the LFG with the goal of avoiding emissions of methane (CH₄) into the atmosphere.

LFG (which is rich in CH₄) has been historically, is recently and will be in future generated at the Bantargebang landfill as result of the anaerobic decomposition of municipal solid waste (MSW) disposed continuously on this landfill.

The Project Activity comprises the implementation and operation of a Landfill Degassing system in order to capture, flare and utilise as energy source the Landfill Gas (LFG) which is rich in methane. LFG is captured on the landfill by a system of vertical and horizontal wells, which are connected to a blower by a pipeline grid (pipelines constructed in polyethylene). The blower creates a negative pressure in the pipeline system; a transport of the LFG to the blower is so initiated. Before passing the blower, the LFG is conditioned in a "Condensate Filter" in order to remove water drops and contaminants. In a second step, the landfill gas is dried in a chilling unit, before it is fed either to the enclosed high-temperature flare where it is flared at combustion temperatures between 1,000 and 1,200 °C or to a set of gas engines converting the LFG into electrical energy (including injection in the national power grid). While being transported to the flare resp. the engines, the LFG passes several monitoring devices such as flow-meters (to determine the total amount and the amounts of gas being sent to the flare resp. the electricity generation facility, including temperature and pressure sensors) and a gas analyser (to determine the methane content of the gas). The electrical power, generated at the electricity generation facility, is send injected via transformers and a medium voltage panel into the national grid (in two different transmission lines). The energy needs of the Project Activity are either fed by own production or by consumption from the public grid.

During the monitoring period from 2011-02-26 to 2012-07-31, the Project Activity comprised the following technical installations:

Installed power capacity:	Stepwise increased up to 12.782 MW _{el} (this capacity has been reached on 2012-05-25)
Installed flaring capacity:	Designed for a maximum flow of 2,500 Nm ³ /h of LFG
Blower capacity	Designed for a maximum flow of 8,000 Nm ³ /h of LFG

During the conducted site visit (2012-10-09), the average LFG flow was between 4,000 and 5,000 Nm³/h with a methane content of approx. 50 % CH₄.

As per the registered PDD ^{/15/} of the CDM Project Activity, baseline emissions are calculated from two different sources:

1. Methane captured and destroyed by the Project Activity by combustion in an enclosed high-temperature flare and in gas engines. This amount of Methane captured and destroyed is calculated by the multiplication of
 - the total amount being delivered to the project (resp. the sum of amount being sent to the flare and of amount being sent to the electricity generation facility; whatever is lower will be taken for conservativeness reason),
 - methane concentration of the LFG,
 - methane density.
2. CO₂-displacement due to injection of electricity into the national grid, which is dominated by energy produced with fossil fuels. This CO₂-displacement is calculated by the multiplication of
 - The amount of electricity injected into the grid less the amount of electricity consumed from the grid by the Project Activity,
 - Emission Factor of the Power Transmission Grid.

Project emissions are calculated as the methane emissions from the flaring activity and the consumption of fossil fuels (if applicable). The emission reduction is calculated as the difference of baseline emissions and the project emissions as the leakage emissions are considered as zero for the project activity as per the applied methodology.

The total emission reduction for the 1st verification period from 2011-02-26 to 2012-07-31 is 281,951 tCO₂e. The emission reduction is calculated conservatively as per the applied methodology.

Further background information on the project activity can be found in the CDM Project Design document (CDM PDD) as well as the Validation report which are available on the CDM website:

<http://cdm.unfccc.int/Projects/DB/RWTUV1340626014.25/view>

The project activity was implemented at the Bantargebang Final Disposal Site, near the City of Bekasi, West Java Province, Republic of Indonesia.

The project characteristics are as mentioned below:

Item	Data
Title of the project activity	Bantargebang Landfill Gas Management & Power Generation
Project Location (specific Address, Country)	No specific address is available except Bantargebang Final Disposal Site facility. Country: Republic of Indonesia

	Province: West Java City: Bekasi The location of the project's electricity generation facility is Latitude: 6° 21' 04.21" South Longitude: 106° 59' 48.33" East
Project proponents	PT Navigat Organic Energy Indonesia PT Zeus Innnavitas
Project Entities	PT Navigat Organic Energy Indonesia Jl. Asia Africa Lot 19 18th Floor Panin Tower Jakarta DKI Jakarta 10270 Indonesia PT Zeus Innnavitas Jl. Gatot Subroto Kav. 22 Lt. 07 Wisma Argo Manunggal Jakarta Selatan DKI Jakarta 12930 Indonesia
Applied Methodologies	ACM0001 version 11 Consolidated baseline and monitoring methodology for landfill gas project activities
Sectoral Scope	Sectoral Scope 13 : Waste handling and disposal Sectoral Scope 1: Energy Industry (renewable/non-renewable source)
Project crediting period	from 2011-02-26 to 2033-07-31
Period verified in this verification	from 2011-02-26 to 2012-07-31

2 VALIDATION PROCESS, FINDINGS AND CONCLUSION

2.1 Validation Process

The overall validation, from Contract Review to Validation Report and Validation Opinion, was conducted by use and application of GLC internal procedures ^{/13/}.

The Project Description (VCS-PD) ^{/1/} and the Monitoring report (VCS-MR) ^{/4/} submitted by PT Navigat Organic Energy Indonesia and additional background documents related to the project design and baseline, i.e. registered CDM-PDD ^{/15/}, CDM validation report ^{/6/}, and the approved consolidated methodology (ACM0001, version 11) ^{/7/} were reviewed.

GLC verified the information contained in the documents reviewed against the requirements set in VCS Standard 3.3 effective as the 2012-10-04 ^{/2/}, the Clean Development Mechanism Validation and Verification Standard version 2.0 ^{/3/} and VCS Validation and Verification Manual 3.0 ^{/61/}.

In addition, from 2012-10-09 to 2012-10-12, GLC's Assessment Team performed an on-site visit and interviews with project stakeholders to confirm information and to resolve issues identified in the document review. Representatives from PT Navigat Organic Energy Indonesia were interviewed.

The main topics of the interviews are summarized in the Table 1 below:

Table 1: Interviewed persons and interview topics (validation process)

Interview Topic	Interviewed persons and source of information
- General aspects of the project - Project background information; - Project technology, operation, maintenance and monitoring capability; - Project monitoring and management plan; - Project approvals (incl. EIA, LOA); - Government policies related to landfill gas projects - Applicability of selected methodology; - Baseline determination; - Emission reductions calculation; - Emission reduction monitoring plan.	Ms. Cynthia Hendrayani, Vice-President Environmental Finance, PT Navigat Organic Energy Indonesia Mr. Dedi Agus Indrasetiawan, Project Engineer, PT Navigat Organic Energy Indonesia Mr. Sandy Cahyo Arsanto, CDM Officer Environmental Finance, PT Navigat Organic Energy Indonesia Mr Nicolas Creber, Manager Electricity Generation Facility, PT Navigat Organic Energy Indonesia Mr B. Roring, Landfill Operation Manager, PT Navigat Organic Energy Indonesia Mr. G Yudhistira, SCF Team, PT Navigat Organic Energy Indonesia Mr. Andry Bester, CDM Team, PT Navigat Organic Energy Indonesia

During the validation process, no material discrepancies were found. However GLC's Assessment Team raised a Corrective Action Request (CAR) regarding the correctness of the project's crediting period.

2.2 Validation Findings

2.2.1 Gap Validation

According to the VCS standard Version 3.3 ^{/2/} for projects validated under the CDM, the cover page and sections 1.2, 1.3, 1.5, 1.6, 1.7, 1.9, 1.10, 1.12.1, 1.12.2, 1.12.3, 1.12.4 and 1.13 of the VCS Project Description Template shall be completed by Project Participant. A validation/verification body shall undertake a validation of same, which shall be accompanied by a validation representation, to provide a gap validation for the project's compliance with the VCS rules.

GLC verified the information reported in the VCS-PD version 02 dated 2012-12-03 ^{/1/} against the CDM-PDD ^{/15/}, CDM validation report ^{/6/}, Project Host country (Indonesia) LOA ^{/55/} and UNFCCC website ^{/15 55/} and found consistent. The detailed validation process is documented in the Table 2 below:

Table 2: Documentation of validation process

CRITERIA OF VCS-PD	Ref	COMMENTS	Concl.
1.2 Sectoral Scope and Project Type	VCS-PD, CDM-PDD, CDM Validation Report	As a landfill gas power generation project, the project falls into the sectoral scopes 13 : Waste handling and disposal and 1: Energy Industry (renewable/non-renewable source).	OK
1.3 Project Proponent	VCS-PD, CDM-PDD, CDM Validation Report, LOA, UNFCCC website, PPA ^{/56/}	PT Navigat Organic Energy Indonesia	OK
1.5 Project Start Date	VCS-PD, Project commercial operation document ^{/19/}	The project starting date in VCS version 3.3 is defined as the date when the project began generating GHG emission reduction or removals. The date 2011-02-25 has been agreed as Commercial Operating Date of the project between the involved parties ^{/19/}. The day following the before mentioned agreement has then been defined as 1st day of the recently assessed monitoring period from 2011-02-26 to 2012-07-31 as it can be seen as first day of normal operation to generate GHG emission reductions.	OK

CRITERIA OF VCS-PD	Ref	COMMENTS	Concl.
1.6 Project Crediting Period	VCS-PD, Project commercial operation document, CDM-PDD, CDM Validation report, UNFCCC website	2011-02-26 to 2033-07-31. Starting date of the Project Activity was 2011-02-26. Under VCS, the crediting period is 10 years (twice renewable). So the VCS crediting period ends on 2041-02-25. For CDM, the 7 year crediting period starts on 2012-08-01 and is twice renewable. So, CDM crediting period end on 2033-07-31. As per VCS Standard, the application of multiple GHG programs requires that the end date is the earliest end date of all applied GHG programs. Thus, crediting period end on 2033-07-31.	OK
1.7 Project Scale and Estimated GHG Emission Reductions or Removals	VCS-PD, CDM-PDD, CDM Validation report	The project is expected to achieve 7,082,998 tCO ₂ e over a 10 years crediting period, this means an average of 708,299 t CO ₂ e. As estimated GHG emission reductions vary from year to year, it can be confirmed, that the smallest amount is 580,477 tCO ₂ e/year. It is thus categorized as "Large Project" as per VCS 3.3 definitions.	OK
1.9 Project Location	Site visit, VCS-PD, CDM-PDD, CDM Validation report	The project site is located within the Bantargebang Final Disposal Site facility. This facility is located near the City of Bekasi, in the Province of West Java, in the Republic of Indonesia. The latitude /longitude of the project's electricity generation facility is 6° 21' 04.21" South and 106° 59' 48.33" East.	OK

CRITERIA OF VCS-PD	Ref	COMMENTS	Concl.
1.10 Condition Prior to Project Initiation	VCS-PD,CDM-PDD,CDM Validation report	The scenario existing prior to the start of the implementation of the project activity is uncontrolled and full emissions of landfill gas (main component: methane) to the atmosphere and power generation based on fossil fuels (by the grid operator).	OK
1.12.1 Right of use	LoA, PPA	The Letter of approval authorizes the participation of PT Navigat Organic Energy Indonesia and PT. Zeus Innavitas as Project Participants in the Project. Additionally, the power purchase agreement (PPA) is signed by PT Navigat Organic Energy Indonesia.	OK
1.12.2 Emission Trading Programs and Other Binding Limits	VCS-PD,CDM-PDD,CDM Validation Report , UNFCCC website	Not applicable.	OK
1.12.3 Participation under Other GHG Programs	VCS-PD,CDM-PDD,CDM Validation Report, UNFCCC website	The project has been registered as a project activity with reference no. 6486 under the CDM (Clean Development Mechanism), on 2012-07-05. The CDM crediting period is 7 years (twice renewable) and starts on 2012-08-01.	OK
1.12.4 Other Forms of Environmental Credit	VCS-PD,CDM-PDD,CDM Validation Report, UNFCCC website	The project neither has nor intends to generate any other form of GHG-related environmental credit for GHG emission reductions or removals claimed under the VCS Program. The project will only generate CDM-	OK

CRITERIA OF VCS-PD	Ref	COMMENTS	Concl.
		CERs from 2012-08-01 on.	
1.13 Additional Information Relevant to the Project - Eligibility Criteria - Leakage Management - Commercially Sensitive Information - Further Information	VCS-PD, CDM-PDD, CDM Validation Report, UNFCCC website	- it is not a grouped project; - Leakage is set to zero, in accordance to the methodology. - Not applicable. - there is not any additional relevant legislative, technical, economic, sectoral, social, environmental, geographic, site-specific and/or temporal information that may have a bearing on the eligibility of the project, the net GHG emission reductions or the quantification of the project's net GHG emission reductions.	OK

2.2.2 Methodology Deviations

No methodology deviation has been applied to the Project.

2.2.3 Project Description Deviations

No deviation in Project Description has been applied to the Project.

2.2.4 New Project Activity Instances

Not applicable since the project is not a grouped project.

2.3 Validation Conclusion

In GLC's opinion, the project conforms to the relevant criteria of VCS Standard Version 3.3.

3 VERIFICATION PROCESS

3.1 Method and Criteria

The verification process was carried out in line with the requirements VCS Version 3 ^{/2/}. In addition, the verification team followed the guidelines of the CDM Validation and Verification Standard ^{/3/} and VCS Validation and Verification Manual ^{/61/}. Standard auditing techniques and GLC's CDM Procedures ^{/13/} were also applied during the verification. The verification team performed first a desk review, followed by an on-site visit to review the project realization.

The verification process was based on assessment of the validated CDM Project Design Document (CDM PDD) ^{/15/}, the monitoring report ^{/4/}, emission reduction calculation spreadsheet (ER sheet) ^{/5/}, supporting documents made available to the verification team and information collected through performing interviews during the on-site assessment. Furthermore publicly available information was considered as far as available and required.

The verification was carried out on the basis of the following requirements, applicable for this project activity:

- Requirements and guidance of VCS Version 3 ^{/2/}.
- Applicable CDM guidelines and requirements, including applicable relevant methodological decisions made by the CDM Executive Board.
- Other relevant rules, including the host country legislation,
- CDM Validation and Verification Standard ^{/3/},
- VCS Validation and Verification Manual ^{/61/},
- Monitoring plan as given in the registered CDM PDD ^{/15/},
- CDM Methodology ACM0001 (version 11): Consolidated baseline and monitoring methodology for landfill gas project activities ^{/7/}

3.2 Document Review

During the desk review all documents initially provided by the client and publicly available documents relevant for the verification were reviewed. The main documents are listed below:

- The validated CDM PDD ^{/15/}, including the monitoring plan and the corresponding validation report ^{/6/};
- Monitoring Report ^{/4/} and Emission reduction calculation sheet ^{/5/};
- The applied monitoring methodologies ^{/7/};

- Requirements and guidance of VCS ^{/2/} as well as applicable methodological relevant decisions, clarifications and guidance from the CDM Executive Board;
- Any other information and references relevant to the project activity's resulting emission reductions (e.g., IPCC reports ^{/16/}, data on electricity generation in the national grid or laboratory analysis and national regulations).

3.3 Interviews

Representatives of PT Navigat Organic Energy Indonesia including the operational staff of the landfill gas plant were interviewed to confirm selected information and to resolve issues identified in the document review. The main topics of the interviews are summarized in below table:

Table 3: Interviewed persons and interview topics (verification process)

Interview Topic	Interviewed persons and source of information
- General aspects of the project	Ms. Cynthia Hendrayani, Vice-President Environmental Finance, PT Navigat Organic Energy Indonesia
- Project equipment operation and technical details	Mr. Dedi Agus Indrasetiawan, Project Engineer, PT Navigat Organic Energy Indonesia
- Changes since validation	Mr. Sandy Cahyo Arsanto, CDM Officer Environmental Finance, PT Navigat Organic Energy Indonesia
- Monitoring and measurement equipment	Mr. Sandy Cahyo Arsanto, CDM Officer Environmental Finance, PT Navigat Organic Energy Indonesia
- Remaining issues from validation	Mr. Sandy Cahyo Arsanto, CDM Officer Environmental Finance, PT Navigat Organic Energy Indonesia
- Calibration procedures	Mr. Sandy Cahyo Arsanto, CDM Officer Environmental Finance, PT Navigat Organic Energy Indonesia
- Quality management system	Mr. Nicolas Creber, Manager Electricity Generation Facility, PT Navigat Organic Energy Indonesia
- Involved personnel and responsibilities	Mr. Nicolas Creber, Manager Electricity Generation Facility, PT Navigat Organic Energy Indonesia
- Training and practice of the operational personnel	Mr. B. Roring, Landfill Operation Manager, PT Navigat Organic Energy Indonesia
- Implementation of the monitoring plan	Mr. G Yudhistira, SCF Team, PT Navigat Organic Energy Indonesia
- Monitoring data management	Mr. G Yudhistira, SCF Team, PT Navigat Organic Energy Indonesia
- Data uncertainty and residual risks	Mr. Andry Bester, CDM Team, PT Navigat Organic Energy Indonesia
- GHG calculation	Mr. Andry Bester, CDM Team, PT Navigat Organic Energy Indonesia

Interview Topic	Interviewed persons and source of information
- Procedural aspects of the verification - Maintenance - Environmental aspects	

3.4 Site Inspections

From 2012-10-09 to 2012-10-12 GLC carried out an on-site verification of the project activity. During the visit, GLC's verification team verified that the actual implementation of the project was as described in the registered CDM PDD. Refer to 4.1 for details.

On-site inspection included review of data and records, interviews with the plant manager, project engineer, shift engineers, operations and maintenance personnel confirming all data sources used in the Monitoring Report.

The main tasks covered during the on-site visit included, but were not limited to:

- Verified whether the project was implemented as per the description in the CDM PDD^{/15/}.
- Confirmation that all monitoring equipment have been installed and operated as per the monitoring plan of the CDM PDD^{/15/}.
- The operating staff was interviewed and observed in order to check the risks of inappropriate operation and data collection procedures.
- Information processes for generating, aggregating and reporting the selected monitored parameters were reviewed.
- Calibration records of all metering equipment were checked and calibration procedure was verified.
- The monitored data was checked completely.

3.5 Resolution of Any Material Discrepancy

On the basis of the desk review, the on-site visit, follow-up interviews and further background investigation the verification report is completed.

Findings established during the verification may be that:

- i) The verification is not able to obtain sufficient evidence for the reported emission reduction or part of the reported emission reduction. In this case these emission reduction shall not be verified and certified;
- ii) The verification has identified significant misstatements in the reported emission reduction.

A detailed list of verification findings form the draft verification report, which was sent to the client for resolution of raised CARs and CLs.

Nonconformities raised during the verification can either be seen as a non-fulfilment of criteria ensuring the proper implementation of a project or where a risk to deliver high quality emission reduction is identified.

Corrective Action Requests (CARs) are issued, if:

- the project participants have made mistakes that will influence the ability of the project activity to achieve real, measurable additional emission reductions;
- the CDM or VCS requirements have not been met;
- there is a risk that emission reductions cannot be monitored or calculated.

Forward Action Requests (FAR) indicate essential risks for further periodic verifications. Forward Action Requests are issued, if:

- The actual status requires a special focus on this item for the next consecutive verification, or an adjustment of the monitoring plan is recommended.

The verification team uses the term Clarification Request (CL), which is issued if:

- Information is insufficient or not clear enough to determine whether the applicable CDM or VCS requirements have been met.

For a detailed list of all CARs, CLs and FARs raised in the course of the verification please refer to Annex 3.

4 VERIFICATION FINDINGS

4.1 Project Implementation Status

On the basis of the on-site visit and the reviewed project documentation, the verification team confirms that the project was implemented as described in the registered CDM PDD. All the physical components and project boundary are in conformity with the description in registered CDM PDD.

According to the validated CDM PDD, the CDM start date of the project was 2008-12-05^{/15/17/} as on this date the contractual legal and financial obligations of the project developer entered into force. The physical implementation of the project reached its first milestone on 2010-06-30, as two gas engines

(internal IDs: GE-01 and GE-02) have been successfully commissioned ^{/18/}. The 2nd big milestone has been reached on 2011-02-25, as this date has been agreed as Commercial Operating Date of the project between the involved parties ^{/19/}. The day following the before mentioned agreement has then been defined as 1st day of the recently assessed monitoring period from 2011-02-26 to 2012-07-31. On the dates below the following milestones have been reached within the monitoring period:

2011-03-04	Successfully commissioning of 2 gas engines (internal IDs: GE-03 and GE-04) ^{/20/}
2011-06-25	Successfully commissioning of flare unit and degassing station ^{/21/}
2011-07-16	Successfully commissioning of 2 gas engines (internal IDs: GE-09 and GE-10) ^{/22/}
2011-07-27	Successfully commissioning of 2 gas engines (internal IDs: GE-07 and GE-08) ^{/23/}
2012-02-13	Successfully commissioning of 2 gas engines (internal IDs: GE-11 and GE-12) ^{/24/}

Thus, in total at the end of the monitoring period (2012-07-31), the following capacities and devices have inter alia been installed and implemented on the project site:

- A set of 10 gas engines (internal IDs: GE-01, GE-02, GE-03, GE-04, GE-07, GE-08, GE-09, GE-10, GE-11, GE-12), providing a total installed capacity of 12.782 MWel.
- An enclosed high-temperature flare with a maximum capacity of 2,500 Nm³/h.
- Two parallel lines of gas conditioning units, with a maximum capacity of 8,000 Nm³/h.
- Several sets of flowmeters, pressure and temperature sensors.
- A continuous stationary gas analyser.
- A set of electricity meters.
- Devices as transformers and medium voltage panels to inject the electricity in the national power grid.

The project activity involves capturing of methane generated on the landfill. The captured methane (landfill gas) is utilized as fuel in onsite gas engines. Any excess landfill gas is combusted in the project's high temperature enclosed flare. This is in line with the description in registered CDM PDD.

GLC's verification team has also verified that all the parameters are monitored as described in the monitoring plan of the CDM PDD in order to estimate the emission reduction.

It is noteworthy that due to commissioning, not all instruments and devices have been in place and fully operational on the starting date of this monitoring period (2011-02-26). On this date, the electricity generation facility was able to export electricity to the grid. This was monitored from the early beginning on by the use of electricity meters. The remaining instruments related to landfill gas flow, landfill gas methane content and flare's exhaust gas temperature have been commissioned on 25th June 2011. Thus, for the months of February, March, April and May 2011, emission reductions from capture and destruction of methane contained in landfill gas are not claimed. For these months, only the emission reductions for CO₂-displacement (through electricity exported to the grid) are claimed and verified.

As per the applied methodology ACM0001 (version 11) ^{/7/}, the project boundary is defined as below:

“The project boundary is the site of the project activity where the gas is captured and destroyed/used. If the electricity for project activity is sourced from grid or electricity generated by the LFG captured would have been generated by power generation sources connected to the grid, the project boundary shall include all the power generation sources connected to the grid to which the project activity is connected”.

Based on document review and onsite verification, it is confirmed that all the emission sources within the boundary have been considered appropriately.

Implementation status of the monitoring plan and the completeness of monitoring

The verification team has followed two steps to assess the completeness of the monitoring. The first step is document review and the second step comprises on-site interviews and observations. As a part of step 1, the verification team has reviewed the submitted documents such as MR ^{/4/}, ER sheets ^{/5/} and supporting documents stated in Annex 2. The monitoring parameters have been monitored with calibrated monitoring equipment as described in the monitoring plan of the registered CDM PDD ^{/15/}. It is confirmed that the monitoring mechanism is effective and reliable². Wherever a parameter was not sufficiently monitored as per the monitoring plan of the CDM PDD, a deviation was applied by the Project Participant and the emission reduction was calculated by a conservative approach. Please see below sections for details.

Therefore, based on document review and onsite verification, it is confirmed that all the parameters were monitored in accordance with the registered monitoring plan during the monitoring period. In the following section, the details of monitoring in accordance with the monitoring plan of the registered CDM PDD are assessed.

It is noteworthy that the names of monitored parameters for the flow rates of LFG have been changed. The reason for this is a change of the recording time interval. The PDD stated that LFG flow is to be reported hourly, while the Project uses a much higher data recording resolution (per minute). Thus, the index has been changed from “_h” (hour) to “_t” (minute). This is assessed in compliance with monitoring

² It is noteworthy that due to commissioning, not all instruments and devices have been in place and fully operational on the starting date of this monitoring period (2011-02-26). On this date, the electricity generation facility was able to export electricity to the grid. This was monitored from the early beginning on by the use of electricity meters. The remaining instruments related to landfill gas flow, landfill gas methane content and flare’s exhaust gas temperature have been commissioned in June 2011. Thus, for the months of February, March, April and May 2011, emission reductions from capture and destruction of methane contained in landfill gas are not claimed. For these months, only the emission reductions for CO₂-displacement (through electricity exported to the grid) are claimed and verified.

plan since hourly value can still be obtained from the minute values; change in parameter name is only for clearer presentation.

Table 4 Assessment of monitored parameters

Monitored Parameter	Assessment/ Observation
Data / Parameter: (as in monitoring plan of the CDM PDD):	LFG _{total,t} Total amount of landfill gas captured in Normal Volume on each time t [Nm ³]
Measuring frequency:	LFG _{total,t} has been continuously measured, where continuous measurements have been recorded and reported with an every minute frequency.
Reporting frequency:	Every minute.
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes. As per ACM0001 (version 11) ^{/7/} , “continuous” is indicated in the row “ <i>Monitoring Frequency</i> ” of the applicable table in the monitoring methodology and it does not specify the reporting frequency. However, since “(<i>average value in a time interval not greater than an hour shall be used in the calculations of emission reductions</i>)” is indicated, it is reasonable to believe that hourly reporting would be sufficient to fulfil methodology requirements. As per the monitoring plan of the registered PDD ^{/15/} , “ <i>data from this flow-meter is continuously logged</i> ” and PDD includes it into the group of parameters which “ <i>are to be updated on hourly basis</i> ”. Thus, the adopted measuring and reporting frequencies for this parameter, i.e. continuous measuring, minute-by-minute reporting, are assumed as in accordance with the monitoring plan of the registered PDD ^{/15/} and monitoring methodology ACM0001 (version 11) ^{/7/} .
Type of monitoring equipment:	Two LFG flow meters of the same model were used to monitor LFG _{total,t} during the considered monitoring period. The installed LFG flow meters are manufactured by Endress and Hauser and are of differential pressure flow-meter type, model Deltabar S PMD 70. Each Deltabar S PMD 70 flow meter consists of a flow element and a signal transmitter unit. The flow element uses the differential pressure principle: A Pitot tube (as primary element) creates a pressure difference inside the pipe, which is a direct measure of volume flow. Two impulse lines carry the differential pressure to the transmitter, where it is converted into the corresponding output signals. The specifications of the installed LFG flow meters are presented below:

	Specifications of the installed LFG flow meter	
	Internal ID	BGB.FIR.0002
	Position Tag	FIR70.0
	Period installed	2011-02-24 to 2011-10-05
	Manufacturer	Endress and Hauser
	Model	Deltabar S PMD 70
	Serial Number	E105920109D
	Measuring principle	differential pressure
	Operating temperature	-20...+85°C
	Maximum permissible error:	±0.5 %
	Source: ^{125/ 126/}	
	Due to changes in pipeline diameter (from initially 210 mm to 312 mm), the following flow meter has been installed on 2011-10-05 and was operational until the end of the monitoring period	
	Specifications of the installed LFG flow meter	
	Internal ID	BGB.FIR.0003
	Position Tag	FIR70.0
	Period installed	2011-10-05 to 2012-07-31
	Manufacturer	Endress and Hauser
	Model	Deltabar S PMD 70
	Serial Number	E105910109D
	Measuring principle	differential pressure
	Operating temperature	-20...+85°C
	Maximum permissible error:	±0.5 %
	Source: ^{125/ 126/}	
	In order to measure the LFG flow in normalized volume that means at standard temperature and pressure conditions (0 degree Celsius and 1.013 bar), the above mentioned flow measuring unit is equipped with a temperature sensor and an absolute pressure sensor. The values of these two devices are combined with the values as measured with the flow meters in the project's data acquisition system. The recorded and reported values are then expressed in normalized cubic meters (Nm³).	

The specifications of the installed thermometer are presented below:

Resistance thermometers are sensors with the task to measure temperature by correlating the resistance of the sensor element with temperature. The designation "Pt" refers to the material "Platinum" used as core material to measure the resistance. The designation of "100" refers to nominal resistance of 100 ohms at 0 °C, thus these sensors are called Pt100 sensors. The temperature sensing unit consists of the following components:

- platinum coil fitted into two bores of a ceramic rod
- wires connecting the platinum coil with a transmitter

Specifications of the installed LFG temperature sensor	
Internal ID	BGB.TIR.0004
Position Tag	TIR 70.0
Manufacturer	Endress and Hauser
Model	Omnigrad M TR10
Serial Number	E1046A14152
Measuring principle	Resistance thermometer
Operating range	-20...+70°C
Maximum permissible error:	±0.5 %

Source: ^{126/ 127}

The specifications of the installed pressure meter are presented below:

The pressure meters use silicon sensors. This sensor measures the expansion created by the recent pressure and converts it into an electrical signal.

Specifications of the utilized pressure meters	
Internal ID	BGB.PIR.0001
Position Tag	PIR 70.0
Manufacturer	Endress and Hauser
Model	Cerabar M PMC 51
Serial Number	E101AD01128
Maximum permissible error:	±0,5 %

	Source: ^{/26/ /28/}
Is accuracy of the monitoring equipment as stated in the CDM PDD? If the CDM PDD does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practice?	Yes. The registered PDD and monitoring methodology do not specify any accuracy requirement for the LFG flow meter, the LFG temperature sensor and the LFG pressure sensor installed at the project site to monitor LFG_{total,t}. The accuracy range for the installed instruments is $\pm 0.5\%$ as per manufacturer's information. It is GLC's contention based on its Sectoral expertise that the use of the installed instrument represents good practise for monitoring of LFG flow, LFG temperature and LFG pressure.
Calibration frequency /interval:	Yearly
Is the calibration interval in line with the monitoring plan of the CDM PDD? If the CDM PDD does not specify the frequency of calibration, does the selected frequency represent good monitoring practice?	The monitoring plan of the registered PDD does not clearly specify calibration intervals. However, the following is stated: <i>"Flowmeter is to be maintained and calibrated periodically according to manufacturer recommendation"</i>. The monitoring methodology states that flowmeters <i>"should be subject to a regular maintenance and testing regime to ensure accuracy"</i>. As the manufacturer is not giving a detailed recommendation ^{/26/}, Project Participant has decided to perform yearly a service and inspection of all devices to monitor LFG flow (including thermometer and pressure sensor). If during this check the instrument is found outside the maximum permissible error, a calibration will be executed. Thus, the interval applied by Project Participant can be deemed correct.
Company performing the calibration:	<u>Factory inspection / calibration</u> E105920109D, E105910109D and E101AD01128: Endress + Hauser GmbH + Co.KG, Maulburg / Germany ^{/29/ /30/ /32/} E1046A14152: Endress + Hauser Sigestherm S.r.l., Pessano / Italy ^{/31/} <u>On site inspection / calibration</u> E1046A14152, E105910109D, E101AD01128: Endress + Hauser Messtechnik GmbH + Co.KG, Weil am Rhein/Germany ^{/30/ /31/ /32/}
Did calibration confirm proper functioning of monitoring equipment? (Yes / No):	Yes. The calibration certificates were checked and the meters were found functioning properly.
Is (are) calibration(s) valid for the whole reporting period?	Yes. The calibrations are valid for the whole reporting period. Calibrations have been carried out on the following dates: Flow Meter Deltabar S PMD 70 with serial number

	E105920109D: 2011-02-01 ^{/29/} Flow Meter Deltabar S PMD 70 with serial number E105910109D: 2011-02-01 and 2011-12-09 ^{/30/} Temperature sensor Omnigrad M TR10 with serial number: E1046A14152: 2011-01-25 and 2011-12-21 ^{/31/} Pressure sensor Cerabar M PMC 51 with serial number E101AD01128: 2011-01-25 and 2011-12-21 ^{/32/}
If applicable, has the reported data been cross-checked with other available data?	Not applicable.
How were the values in the monitoring report verified?	During the on-site visit, Project Participant downloaded the electronic data (for the time covering the recent monitoring period) as available on the project's data acquisition system. This raw data ^{/14/} has been handed over to GLC's Assessment Team. It is noteworthy that raw data is downloaded in an encrypted file and must be converted into spreadsheets using special software. GLC's Assessment Team has converted the complete set of raw data and compared the so created data against the data provided in the ER spreadsheets ^{/5/}. No quantitative deviations or differences were identified while comparing the two data bases as explained above. Moreover, it is noteworthy that the instruments related to landfill gas flow, landfill gas methane content and flare's exhaust gas temperature have been commissioned on 25th June 2011. Thus, for the months of February, March, April and May 2011, emission reductions from capture and destruction of methane contained in landfill gas are not claimed. (For these months, only the emission reductions for CO₂-displacement (through electricity exported to the grid) are claimed and verified). This is assessed as reasonable and conservative.
Does the data management (from monitoring equipment to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes, it is confirmed from document review and onsite verification that the data management ensures correct transfer of data and reporting of emission reductions and the necessary QA/QC procedures are in place and are continuously improved.

Monitored Parameter	Assessment/ Observation
Data / Parameter: (as in monitoring plan of the CDM PDD):	LFG_{electricity,t} Total amount of landfill gas fed to power plant on Normal Volume on each time t [Nm³]
Measuring frequency:	LFG _{electricity,t} has been continuously measured, where continuous measurements have been recorded and

	reported with an every minute frequency.																		
Reporting frequency:	Every minute.																		
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	As per ACM0001 (version 11) ^{/7/}, “continuous” is indicated in the row “<i>Monitoring Frequency</i>” of the applicable table in the monitoring methodology and it does not specify the reporting frequency. However, since “(average value in a time interval not greater than an hour shall be used in the calculations of emission reductions)” is indicated, it is reasonable to believe that hourly reporting would be sufficient to fulfil methodology requirements. As per the monitoring plan of the registered PDD ^{/15/}, “data from this flow-meter is continuously logged” and PDD includes it into the group of parameters which “are to be updated on hourly basis”. Thus, the adopted measuring and reporting frequencies for this parameter, i.e. continuous measuring, minute-by-minute reporting, are assumed as in accordance with the monitoring plan of the registered PDD ^{/15/} and monitoring methodology ACM0001 (version 11) ^{/7/}.																		
Type of monitoring equipment:	Two LFG flow meters of the same model were used to monitor LFG_{electricity,t} during the considered monitoring period. The installed LFG flow meters are manufactured by Endress and Hauser and are of differential pressure flow-meter type, model Deltabar S PMD 70. Each Deltabar S PMD 70 flow meter consists of a flow element and a signal transmitter unit. The flow element uses the differential pressure principle: A Pitot tube (as primary element) creates a pressure difference inside the pipe, which is a direct measure of volume flow. Two impulse lines carry the differential pressure to the transmitter, where it is converted into the corresponding output signals. The specifications of the installed LFG flow meters are presented below: <table border="1"> <thead> <tr> <th colspan="2">Specifications of the installed LFG flow meter</th></tr> </thead> <tbody> <tr> <td>Internal ID</td><td>BGB.FIR.0006</td></tr> <tr> <td>Position Tag</td><td>FIR90.0</td></tr> <tr> <td>Period installed</td><td>2011-02-24 to 2011-10-05</td></tr> <tr> <td>Manufacturer</td><td>Endress and Hauser</td></tr> <tr> <td>Model</td><td>Deltabar S PMD 70</td></tr> <tr> <td>Serial Number</td><td>E1058F0109D</td></tr> <tr> <td>Measuring principle</td><td>differential pressure</td></tr> <tr> <td>Operating temperature</td><td>-20...+85°C</td></tr> </tbody> </table>	Specifications of the installed LFG flow meter		Internal ID	BGB.FIR.0006	Position Tag	FIR90.0	Period installed	2011-02-24 to 2011-10-05	Manufacturer	Endress and Hauser	Model	Deltabar S PMD 70	Serial Number	E1058F0109D	Measuring principle	differential pressure	Operating temperature	-20...+85°C
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Position Tag	FIR90.0																						
Period installed	2011-10-05 to 2012-07-31																						
Manufacturer	Endress and Hauser																						
Model	Deltabar S PMD 70																						
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Calibration frequency /interval:	Yearly																																
Is the calibration interval in line with the monitoring plan of the CDM PDD? If the CDM PDD does not specify the frequency of calibration, does the selected frequency	The monitoring plan of the registered PDD does not clearly specify calibration intervals. However, the following is stated: <i>"Flowmeter is to be maintained and calibrated</i>																																

represent good monitoring practice?	<i>periodically according to manufacturer recommendation</i>". The monitoring methodology states that flowmeters "should be subject to a regular maintenance and testing regime to ensure accuracy". As the manufacturer is not giving a detailed recommendation ^{/26/}, Project Participant has decided to perform yearly a service and inspection of all devices to monitor LFG flow (including thermometer and pressure sensor). If during this check the instrument is found outside the maximum permissible error, a calibration will be executed. Thus, the interval applied by Project Participant can be deemed correct.
Company performing the calibration:	<u>Factory inspection / calibration</u> E1058F0109D, E1058E0109D, E101AE01128: Endress + Hauser GmbH + Co.KG, Maulburg / Germany ^{/33/ /34/ /36/} E1046B14152: Endress + Hauser Sigestherm S.r.l., Pessano / Italy ^{/35/} <u>On site inspection / calibration</u> E1058E0109D, E1046B14152, E101AE01128: Endress + Hauser Messtechnik GmbH + Co.KG, Weil am Rhein/Germany ^{/34/ /35/ /36/}
Did calibration confirm proper functioning of monitoring equipment? (Yes / No):	Yes. The calibration certificates were checked and the meters were found functioning properly.
Is (are) calibration(s) valid for the whole reporting period?	Yes. The calibrations are valid for the whole reporting period. Calibrations have been carried out on the following dates: Flow Meter Deltabar S PMD 70 with serial number E1058F0109D: 2011-02-01 ^{/33/} Flow Meter Deltabar S PMD 70 with serial number E1058E0109D: 2011-02-01 and 2011-12-14 ^{/34/} Temperature sensor Omnigrad M TR10 with serial number E1046B14152: 2011-01-25 and 2011-12-21 ^{/35/} Pressure sensor Cerabar M PMC 51 with serial number E101AE01128: 2011-01-25 and 2011-12-21 ^{/36/}
If applicable, has the reported data been cross-checked with other available data?	Not applicable.
How were the values in the monitoring report verified?	During the on-site visit, Project Participant downloaded the electronic data (for the time covering the recent monitoring period) as available on the project's data acquisition system. This raw data ^{/14/} has been handed over to GLC's Assessment Team. It is noteworthy that raw data is downloaded in an encrypted file and must be converted into spreadsheets using special software. GLC's Assessment Team has converted the complete set of raw

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Does the data management (from monitoring equipment to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes, it is confirmed from document review and onsite verification that the data management ensures correct transfer of data and reporting of emission reductions and the necessary QA/QC procedures are in place and are continuously improved.

Monitored Parameter	Assessment/ Observation
Data / Parameter: (as in monitoring plan of the CDM PDD):	LFG_{flare,t} Total amount of landfill gas fed to flare unit on Normal Volume on each time t [Nm³]
Measuring frequency:	LFG _{flare,t} has been continuously measured, where continuous measurements have been recorded and reported with an every minute frequency.
Reporting frequency:	Every minute.
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	As per ACM0001 (version 11) ^{/11/}, "continuous" is indicated in the row "<i>Monitoring Frequency</i>" of the applicable table in the monitoring methodology and it does not specify the reporting frequency. However, since "<i>(average value in a time interval not greater than an hour shall be used in the calculations of emission reductions)</i>" is indicated, it is reasonable to believe that hourly reporting would be sufficient to fulfil methodology requirements. As per the monitoring plan of the registered PDD ^{/15/}, "<i>data from this flow-meter is continuously logged</i>" and PDD includes it into the group of parameters which "<i>are to be updated on hourly basis</i>". Thus, the adopted measuring and reporting frequencies for this parameter, i.e. continuous measuring, minute-by-minute reporting, are assumed as in accordance with the monitoring plan of the registered PDD ^{/15/} and monitoring

	methodology ACM0001 (version 11) ¹⁷⁷ .																		
Type of monitoring equipment:	A LFG flow meter was used to monitor LFG_{flare,t} during the considered monitoring period. The installed LFG flow meter is manufactured by Endress and Hauser and is of differential pressure flow-meter type, model Deltabar S PMD 70. The Deltabar S PMD 70 flow meter consists of a flow element and a signal transmitter unit. The flow element uses the differential pressure principle: A Pitot tube (as primary element) creates a pressure difference inside the pipe, which is a direct measure of volume flow. Two impulse lines carry the differential pressure to the transmitter, where it is converted into the corresponding output signals. The specifications of the installed LFG flow meter are presented below: <table border="1"> <thead> <tr> <th colspan="2">Specifications of the installed LFG flow meter</th></tr> </thead> <tbody> <tr> <td>Internal ID</td><td>BGB.FIR.0010</td></tr> <tr> <td>Position Tag</td><td>FIR81.3</td></tr> <tr> <td>Manufacturer</td><td>Endress and Hauser</td></tr> <tr> <td>Model</td><td>Deltabar S PMD 70</td></tr> <tr> <td>Serial Number</td><td>E105900109D</td></tr> <tr> <td>Measuring principle</td><td>differential pressure</td></tr> <tr> <td>Operating temperature</td><td>-20...+85°C</td></tr> <tr> <td>Maximum permissible error:</td><td>±0.5 %</td></tr> </tbody> </table> Source: ^{125/ 126/} In order to measure the LFG flow in normalized volume that means at standard temperature and pressure conditions (0 degree Celsius and 1.013 bar), the above mentioned flow measuring unit is equipped with a temperature sensor and an absolute pressure sensor. The values of these two devices are combined with the values as measured with the flow meters in the project's data acquisition system. The recorded and reported values are then expressed in normalized cubic meters (Nm³). The specifications of the installed thermometer are presented below: Resistance thermometers are sensors with the task to measure temperature by correlating the resistance of the sensor element with temperature. The designation "Pt" refers to the material "Platinum" used as core material to measure the resistance. The designation of "100" refers to nominal resistance of 100 ohms at 0 °C, thus these	Specifications of the installed LFG flow meter		Internal ID	BGB.FIR.0010	Position Tag	FIR81.3	Manufacturer	Endress and Hauser	Model	Deltabar S PMD 70	Serial Number	E105900109D	Measuring principle	differential pressure	Operating temperature	-20...+85°C	Maximum permissible error:	±0.5 %
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	sensors are called Pt100 sensors. The temperature sensing unit consists of the following components: - platinum coil fitted into two bores of a ceramic rod - wires connecting the platinum coil with a transmitter <table border="1"> <thead> <tr> <th colspan="2">Specifications of the installed LFG temperature sensor</th></tr> </thead> <tbody> <tr> <td>Internal ID</td><td>BGB.TIR.0011</td></tr> <tr> <td>Position Tag</td><td>TIR 81.3</td></tr> <tr> <td>Manufacturer</td><td>Endress and Hauser</td></tr> <tr> <td>Model</td><td>Omnigrad M TR10</td></tr> <tr> <td>Serial Number</td><td>E1046914152</td></tr> <tr> <td>Measuring principle</td><td>Resistance thermometer</td></tr> <tr> <td>Operating range</td><td>-20...+70°C</td></tr> <tr> <td>Maximum permissible error:</td><td>±0.5 %</td></tr> </tbody> </table> Source: /26/ /27/ The specifications of the installed pressure meter are presented below: The pressure meters use silicon sensors. This sensor measures the expansion created by the recent pressure and converts it into an electrical signal. <table border="1"> <thead> <tr> <th colspan="2">Specifications of the utilized pressure meters</th></tr> </thead> <tbody> <tr> <td>Internal ID</td><td>BGB.PIR.0009</td></tr> <tr> <td>Position Tag</td><td>PIR 81.3</td></tr> <tr> <td>Manufacturer</td><td>Endress and Hauser</td></tr> <tr> <td>Model</td><td>Cerabar M PMC 51</td></tr> <tr> <td>Serial Number</td><td>E101AC01128</td></tr> <tr> <td>Maximum permissible error:</td><td>±0,5 %</td></tr> </tbody> </table> Source: /26/ /28/	Specifications of the installed LFG temperature sensor		Internal ID	BGB.TIR.0011	Position Tag	TIR 81.3	Manufacturer	Endress and Hauser	Model	Omnigrad M TR10	Serial Number	E1046914152	Measuring principle	Resistance thermometer	Operating range	-20...+70°C	Maximum permissible error:	±0.5 %	Specifications of the utilized pressure meters		Internal ID	BGB.PIR.0009	Position Tag	PIR 81.3	Manufacturer	Endress and Hauser	Model	Cerabar M PMC 51	Serial Number	E101AC01128	Maximum permissible error:	±0,5 %
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Is accuracy of the monitoring equipment as stated in the CDM PDD? If the CDM PDD does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practice?	Yes. The registered PDD and monitoring methodology do not specify any accuracy requirement for the LFG flow meter, the LFG temperature sensor and the LFG pressure sensor installed at the project site to monitor LFG_{flare,t}. The accuracy range for the installed instruments is ±0.5% as per manufacturer's information. It is GLC's contention based on its Sectoral expertise that the use of the installed instrument represents good practise for monitoring of LFG																																

	flow, LFG temperature and LFG pressure.
Calibration frequency /interval:	Yearly
Is the calibration interval in line with the monitoring plan of the CDM PDD? If the CDM PDD does not specify the frequency of calibration, does the selected frequency represent good monitoring practice?	The monitoring plan of the registered PDD does not clearly specify calibration intervals. However, the following is stated: <i>"Flowmeter is to be maintained and calibrated periodically according to manufacturer recommendation"</i>. The monitoring methodology states that flowmeters <i>"should be subject to a regular maintenance and testing regime to ensure accuracy"</i>. As the manufacturer is not giving a detailed recommendation ^{/26/}, Project Participant has decided to perform yearly a service and inspection of all devices to monitor LFG flow (including thermometer and pressure sensor). If during this check the instrument is found outside the maximum permissible error, a calibration will be executed. Thus, the interval applied by Project Participant can be deemed correct.
Company performing the calibration:	<u>Factory inspection / calibration</u> E105900109D, E101AC01128: Endress + Hauser GmbH + Co.KG, Maulburg / Germany ^{/37/ /39/} E1046914152: Endress + Hauser Sigestherm S.r.l., Pessano / Italy ^{/38/} <u>On site inspection / calibration</u> E105900109D, E101AC01128, E1046914152: Endress + Hauser Messtechnik GmbH + Co.KG, Weil am Rhein/Germany ^{/37/ /38/ /39/}
Did calibration confirm proper functioning of monitoring equipment? (Yes / No):	Yes. The calibration certificates were checked and the meters were found functioning properly.
Is (are) calibration(s) valid for the whole reporting period?	Yes. The calibrations are valid for the whole reporting period. Calibrations have been carried out on the following dates: Flow Meter Deltabar S PMD 70 with serial number: 2011-02-01 ^{/33/} Flow Meter Deltabar S PMD 70 with serial number : 2011-02-01 and 2011-12-14 ^{/34/} Temperature sensor Omnigrad M TR10 with serial number: 2011-01-25 and 2011-12-21 ^{/35/} Pressure sensor Cerabar M PMC 51 with serial number: 2011-01-25 and 2011-12-21 ^{/36/}
If applicable, has the reported data been cross-checked with other available data?	Not applicable.
How were the values in the monitoring report	During the on-site visit, Project Participant downloaded the electronic data (for the time covering the recent monitoring

verified?	period) as available on the project's data acquisition system. This raw data ^{/14/} has been handed over to GLC's Assessment Team. It is noteworthy that raw data is downloaded in an encrypted file and must be converted into spreadsheets using special software. GLC's Assessment Team has converted the complete set of raw data and compared the so created data against the data provided in the ER spreadsheets ^{/5/}. No quantitative deviations or differences were identified while comparing the two data bases as explained above. Moreover, it is noteworthy that the instruments related to landfill gas flow, landfill gas methane content and flare's exhaust gas temperature have been commissioned on 25th June 2011. Thus, for the months of February, March, April and May 2011, emission reductions from capture and destruction of methane contained in landfill gas are not claimed. (For these months, only the emission reductions for CO₂-displacement (through electricity exported to the grid) are claimed and verified). This is assessed as reasonable and conservative.
Does the data management (from monitoring equipment to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes, it is confirmed from document review and onsite verification that the data management ensures correct transfer of data and reporting of emission reductions and the necessary QA/QC procedures are in place and are continuously improved.

Monitored Parameter	Assessment/ Observation
Data / Parameter: (as in monitoring plan of the CDM PDD):	w_{CH4} Volumetric fraction of CH₄ in landfill gas [Nm³ (CH₄)/ Nm³ (LFG)]
Measuring frequency:	w_{CH4} has been continuously measured, where continuous measurements have been recorded and reported with an every minute frequency.
Reporting frequency:	Every minute.
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes. As per ACM0001 (version 11) ^{/7/}, "Continuous (average value in a time interval not greater than an hour shall be used in the calculations of emission reductions)" is indicated in the row "Monitoring Frequency" of the applicable table in the monitoring methodology. The methodology does not explicitly indicate reporting frequency. As per the monitoring plan of the registered PDD ^{/15/}, this parameter is categorized into the group of parameters which "are to be updated on hourly basis". Additionally, the

	PDD states that gas “is to be sampled on hourly basis at a minimum”. Thus, the adopted measuring and reporting frequencies for this parameter, i.e. continuous measuring, minute-by-minute reporting, are assumed as in accordance with the monitoring plan of the registered PDD ^{/15/} and monitoring methodology ACM0001 (version 11) ^{/7/}.																				
Type of monitoring equipment:	A continuous CH₄/O₂ content gas analyzer unit was used to monitor w_{CH₄} during the considered monitoring period. The installed LFG gas analyzer is manufactured by NUK NENNING UND KRUMM GMBH (Germany). The landfill gas to be analysed and measured is taken direct from the landfill gas pipeline and fed via the internal process gas pump to the CH₄ sensor. The methane gas analyser is an infrared-sensor (IR). It records the absorption of infrared rays caused by the measuring gas. The wavelengths of the absorption bands characterise the type of gas, whilst the intensity of the absorption is a measurement for the concentration of the measuring components. In contrary to the analysing cells for methane, the oxygen analyser uses a paramagnetic sensor. The main specifications of the utilized CH₄/O₂ content gas analyser unit are presented below: <table border="1"> <thead> <tr> <th colspan="2">Specifications of the installed continuous CH₄/O₂ content gas analyzer</th></tr> </thead> <tbody> <tr> <td>Internal ID</td><td>BGB.GAE.0016</td></tr> <tr> <td>Position Tag</td><td>QIAL-CH₄, GA141</td></tr> <tr> <td>Manufacturer</td><td>NUK NENNING UND KRUMM GMBH, Germany</td></tr> <tr> <td>Model</td><td>NUK NGA5 CH₄</td></tr> <tr> <td>Serial Number System</td><td>N11032</td></tr> <tr> <td>Serial Number Methane analyser</td><td>A1561</td></tr> <tr> <td>Measuring principle</td><td>Infrared absorption</td></tr> <tr> <td>Measuring range</td><td>0-100 % by volume</td></tr> <tr> <td>Accuracy:</td><td>±2.0 % of full scale</td></tr> </tbody> </table> Source: ^{/40/}	Specifications of the installed continuous CH ₄ /O ₂ content gas analyzer		Internal ID	BGB.GAE.0016	Position Tag	QIAL-CH ₄ , GA141	Manufacturer	NUK NENNING UND KRUMM GMBH, Germany	Model	NUK NGA5 CH ₄	Serial Number System	N11032	Serial Number Methane analyser	A1561	Measuring principle	Infrared absorption	Measuring range	0-100 % by volume	Accuracy:	±2.0 % of full scale
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Accuracy:	±2.0 % of full scale																				
Is accuracy of the monitoring equipment as stated in the CDM PDD? If the CDM PDD does not specify the accuracy of the monitoring equipment, does the monitoring equipment	Yes. The registered PDD and monitoring methodology do not specify any accuracy requirement for the CH₄/O₂ content																				

represent good monitoring practice?	gas analyser unit. The accuracy range for the installed instruments is $\pm 2.0\%$ as per manufacturer's information. It is GLC's contention based on its Sectoral expertise that the use of the installed instrument represents good practise for monitoring of Volumetric fraction of CH ₄ in landfill gas.
Calibration frequency /interval:	Weekly ^{/41/}
Is the calibration interval in line with the monitoring plan of the CDM PDD? If the CDM PDD does not specify the frequency of calibration, does the selected frequency represent good monitoring practice?	The monitoring plan of the registered PDD does not clearly specify calibration intervals. However, the following is stated: <i>"Gas analysers are to be calibrated in accordance to manufacturer recommendation. A zero check and a typical value check should be performed by comparison with a standard gas as recommended by manufacturer"</i>. The monitoring methodology states that the <i>"gas analyser should be subject to a regular maintenance and testing regime to ensure accuracy"</i>. According to manufacturer's recommendation ^{/41/}, the calibration interval for this analyser unit is weekly. Thus, the interval applied by Project Participant can be deemed correct. It is noteworthy that during the monitoring period (from 2011-02-26 to 2012-07-31) the gas analyser has initially not been calibrated in a documented way. In fact, first documented calibration was performed on 2012-04-12. See below for details (point <i>"Did calibration confirm proper functioning of monitoring equipment"</i>)
Company performing the calibration:	<u>Factory inspection / calibration</u> NUK NENNING UND KRUMM GMBH / Germany ^{/42/} <u>On site inspection / calibration</u> Operational Staff of the electricity generation facility. Staff performing the inspections and calibrations has been trained by the supplier of the flaring system, of which the gas analysing unit is part ^{/43/}.
Did calibration confirm proper functioning of monitoring equipment? (Yes / No):	The calibration certificates were checked. Some of the calibrations show deviations higher than the accuracy as defined by the manufacturer. As a reaction to the detected deviations, Project Participant applied a correction due to gas calibration drifts above 1% accuracy tolerance. To be conservative, Project Participant assumes an accuracy of +1 % (instead of +2% as defined by the manufacturer). If during an inspection the reading of the methane analyser (while providing a certified span gas ^{/45/} to the gas analyser covering the range of methane values to be expected during normal operation) shows values $\geq 1.0\%$ than the value of the certified gas, the difference between 1.0 % and the deviation is deducted from all reported methane values for the time between the actual reading and the last inspection. This routine is only applied in cases the

analyser shows a higher value than the standard gas (positive drift), when the analyser is over-reporting. The available calibration events, the detected deviation and the application of a deduction due to detected deviation in case of over reporting is listed below:

Calibration Dates of methane analyser and its deviation		
Calibration date ^{/44/}	Deviation detected ^{/44/} in [%]	Deviation [%] applied* in ER-sheets ^{/5/}
2012-04-12	No. See separate assessment below.	-
2012-04-18	No	-
2012-04-25	No	-
2012-05-02	No	-
2012-05-14	No	-
2012-05-21	No	-
2012-05-28	No.	-
2012-06-04	Yes, 3.0	2.0
2012-06-12	No.	-
2012-06-15	Yes, 2.6	1.6
2012-06-19	Yes, 3.0	2.0
2012-06-25	Yes, 2.7	1.7
2012-06-29	Yes, 2.4	0.2
2012-07-02	Yes, 1.2	0.2
2012-07-05	Yes, 2.6	1.6
2012-07-09	Yes, 2.7	1.7
2012-07-13	Yes, 2.7	1.7
2012-07-16	No.	-
2012-07-20	Yes, 2.7	1.7
2012-07-24	Yes, 2.0	1.0
2012-07-27	Yes, 1.9	0.9
2012-07-31	Yes, 2.2	1.2
2012-08-06	Yes, 2.3	1.3

**for the period between the calibration in question and the last performed calibration / inspection event.*

	As noted by Project Participant in the Monitoring Report and as verified by check of documents and conducted interviews by GLC's Assessment Team, between commissioning of the degassing unit 2011-06-25 and 2012-04-12, no calibrations are documented. As per calibration protocol of 2012-04-12 ^{/44/}, the methane analyser displayed a value of 55.9 % while being provided a certified span gas ^{/45/} with a methane concentration of 60.1 %. This means, that the analyser is reading 4.2 % lower than the span gas. An under-reporting (negative drift) can be stated. Thus, Project Participant did not apply any deduction from the read values until 2012-04-12. During the on-site visit, GLC's Assessment Team conducted interviews with technical staff on project site. While calibrations have not been documented, it has been confirmed by technical staff that the analysing system has been maintained according to manufacturer's recommendation (i.e.change of filters, functional checks, removal of condensate). This is plausible as a visual check of the analysing system (installed in a switchboard) by GLC's Assessment Team during the on-site visit showed the system to be in a good shape. Additionally, a part of the installed gas engines rely on the signal of CH₄-content (the power load is a function of CH₄-content), so it's the own best interest of Project Participant to have a proper signal of actual methane content in LFG sent to the gas engines. Given the circumstances that the analysing system has been properly maintained during commissioning till 2012-04-12 (period without any documented calibration event) and that it has been found underreporting during the first calibration on 2012-04-12, it is acceptable that Project Participant does not apply any deduction for recorded methane values for the period in question. For all further calibration events, the deduction (if necessary) as per table above has been applied.
Is (are) calibration(s) valid for the whole reporting period?	Yes. It is noteworthy that the first calibration after commissioning (on 2011-06-25) has been performed on 2012-04-12 ^{/44/}. From this date on till the end of the monitoring period, calibrations have been done weekly or in an even shorter frequency. Given the fact that for the period without any documented calibration (2011-06-25 to 2012-04-12) the instrument has been underreporting (details see above) and that for the period from 2012-04-12 to the end of the monitoring period the calibration frequency as per manufacturer's recommendation has been applied, it is the opinion of GLC's Assessment Team that the calibrations are valid for the whole reporting period.

If applicable, has the reported data been cross-checked with other available data?	Not applicable.
How were the values in the monitoring report verified?	During the on-site visit, Project Participant downloaded the electronic data (for the time covering the recent monitoring period) as available on the project's data acquisition system. This raw data ^{/14/} has been handed over to GLC's Assessment Team. It is noteworthy that raw data is downloaded in an encrypted file and must be converted into spreadsheets using special software. GLC's Assessment Team has converted the complete set of raw data and compared the so created data against the data provided in the ER spreadsheets ^{/15/}. No quantitative deviations or differences were identified while comparing the two data bases as explained above. Moreover, it is noteworthy that the instruments related to landfill gas flow, landfill gas methane content and flare's exhaust gas temperature have been commissioned on 25th June 2011. Thus, for the months of February, March, April and May 2011, emission reductions from capture and destruction of methane contained in landfill gas are not claimed. (For these months, only the emission reductions for CO₂-displacement (through electricity exported to the grid) are claimed and verified). This is assessed as reasonable and conservative.
Does the data management (from monitoring equipment to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes, it is confirmed from document review and onsite verification that the data management ensures correct transfer of data and reporting of emission reductions and the necessary QA/QC procedures are in place and are continuously improved.

Monitored Parameter	Assessment/ Observation
Data / Parameter: (as in monitoring plan of the CDM PDD):	EL _{im,y} Amount of electricity imported by the Project from the grid [MWh]
Measuring frequency:	EL _{im,y} has been continuously measured.
Reporting frequency:	Every month.
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes. As per ACM0001 (version 11) ^{/7/}, "continuous" is indicated in the row "<i>Monitoring Frequency</i>" of the applicable table in the monitoring methodology; reporting frequency is not indicated As per the monitoring plan of the registered PDD ^{/15/}, this parameter is categorized into the group of parameters

	which “are to be updated on hourly basis”. Additionally, it is stated that data “is logged consolidated monthly to be benchmarked against receipt from buyer (PLN)”. Thus, the adopted measuring and reporting frequencies for this parameter, i.e. continuous measuring, monthly reporting, are assumed as in accordance with the monitoring plan of the registered PDD ^{/15/} and monitoring methodology ACM0001 (version 11) ^{/7/}.																																
Type of monitoring equipment:	Different electricity meters were used to monitor EL_{im,y} during the considered monitoring period. It is noteworthy that the electricity meters installed are owned and operated by the power grid operator and electricity buyer “PLN” (PT. PLN (Persero)). The specifications of the installed electricity meters are presented below: <table border="1"> <thead> <tr> <th colspan="2">Specifications of the installed electricity meter</th></tr> </thead> <tbody> <tr> <td>Internal ID</td><td>Instrument I</td></tr> <tr> <td>Position Tag</td><td>EM-02</td></tr> <tr> <td>Period installed</td><td>2010-09-23 to 2012-01-12</td></tr> <tr> <td>Manufacturer</td><td>Actaris</td></tr> <tr> <td>Model</td><td>SL761B071</td></tr> <tr> <td>Serial Number</td><td>53081963</td></tr> <tr> <td>Multiplying factor</td><td>8,000</td></tr> <tr> <td>Accuracy Class</td><td>0.5</td></tr> </tbody> </table> Source: ^{/46/} Due to changes implemented by the owner of the electricity meter (PLN), the following meter has been installed: <table border="1"> <thead> <tr> <th colspan="2">Specifications of the installed electricity meter</th></tr> </thead> <tbody> <tr> <td>Internal ID</td><td>Instrument II</td></tr> <tr> <td>Position Tag</td><td>EM-02</td></tr> <tr> <td>Period installed</td><td>2012-01-12 to 2012-07-31</td></tr> <tr> <td>Manufacturer</td><td>EDMI</td></tr> <tr> <td>Model</td><td>Mk6E-N680</td></tr> <tr> <td>Serial Number</td><td>211556670</td></tr> </tbody> </table>	Specifications of the installed electricity meter		Internal ID	Instrument I	Position Tag	EM-02	Period installed	2010-09-23 to 2012-01-12	Manufacturer	Actaris	Model	SL761B071	Serial Number	53081963	Multiplying factor	8,000	Accuracy Class	0.5	Specifications of the installed electricity meter		Internal ID	Instrument II	Position Tag	EM-02	Period installed	2012-01-12 to 2012-07-31	Manufacturer	EDMI	Model	Mk6E-N680	Serial Number	211556670
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	<table><tr><td>Multiplying factor</td><td>16,000</td></tr><tr><td>Accuracy Class</td><td>0.2s</td></tr></table> Source: ^{/47/} As the power grid showed instabilities, the operator PLN decided to implement a new line in addition to the existing single line. This change was implemented on 2012-05-25. The already existing line is named “JAGUAR”, while the new line is named “MERCY”. The electricity meter of the existing line “JAGUAR” remained in place on position tag EM-02, while for the new line “MERCY” an own electricity meter on position tag EM-04 has been installed. The specifications of this new electricity meter are listed below: <table><tr><th colspan="2">Specifications of the installed electricity meter</th></tr><tr><td>Position Tag</td><td>EM-04</td></tr><tr><td>Period installed</td><td>2012-05-25 to 2012-07-31</td></tr><tr><td>Manufacturer</td><td>EDMI</td></tr><tr><td>Model</td><td>Mk6E-N680</td></tr><tr><td>Serial Number</td><td>212063056</td></tr><tr><td>Multiplying factor</td><td>16,000</td></tr><tr><td>Accuracy Class</td><td>0.2s</td></tr></table> Source: ^{/48/} It is noteworthy, that for the ER calculation, the bills of the grid operator PLN towards PT Navigat Organic Energy Indonesia are taken into account ^{/49/}.	Multiplying factor	16,000	Accuracy Class	0.2s	Specifications of the installed electricity meter		Position Tag	EM-04	Period installed	2012-05-25 to 2012-07-31	Manufacturer	EDMI	Model	Mk6E-N680	Serial Number	212063056	Multiplying factor	16,000	Accuracy Class	0.2s
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Accuracy Class	0.2s																				
Is accuracy of the monitoring equipment as stated in the CDM PDD? If the CDM PDD does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practice?	Yes. The registered PDD and monitoring methodology do not specify any accuracy requirement for the electricity meter, installed at the project site to monitor EL_{im,y}. The accuracy classes for the installed instruments are 0.2s to 0.5 as per manufacturer's information. It is GLC's contention based on its Sectoral expertise that the use of the installed instrument represents good practise for monitoring of electricity imported by the Project from the grid.																				
Calibration frequency /interval:	As the electricity meters are owned and operated by another party (PLN); Project Participant has no influence on calibration regime. Thus, no assessment by GLC's Assessment Team of this point has been performed.																				
Is the calibration interval in line with the monitoring plan of the CDM PDD? If the CDM PDD does not specify the frequency of	As the electricity meters are owned and operated by another party (PLN); Project Participant has no influence on calibration regime. Thus, no assessment by GLC's																				

calibration, does the selected frequency represent good monitoring practice?	Assessment Team of this point has been performed.
Company performing the calibration:	As the electricity meters are owned and operated by another party (PLN); Project Participant has no influence on calibration regime. Thus, no assessment by GLC's Assessment Team of this point has been performed.
Did calibration confirm proper functioning of monitoring equipment? (Yes / No):	As the electricity meters are owned and operated by another party (PLN); Project Participant has no influence on calibration regime. Thus, no assessment by GLC's Assessment Team of this point has been performed.
Is (are) calibration(s) valid for the whole reporting period?	As the electricity meters are owned and operated by another party (PLN); Project Participant has no influence on calibration regime. Thus, no assessment by GLC's Assessment Team of this point has been performed.
If applicable, has the reported data been cross-checked with other available data?	Not applicable.
How were the values in the monitoring report verified?	GLC's Assessment Team checked the provided invoices ^{/49/} towards PT Navigat Organic Energy Indonesia. GLC's Assessment Team compared the presented data against the data provided in the ER spreadsheets ^{/5/} . No quantitative deviations or differences were identified while comparing the two data bases as explained above.
Does the data management (from monitoring equipment to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes, it is confirmed from document review and onsite verification that the data management ensures correct transfer of data and reporting of emission reductions and the necessary QA/QC procedures are in place and are continuously improved.

Monitored Parameter	Assessment/ Observation
Data / Parameter: (as in monitoring plan of the CDM PDD):	EL _{ex,y} Amount of electricity exported by the Project to the grid [MWh]
Measuring frequency:	EL _{ex,y} has been continuously measured.
Reporting frequency:	Every month.
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes. As per ACM0001 (version 11) ^{/7/} , "continuous" is indicated in the row " <i>Monitoring Frequency</i> " of the applicable table in the monitoring methodology. Reporting frequency is not indicated. As per the monitoring plan of the registered PDD ^{/15/} , this parameter is categorized into the group of parameters

	which “are to be updated on hourly basis”. Additionally, it is stated that data “is logged consolidated monthly to be benchmarked against receipt from buyer (PLN)”. Thus, the adopted measuring and reporting frequencies for this parameter, i.e. continuous measuring, monthly reporting, are assumed as in accordance with the monitoring plan of the registered PDD ^{/15/} and monitoring methodology ACM0001 (version 11) ^{/7/}.																																				
Type of monitoring equipment:	Different electricity meters were used to monitor EL_{ex,y} during the considered monitoring period. It is noteworthy that the electricity meters installed are owned and operated by the power grid operator and electricity buyer “PLN” (PT. PLN (Persero)). The specifications of the installed electricity meters are presented below: <table border="1"> <thead> <tr> <th colspan="2">Specifications of the installed electricity meter</th></tr> </thead> <tbody> <tr> <td>Internal ID</td><td>Instrument I</td></tr> <tr> <td>Position Tag</td><td>EM-02</td></tr> <tr> <td>Period installed</td><td>2010-09-23 to 2012-01-12</td></tr> <tr> <td>Manufacturer</td><td>Actaris</td></tr> <tr> <td>Model</td><td>SL761B071</td></tr> <tr> <td>Serial Number</td><td>53081963</td></tr> <tr> <td>Multiplying factor</td><td>8,000</td></tr> <tr> <td>Accuracy Class</td><td>0.5</td></tr> </tbody> </table> Source: ^{/46/} Due to changes implemented by the owner of the electricity meter (PLN), the following meter has been installed: <table border="1"> <thead> <tr> <th colspan="2">Specifications of the installed electricity meter</th></tr> </thead> <tbody> <tr> <td>Internal ID</td><td>Instrument II</td></tr> <tr> <td>Position Tag</td><td>EM-02</td></tr> <tr> <td>Period installed</td><td>2012-01-12 to 2012-07-31</td></tr> <tr> <td>Manufacturer</td><td>EDMI</td></tr> <tr> <td>Model</td><td>Mk6E-N680</td></tr> <tr> <td>Serial Number</td><td>211556670</td></tr> <tr> <td>Multiplying factor</td><td>16,000</td></tr> <tr> <td>Accuracy Class</td><td>0.2s</td></tr> </tbody> </table>	Specifications of the installed electricity meter		Internal ID	Instrument I	Position Tag	EM-02	Period installed	2010-09-23 to 2012-01-12	Manufacturer	Actaris	Model	SL761B071	Serial Number	53081963	Multiplying factor	8,000	Accuracy Class	0.5	Specifications of the installed electricity meter		Internal ID	Instrument II	Position Tag	EM-02	Period installed	2012-01-12 to 2012-07-31	Manufacturer	EDMI	Model	Mk6E-N680	Serial Number	211556670	Multiplying factor	16,000	Accuracy Class	0.2s
Specifications of the installed electricity meter																																					
Internal ID	Instrument I																																				
Position Tag	EM-02																																				
Period installed	2010-09-23 to 2012-01-12																																				
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Multiplying factor	16,000																																				
Accuracy Class	0.2s																																				

	Source: ^{/47/} As the power grid showed instabilities, the operator PLN decided to implement a new line in addition to the existing single line. This change was implemented on 2012-05-25. The already existing line is named "JAGUAR", while the new line is named "MERCY". The electricity meter of the existing line "JAGUAR" remained in place on position tag EM-02, while for the new line "MERCY" an own electricity meter on position tag EM-04 has been installed. The specifications of this new electricity meter are listed below: <table border="1"> <thead> <tr> <th colspan="2">Specifications of the installed electricity meter</th></tr> </thead> <tbody> <tr> <td>Position Tag</td><td>EM-04</td></tr> <tr> <td>Period installed</td><td>2012-05-25 to 2012-07-31</td></tr> <tr> <td>Manufacturer</td><td>EDMI</td></tr> <tr> <td>Model</td><td>Mk6E-N680</td></tr> <tr> <td>Serial Number</td><td>212063056</td></tr> <tr> <td>Multiplying factor</td><td>16,000</td></tr> <tr> <td>Accuracy Class</td><td>0.2s</td></tr> </tbody> </table> Source: ^{/48/} It is noteworthy, that for the ER calculation, the bills of the grid operator PLN towards PT Navigat Organic Energy Indonesia are taken into account ^{/49/}.	Specifications of the installed electricity meter		Position Tag	EM-04	Period installed	2012-05-25 to 2012-07-31	Manufacturer	EDMI	Model	Mk6E-N680	Serial Number	212063056	Multiplying factor	16,000	Accuracy Class	0.2s
Specifications of the installed electricity meter																	
Position Tag	EM-04																
Period installed	2012-05-25 to 2012-07-31																
Manufacturer	EDMI																
Model	Mk6E-N680																
Serial Number	212063056																
Multiplying factor	16,000																
Accuracy Class	0.2s																
Is accuracy of the monitoring equipment as stated in the CDM PDD? If the CDM PDD does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practice?	Yes. The registered PDD and monitoring methodology do not specify any accuracy requirement for the electricity meter, installed at the project site to monitor $EL_{ex,y}$. The accuracy classes for the installed instruments are 0.2s to 0.5 as per manufacturer's information. It is GLC's contention based on its Sectoral expertise that the use of the installed instrument represents good practise for monitoring of electricity imported by the Project from the grid.																
Calibration frequency /interval:	As the electricity meters are owned and operated by another party (PLN); Project Participant has no influence on calibration regime. Thus, no assessment by GLC's Assessment Team of this point has been performed.																
Is the calibration interval in line with the monitoring plan of the CDM PDD? If the CDM PDD does not specify the frequency of calibration, does the selected frequency represent good monitoring practice?	As the electricity meters are owned and operated by another party (PLN); Project Participant has no influence on calibration regime. Thus, no assessment by GLC's Assessment Team of this point has been performed.																

Company performing the calibration:	As the electricity meters are owned and operated by another party (PLN); Project Participant has no influence on calibration regime. Thus, no assessment by GLC's Assessment Team of this point has been performed.
Did calibration confirm proper functioning of monitoring equipment? (Yes / No):	As the electricity meters are owned and operated by another party (PLN); Project Participant has no influence on calibration regime. Thus, no assessment by GLC's Assessment Team of this point has been performed.
Is (are) calibration(s) valid for the whole reporting period?	As the electricity meters are owned and operated by another party (PLN); Project Participant has no influence on calibration regime. Thus, no assessment by GLC's Assessment Team of this point has been performed.
If applicable, has the reported data been cross-checked with other available data?	Not applicable.
How were the values in the monitoring report verified?	GLC's Assessment Team checked the provided invoices ^{/49/} towards PT Navigat Organic Energy Indonesia. GLC's Assessment Team compared the presented data against the data provided in the ER spreadsheets ^{/5/} . No quantitative deviations or differences were identified while comparing the two data bases as explained above.
Does the data management (from monitoring equipment to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes, it is confirmed from document review and onsite verification that the data management ensures correct transfer of data and reporting of emission reductions and the necessary QA/QC procedures are in place and are continuously improved.

Monitored Parameter	Assessment/ Observation
Data / Parameter: (as in monitoring plan of the CDM PDD):	$T_{\text{flare},h}$ The flare temperature profile throughout the hour h (temperature of exhaust gas) [°C]
Measuring frequency:	$T_{\text{flare},h}$ has been continuously measured, where continuous measurements have been recorded and reported with an every minute frequency.
Reporting frequency:	Every minute.
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes. It is noteworthy that the parameter $T_{\text{flare},h}$ belongs to the determination of $PE_{\text{flare},y}$ (Project emissions from flaring of the residual gas stream in year y). The monitoring methodology ACM0001 (version 11) ^{/7/} refers for the determination of $PE_{\text{flare},y}$ to the methodological "Tool to determine project emissions from flaring gases containing

	methane”^{/9/}. As per referred tool, “continuous” is indicated in the row “Monitoring Frequency” of the applicable table in the tool. Reporting frequency is not indicated. As per the monitoring plan of the registered PDD^{/15/}, “Temperature is to be logged minimum per minute basis to be able to establish its hourly profile”. Thus, the adopted measuring and reporting frequencies for this parameter, i.e. continuous measuring, minute-by-minute reporting, are assumed as in accordance with the monitoring plan of the registered PDD^{/15/} and monitoring methodology ACM0001 (version 11)^{/7/}.																						
Type of monitoring equipment:	A thermocouple of type “N” is used to monitor $T_{\text{flare,h}}$ during the considered monitoring period. This thermocouple consists of a ceramic protection tube, in which two thermocouple wires are inserted. These wires consist of different materials (conductors), in which a temperature difference between these conductors produces a voltage difference between the two substances, which can be measured and transferred into a temperature reading. The specifications of the thermocouple are presented below: <table border="1"> <thead> <tr> <th colspan="2">Specifications of the installed thermocouple</th></tr> </thead> <tbody> <tr> <td>Internal ID</td><td>BGB.TIC.0015</td></tr> <tr> <td>Position Tag</td><td>TICAH 81.26</td></tr> <tr> <td>Period installed</td><td>2011-06-05 to 2012-08-30</td></tr> <tr> <td>Manufacturer</td><td>Jumo GmbH & Co.KG, Germany</td></tr> <tr> <td>Model</td><td>Thermocouple Type N</td></tr> <tr> <td>Serial Number</td><td>Without any serial number</td></tr> <tr> <td>Measuring principle</td><td>Seebeck effect</td></tr> <tr> <td>Maximum temperature</td><td>1,200 °C</td></tr> <tr> <td>Defined up to</td><td>1,300 °C</td></tr> <tr> <td>Tolerance class</td><td>±2.5 °C</td></tr> </tbody> </table> Source: ^{/50/} ^{/51/}	Specifications of the installed thermocouple		Internal ID	BGB.TIC.0015	Position Tag	TICAH 81.26	Period installed	2011-06-05 to 2012-08-30	Manufacturer	Jumo GmbH & Co.KG, Germany	Model	Thermocouple Type N	Serial Number	Without any serial number	Measuring principle	Seebeck effect	Maximum temperature	1,200 °C	Defined up to	1,300 °C	Tolerance class	±2.5 °C
Specifications of the installed thermocouple																							
Internal ID	BGB.TIC.0015																						
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Defined up to	1,300 °C																						
Tolerance class	±2.5 °C																						
Is accuracy of the monitoring equipment as stated in the CDM PDD? If the CDM PDD does not specify the accuracy of the monitoring	Yes. The registered PDD and monitoring methodology do not specify any accuracy requirement for the thermocouple																						

equipment, does the monitoring equipment represent good monitoring practice?	installed at the project site to monitor $T_{\text{flare,h}}$. The accuracy range for the installed instruments is $\pm 2.5^{\circ}\text{C}$ as per manufacturer's information. It is GLC's contention based on its Sectoral expertise that the use of the installed instrument represents good practise for monitoring of flare's exhaust gas temperature.
Calibration frequency /interval:	Yearly
Is the calibration interval in line with the monitoring plan of the CDM PDD? If the CDM PDD does not specify the frequency of calibration, does the selected frequency represent good monitoring practice?	The monitoring plan of the registered PDD does clearly specify calibration intervals. The following is stated: <i>"Thermocouple should be replaced or calibrated annually"</i>. The monitoring methodology resp. the referred tool gives the same statement. It is noteworthy that the thermocouple (commissioning on 2011-06-25) was scheduled for calibration resp. replacement with a new unit no later than 2012-06-24. As per statement of Project Participant (and confirmed by interviews with operational staff during the on-site visit) the replacement took place on 2012-08-30. This delay bases on sourcing and transport problems (unit has to be ordered in Europe). This delay in calibration resp. replacement can be accepted for the following reasons: Between 2012-06-24 (date the calibration/replacement was due) and 2012-07-31 (last day of the monitoring period), the flare was operational for 47 min (of 912 minutes for this period). This represents an operation of 5 % for this time. Additionally, in the period mentioned above, 787 Nm³ have been combusted in the enclosed high-temperature flare. This is 1.42 % of all gas sent to the flare and 0.0026 % of all gas captured by the Project Activity during the whole monitoring period. It is noteworthy that the combustion process of the flare is automatically controlled by a second thermocouple (installed in the center of the flare stack). This control allows a high-efficient and low-emission combustion according to various strong european standards. Furthermore, no indication has been found that the flare has not been properly working. While the flare was not operating continuously for the whole monitoring period, the thermocouple in question is likely suffering less wear and tear than in continuous operation. Finally, given the fact that flare operated only for a limited time (for 2,026 minutes the flare has been operative) and within a small period, it is reasonable to accept this delay. Thus, the interval applied by Project Participant can be deemed correct.
Company performing the calibration:	Project Participant decided to replace the used unit with a new unit.

Did calibration confirm proper functioning of monitoring equipment? (Yes / No):	Not applicable as a new unit has been installed.
Is (are) calibration(s) valid for the whole reporting period?	Not applicable as a new unit has been installed.
If applicable, has the reported data been cross-checked with other available data?	Not applicable.
How were the values in the monitoring report verified?	During the on-site visit, Project Participant downloaded the electronic data (for the time covering the recent monitoring period) as available on the project's data acquisition system. This raw data ^{/14/} has been handed over to GLC's Assessment Team. It is noteworthy that raw data is downloaded in an encrypted file and must be converted into spreadsheets using special software. GLC's Assessment Team has converted the complete set of raw data and compared the so created data against the data provided in the ER spreadsheets ^{/5/}. No quantitative deviations or differences were identified while comparing the two data bases as explained above.
Does the data management (from monitoring equipment to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes, it is confirmed from document review and onsite verification that the data management ensures correct transfer of data and reporting of emission reductions and the necessary QA/QC procedures are in place and are continuously improved.

Monitored Parameter	Assessment/ Observation
Data / Parameter: (as in monitoring plan of the CDM PDD):	Other flare operational parameters [no data unit] The monitoring is done by defining if operational conditions are met. As per manufacturer's specification, there are four operational conditions: (a) gas flow rate to FL-01 up to 2,500Nm³/hr (b) combustion temperature 1,000 to 1,200degC; (c) flare auto start-up or shut-down upon detection of landfill gas pressure (equivalent to flow); (d) burner temperature regulator to maintain combustion range temperature stable within the range. Based on document review and on-site interview, manufacturer has set during commissioning the setpoints for an optimized combustion and this setting is password protected. The assessment team thus assumes that flare is operating acc. to manufacturer's recommendation as long

	as there is no malfunction in the PLC and the maintenance is correctly done. One S-type thermocouple is installed to monitor the combustion temperature. As per manufacturer's specification, the flare can be seen as operational if combustion temperature is in a range between 1,000 and 1,200°C. An automatic burner control regulates combustion temperature to make sure it is always within the range. The flare has been maintained as per manufacturer specifications and there are no malfunctions reported of the automatic combustion. Checking the data reveals that there are in total 9 minutes where LFG_{flare} exceeds the benchmark of 2,500 Nm³/h, among which 4 minutes are on 2011-06-24 (during commissioning) and 5 minutes are on 2011-12-04 (on which date the calibration was done). No corrections is deemed necessary due to the following reasons: -The occasions only happened in evidenced commissioning phase and calibration date; under "normal" operating conditions, such an exceeding did not occur. - The occasions are very short time, which can be deemed as within the 'materiality limits'; -If we correct this flowrate to its maximum capacity of 2,500Nm³, the results would be lower Project Emission which is less conservative than taking the real value: a) The high value for $LFG_{flare,y}$ is not considered in calculation of $MD_{project,y}$ as the formula is taking the lower value (comparison between $LFG_{total,y}$ and sum of $LFG_{flare,y}$ and $LFG_{electricity,y}$). b) Additionally, the high value, combined with a 0% efficiency and roundabout 50% of methane creates high $PE_{flare,y}$ which are finally deducted. As for the period of 2011-12-09, roundabout 260 tCO₂ are calculated as PE_{flare} and finally deducted what is more than conservative.
Measuring frequency:	N/A. Combustion temperature is automatically controlled to ensure the compliance.
Reporting frequency:	N/A. Combustion temperature is automatically controlled to ensure the compliance.
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	N/A. Combustion temperature is automatically controlled to ensure the compliance.
Type of monitoring equipment:	S-type thermocouple. Combustion temperature is automatically controlled to ensure the compliance.

Is accuracy of the monitoring equipment as stated in the CDM PDD? If the CDM PDD does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practice?	Combustion temperature is automatically controled to ensure the compliance.
Calibration frequency /interval:	N/A. Combustion temperature is automatically controled to ensure the compliance.
Is the calibration interval in line with the monitoring plan of the CDM PDD? If the CDM PDD does not specify the frequency of calibration, does the selected frequency represent good monitoring practice?	N/A. Combustion temperature is automatically controled to ensure the compliance.
Company performing the calibration:	N/A. Combustion temperature is automatically controled to ensure the compliance.
Did calibration confirm proper functioning of monitoring equipment? (Yes / No):	N/A. Combustion temperature is automatically controled to ensure the compliance.
Is (are) calibration(s) valid for the whole reporting period?	N/A. Combustion temperature is automatically controled to ensure the compliance.
If applicable, has the reported data been cross-checked with other available data?	N/A. Combustion temperature is automatically controled to ensure the compliance, and its value is not directly used in ER calculation.
How were the values in the monitoring report verified?	N/A. Combustion temperature is automatically controled to ensure the compliance, and its value is not directly used in ER calculation.
Does the data management (from monitoring equipment to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	N/A. Combustion temperature is automatically controled to ensure the compliance, and its value is not directly used in ER calculation.

Monitored Parameter	Assessment/ Observation
Data / Parameter: (as in monitoring plan of the CDM PDD):	FC _{i,m,y} Consumption of fuel type I in year y for power plant in group m respectively (applied in EF calculation) [kJ]
Measuring frequency:	Annually
Reporting frequency:	Annually
Is measuring and reporting frequency in	Yes. As per the monitoring plan and methodology, this

accordance with the monitoring plan and monitoring methodology? (Yes / No)	parameter should be updated yearly for OM calculation and updated during crediting period renewal for BM calculation. Thus the annual monitoring and reporting in actual practice is in accordance with the monitoring plan and monitoring methodology.
Type of monitoring equipment:	N/A. Monitored values of this project come from EF sheets issued by Ministry of Energy and Mineral Resources ^{/58/} . This is assessed as official and reliable source.
Is accuracy of the monitoring equipment as stated in the CDM PDD? If the CDM PDD does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practice?	N/A. Monitored values of this project come from EF sheets issued by Ministry of Energy and Mineral Resources ^{/58/} . This is assessed as official and reliable source.
Calibration frequency /interval:	N/A
Is the calibration interval in line with the monitoring plan of the CDM PDD? If the CDM PDD does not specify the frequency of calibration, does the selected frequency represent good monitoring practice?	N/A
Company performing the calibration:	N/A
Did calibration confirm proper functioning of monitoring equipment? (Yes / No):	N/A
Is (are) calibration(s) valid for the whole reporting period?	N/A
If applicable, has the reported data been cross-checked with other available data?	N/A.
How were the values in the monitoring report verified?	Values of this parameter in EF sheets ^{/57/} are compared with values of this parameter in official sheets issued by Ministry of Energy and Mineral Resources ^{/58/} , and the verifiers hereby confirm that all values used in EF calculation of this project are the same as relevant values contained in official sheets ^{/58/} .
Does the data management (from monitoring equipment to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes. Comparison is done between NOEI's EF sheets ^{/57/} and official sheets issued by Ministry of Energy and Mineral Resources ^{/58/} . Data transfer is assessed as correct. No inconsistency is found.

Monitored Parameter	Assessment/ Observation
Data / Parameter: (as in monitoring plan of the CDM PDD):	NCV _i Net calorific value of fuel type i used by power plant in the grid [TJ/kt or GJ/t]

Measuring frequency:	Annually
Reporting frequency:	Annually
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes. As per the monitoring plan and methodology, this parameter should be updated yearly for OM calculation and updated during crediting period renewal for BM calculation. Thus the annual monitoring and reporting in actual practice is in accordance with the monitoring plan and monitoring methodology.
Type of monitoring equipment:	N/A. Monitored values of this project come from EF sheets issued by Ministry of Energy and Mineral Resources ^{/58/} . This is assessed as official and reliable source.
Is accuracy of the monitoring equipment as stated in the CDM PDD? If the CDM PDD does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practice?	N/A. Monitored values of this project come from EF sheets issued by Ministry of Energy and Mineral Resources ^{/58/} . This is assessed as official and reliable source.
Calibration frequency /interval:	N/A
Is the calibration interval in line with the monitoring plan of the CDM PDD? If the CDM PDD does not specify the frequency of calibration, does the selected frequency represent good monitoring practice?	N/A
Company performing the calibration:	N/A
Did calibration confirm proper functioning of monitoring equipment? (Yes / No):	N/A
Is (are) calibration(s) valid for the whole reporting period?	N/A
If applicable, has the reported data been cross-checked with other available data?	N/A.
How were the values in the monitoring report verified?	Values of this parameter in EF sheets ^{/57/} are compared with values of this parameter in official sheets issued by Ministry of Energy and Mineral Resources ^{/58/} , and the verifiers hereby confirm that all values used in EF calculation of this project are the same as relevant values contained in official sheets ^{/58/} .
Does the data management (from monitoring equipment to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes. Comparison is done between NOEI's EF sheets ^{/57/} and official sheets issued by Ministry of Energy and Mineral Resources ^{/58/} . Data transfer is assessed as correct. No inconsistency is found.

Monitored Parameter	Assessment/ Observation
Data / Parameter:	EF _{CO2,i}

(as in monitoring plan of the CDM PDD):	Emission factor of fuel type i [tCO ₂ /TJ]
Measuring frequency:	Annually
Reporting frequency:	Annually
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes. As per the monitoring plan and methodology, this parameter should be updated yearly for OM calculation and updated during crediting period renewal for BM calculation. Thus the annual monitoring and reporting in actual practice is in accordance with the monitoring plan and monitoring methodology.
Type of monitoring equipment:	N/A. Monitored values of this project come from EF sheets issued by Ministry of Energy and Mineral Resources ^{/58/} . This is assessed as official and reliable source.
Is accuracy of the monitoring equipment as stated in the CDM PDD? If the CDM PDD does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practice?	N/A. Monitored values of this project come from EF sheets issued by Ministry of Energy and Mineral Resources ^{/58/} . This is assessed as official and reliable source.
Calibration frequency /interval:	N/A
Is the calibration interval in line with the monitoring plan of the CDM PDD? If the CDM PDD does not specify the frequency of calibration, does the selected frequency represent good monitoring practice?	N/A
Company performing the calibration:	N/A
Did calibration confirm proper functioning of monitoring equipment? (Yes / No):	N/A
Is (are) calibration(s) valid for the whole reporting period?	N/A
If applicable, has the reported data been cross-checked with other available data?	N/A.
How were the values in the monitoring report verified?	Values of this parameter in EF sheets ^{/57/} are compared with values of this parameter in official sheets issued by Ministry of Energy and Mineral Resources ^{/58/} , and the verifiers hereby confirm that all values used in EF calculation of this project are the same as relevant values contained in official sheets ^{/58/} .
Does the data management (from monitoring equipment to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes. Comparison is done between NOEI's EF sheets ^{/57/} and official sheets issued by Ministry of Energy and Mineral Resources ^{/58/} . Data transfer is assessed as correct. No inconsistency is found.

Monitored Parameter	Assessment/ Observation
Data / Parameter: (as in monitoring plan of the CDM PDD):	$\sum_m EG_{m,y}$ Net electricity delivered to power plants in group m in year y [MWH]
Measuring frequency:	Annually
Reporting frequency:	Annually
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes. As per the monitoring plan and methodology, this parameter should be updated yearly for OM calculation and updated during crediting period renewal for BM calculation. Thus the annual monitoring and reporting in actual practice is in accordance with the monitoring plan and monitoring methodology.
Type of monitoring equipment:	N/A. Monitored values of this project come from EF sheets issued by Ministry of Energy and Mineral Resources ^{/58/} . This is assessed as official and reliable source.
Is accuracy of the monitoring equipment as stated in the CDM PDD? If the CDM PDD does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practice?	N/A. Monitored values of this project come from EF sheets issued by Ministry of Energy and Mineral Resources ^{/58/} . This is assessed as official and reliable source.
Calibration frequency /interval:	N/A
Is the calibration interval in line with the monitoring plan of the CDM PDD? If the CDM PDD does not specify the frequency of calibration, does the selected frequency represent good monitoring practice?	N/A
Company performing the calibration:	N/A
Did calibration confirm proper functioning of monitoring equipment? (Yes / No):	N/A
Is (are) calibration(s) valid for the whole reporting period?	N/A
If applicable, has the reported data been cross-checked with other available data?	N/A.
How were the values in the monitoring report verified?	Values of this parameter in EF sheets ^{/57/} are compared with values of this parameter in official sheets issued by Ministry of Energy and Mineral Resources ^{/58/} , and the verifiers hereby confirm that all values used in EF calculation of this project are the same as relevant values contained in official sheets ^{/58/} .
Does the data management (from monitoring equipment to emission reduction calculation)	Yes. Comparison is done between NOEI's EF sheets ^{/57/} and official sheets issued by Ministry of Energy and

ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Mineral Resources ^{/58/} . Data transfer is assessed as correct. No inconsistency is found.
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Monitored Parameter	Assessment/ Observation
Data / Parameter: (as in monitoring plan of the CDM PDD):	η_m Average net energy conversion efficiency of a power unit included in sub-group "Group A2" of group <i>m</i> power plant. [No unit]
Measuring frequency:	N/A
Reporting frequency:	N/A
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	According to monitoring plan, this parameter applies methodology default value as stipulated in Annex 1 of "Tool to calculate emission factor of an electricity system", Version 2.2.1.
Type of monitoring equipment:	N/A
Is accuracy of the monitoring equipment as stated in the CDM PDD? If the CDM PDD does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practice?	N/A
Calibration frequency /interval:	N/A
Is the calibration interval in line with the monitoring plan of the CDM PDD? If the CDM PDD does not specify the frequency of calibration, does the selected frequency represent good monitoring practice?	N/A
Company performing the calibration:	N/A
Did calibration confirm proper functioning of monitoring equipment? (Yes / No):	N/A
Is (are) calibration(s) valid for the whole reporting period?	N/A
If applicable, has the reported data been cross-checked with other available data?	N/A
How were the values in the monitoring report verified?	Values used in EF calculation is compared against the default value contained in methodology tool and no inconsistency is found.
Does the data management (from monitoring equipment to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes, comparison is done between value used in EF calculation and value contained in methodology tool. Data transfer is assessed as correct.

Monitored Parameter	Assessment/ Observation
Data / Parameter: (as in monitoring plan of the CDM PDD):	MG _{PR,y} Amount of (theoretical) methane generated during year y of the project activity [tCH ₄]
Measuring frequency:	Annually
Reporting frequency:	Annually
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes, annual updating is in accordance with MP and methodology.
Type of monitoring equipment:	N/A. This parameter is a theoretical calculation based on existing waste already deposited in the landfill, incoming amount of waste actually disposed, and default factors, using Methodological Tool. The only monitored sub-parameter for calculation of MG _{PR,y} is Annual quantity of waste delivered to the landfill. Refer to the next table for A _{T,x} for assessment.
Is accuracy of the monitoring equipment as stated in the CDM PDD? If the CDM PDD does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practice?	N/A
Calibration frequency /interval:	N/A
Is the calibration interval in line with the monitoring plan of the CDM PDD? If the CDM PDD does not specify the frequency of calibration, does the selected frequency represent good monitoring practice?	N/A
Company performing the calibration:	N/A
Did calibration confirm proper functioning of monitoring equipment? (Yes / No):	N/A
Is (are) calibration(s) valid for the whole reporting period?	N/A
If applicable, has the reported data been cross-checked with other available data?	N/A. This parameter is not involved in ER calculation at all and it is not necessary to conduct cross check.
How were the values in the monitoring report verified?	The parameter is calculated and the verifiers hereby confirm that the calculation is in accordance with methodology and registered PDD.
Does the data management (from monitoring equipment to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC	The parameter is calculated and the verifiers hereby confirm that the calculation is in accordance with methodology and registered PDD.

processes in place?	
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Monitored Parameter	Assessment/ Observation
Data / Parameter: (as in monitoring plan of the CDM PDD):	$A_{T,x}$ Annual quantity of waste delivered to the landfill in year x; [tonnes]
Measuring frequency:	Per incident at receiving station
Reporting frequency:	Monthly
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	According to the methodology tool ^{/10/} , this parameter should be monitored "Continuously, aggregated at least annually for year x or monthly for month i" According to MP, this parameter should be reported monthly. Thus the actual practice, i.e. measuring per incident and reporting monthly, is in accordance with the monitoring plan and methodology tool ^{/10/} .
Type of monitoring equipment:	Weighbridge Class III KUBOTA KLD1000, Serial Number CC 049461032, Accuracy +/-20kg ^{/60/}
Is accuracy of the monitoring equipment as stated in the CDM PDD? If the CDM PDD does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practice?	No accuracy requirement is indicated in CDM PDD. The actual monitoring equipment's accuracy is +/-20kg, which is assessed as representing good monitoring practice.
Calibration frequency /interval:	Annually. However, since start of operation only one calibration was done, which is dated 2012-07-02.
Is the calibration interval in line with the monitoring plan of the CDM PDD? If the CDM PDD does not specify the frequency of calibration, does the selected frequency represent good monitoring practice?	Methodology tool ^{/10/} does not indicate the calibration interval. As per monitoring plan in PDD, Weighbridge is subject to regular calibration as required by national regulation for trade. The actual designed calibration frequency, i.e. annually, is assessed as representing good monitoring practice. The delayed calibration would in principle result in discounted emission reduction, following applicable guideline in VVS ^{/2/} . However, since monitoring result of this parameter is not involved in ER calculation, the delayed calibration has no impact on ER.
Company performing the calibration:	KARAWANG METROLOGY CENTER
Did calibration confirm proper functioning of monitoring equipment? (Yes / No):	Yes.
Is (are) calibration(s) valid for the whole reporting period?	No. Only one calibration was done, which was on 2012-07-02 and is valid until 2013-07-02. The period before 2012-

	07-02 is not covered. However, since monitoring result of this parameter is not involved in ER calculation, the delayed calibration has no impact on ER.
If applicable, has the reported data been cross-checked with other available data?	Not applicable. The reported data are already provided by 3 rd party operator. Moreover, this parameter is for reference only; it is not involved in the ER calculation.
How were the values in the monitoring report verified?	Values are compared against the waste reports provided by 3 rd party operator ^{/59/} .
Does the data management (from monitoring equipment to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Values are compared against the waste reports provided by 3 rd party operator ^{/59/} and no inconsistency is found.

Monitored Parameter	Assessment/ Observation
Data / Parameter: (as in monitoring plan of the CDM PDD):	OH _{PP} Operation hour of the power plant [hours] For the considered monitoring period, monitoring of "Operational Hours of the power plant" was performed on the basis of the parameter LFG _{electricity,t} . The power plant is counted as "not operational" if for more than 59 minutes the measured flow of the parameter LFG _{electricity,t} (internal position tag FIR 90.0) is ≤1Nm ³ /h. In case the before described condition is not met, the power plant is counted as operational.
Measuring frequency:	Every minute.
Reporting frequency:	Monthly.
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	Yes. Monitoring plan of the registered PDD ^{/15/} does not mention any frequency of monitoring/recording. As per ACM0001 (version 11) ^{/7/} , "annually" is indicated in the row "Monitoring Frequency" of the applicable table in the monitoring methodology and it does not specify the reporting frequency. Thus, the adopted measuring and reporting frequencies for this parameter, i.e. minute-by-minute measuring and monthly reporting, are assumed as in accordance with the monitoring plan of the registered PDD ^{/15/} and monitoring methodology ACM0001 (version 11) ^{/7/} .
Type of monitoring equipment:	N/A. It is electronically recorded from flow meter,

	calculated from duration of which the power plant is receiving gas. For further details see assessment of parameter $LFG_{electricity,t}$.
Is accuracy of the monitoring equipment as stated in the CDM PDD? If the CDM PDD does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practice?	N/A. For further details see assessment of parameter $LFG_{electricity,t}$.
Calibration frequency /interval:	N/A. For further details see assessment of parameter $LFG_{electricity,t}$.
Is the calibration interval in line with the monitoring plan of the CDM PDD? If the CDM PDD does not specify the frequency of calibration, does the selected frequency represent good monitoring practice?	N/A. For further details see assessment of parameter $LFG_{electricity,t}$.
Company performing the calibration:	N/A. For further details see assessment of parameter $LFG_{electricity,t}$.
Did calibration confirm proper functioning of monitoring equipment? (Yes / No):	N/A. For further details see assessment of parameter $LFG_{electricity,t}$.
Is (are) calibration(s) valid for the whole reporting period?	N/A. For further details see assessment of parameter $LFG_{electricity,t}$.
If applicable, has the reported data been cross-checked with other available data?	N/A. For further details see assessment of parameter $LFG_{electricity,t}$.
How were the values in the monitoring report verified?	GLC's Assessment Team checked the application of the above described condition to distinguish between operational and non-operational conditions for the power plant in the monthly CH ₄ abatement spreadsheets ^{/5/} in order to monitor the operational hours of the power plant. No quantitative deviations or differences were identified while the check of the spreadsheets. For further details see assessment of parameter $LFG_{electricity,t}$.
Does the data management (from monitoring equipment to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes, it is confirmed from document review and onsite verification that the data management ensures correct transfer of data and reporting of emission reductions and the necessary QA/QC procedures are in place and are continuously improved.

Monitored Parameter	Assessment/ Observation
Data / Parameter: (as in monitoring plan of the CDM PDD):	FC _{ij,y} Amount of fuel consumed by process type j, where process j is the type of additional units/equipment/machineries that are employed due to project activity which in absence of project will not be used. [tonnes]
Measuring frequency:	Per incident
Reporting frequency:	Per incident
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	N/A. No incident is found in the whole monitoring period and the parameter value is zero. (Refer to CAR 10 for details)
Type of monitoring equipment:	N/A. No incident is found in the whole monitoring period and the parameter value is zero.
Is accuracy of the monitoring equipment as stated in the CDM PDD? If the CDM PDD does not specify the accuracy of the monitoring equipment, does the monitoring equipment represent good monitoring practice?	N/A.
Calibration frequency /interval:	N/A.
Is the calibration interval in line with the monitoring plan of the CDM PDD? If the CDM PDD does not specify the frequency of calibration, does the selected frequency represent good monitoring practice?	N/A.
Company performing the calibration:	N/A.
Did calibration confirm proper functioning of monitoring equipment? (Yes / No):	N/A.
Is (are) calibration(s) valid for the whole reporting period?	N/A.
If applicable, has the reported data been cross-checked with other available data?	N/A.
How were the values in the monitoring report verified?	N/A.
Does the data management (from monitoring equipment to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	N/A.

Thus GLC confirms that

- The monitoring complies with the monitoring plan of the CDM PDD except for some minor deviations. Refer to “Deviations from the Monitoring Plan” for detail.
- All parameters such as baseline, project and leakage emission parameters are monitored as described in the monitoring plan except for some minor deviations. Refer to “Deviations from the Monitoring Plan” for detail.
- The monitoring and reporting frequencies are in line with the monitoring plan.
- The accuracy and the calibration frequency of the measurement equipment are appropriate.

Remaining Issues from Validation or Previous Verification

The verification team has also reviewed validation report available on CDM database at

<http://cdm.unfccc.int/Projects/DB/RWTUV1340626014.25/view>

This is the 1st verification of the project activity. As per CDM validation report ^{/6/}, the assessment team of the validating DOE did not identify any FAR (Forward Action Request). Thus, there are no remaining issues from CDM-Validation or previous verifications.

Deviations from the Monitoring Plan

One deviation from the monitoring plan has been applied by the Project Participant during the monitoring period. Description of this deviation and the assessment on conservativeness of the deviation is provided below:

Description of deviation	Assessment by verification team
<u>Deviation-1</u> As per monitoring plan of the registered PDD, the parameter “flare efficiency” has to be determined on hourly base. Instead of determination of the hourly flare efficiency, Project Participant applies a determination of the minute-by-minute flare efficiency. As Project Participant applies the default value approach (0% / 50 % / 90 %), for calculation of Project Emission from Flaring ($PE_{flare,y}$), for every minute the following is checked: - If flaring temperature is below 500°C, the	<u>Assessment of Deviation-1</u> The “Tool to determine project emission from flaring gas containing methane” ^{/9/} requires the following: <i>“In case of enclosed flares and use of the default value for the flare efficiency, the flare efficiency in the hour h ($\eta_{flare,h}$) is:</i> - 0% if the temperature in the exhaust gas of the flare (T_{flare}) is below 500 °C for more than 20 minutes during the hour h. - 50%, if the temperature in the exhaust gas of the flare (T_{flare}) is above 500 °C for more than 40

default value of 0% flare efficiency is applied. - If flaring temperature is between 1,000 and 1,200°C, the default value of 90% flare efficiency is applied. - If flaring temperature is outside 1,000 and 1,200°C but not below 500°C, the default value of 50% flare efficiency is applied.	<i>minutes during the hour h, but the manufacturer's specifications on proper operation of the flare are not met at any point in time during the hour h.</i> - <i>90%, if the temperature in the exhaust gas of the flare (T_{flare}) is above 500 °C for more than 40 minutes during the hour h and the manufacturer's specifications on proper operation of the flare are met continuously during the hour h. "</i> It is confirmed from the onsite interviews with the operating team and from document check that the enclosed high temperature flare was used during the monitoring period from 2011-02-26 to 2012-07-31 more or less as "emergency unit". This means that under normal conditions, all landfill gas captured on the Bantargebang Landfill is provided to the project's electricity generation facility. Only in a few cases (e.g. if the total electricity generation facility was not operational), gas has been sent to the enclosed high-temperature flare. This events did not occur that often, additionally they last usually only for a short period. This can be seen when comparing the measured values of "Total amount of landfill gas captured" (=29.708.303 Nm³) with the values of "Amount of landfill gas combusted in flare unit" (=55.035 Nm³). The share of "landfill gas combusted in flare unit" represents 0.185 % of the "total amount of landfill gas captured". As it can be seen in this comparison, the amount of LFG flared is very small, thus the flaring time is very small as well. So, it is nearly impossible for Project Participant to apply the requirements as per the tool because flaring events with duration of more than 40 minutes are rare. Due to this, the Project Participant determines flare efficiency on minute-by-minute base, instead of the hourly determination in methodology tool. Since the manufacturer's specifications on proper operation of the flare is upon combustion temperature, and combustion temperature is automatically controlled by PLC, the requirement of combustion temperature between 1,000 and 1,200°C can be regarded as always met, and the flare efficiency should be just dependend on exhaust gas
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	temperature: when it is below or equal to 500°C, the default value of 0% flare efficiency should be applied; and when it is above 500 °C, the default value of 90% flare efficiency should be applied. However, in PP's approach, the 1000-1200°C temperature check is conducted on exhaust gas temperature instead of combustion temperature, so that <i>the flare efficiency in the minute is:</i> - 0% if the temperature in the exhaust gas of the flare (T_{flare}) is below or equal to 500 °C - 50%, if the temperature in the exhaust gas of the flare (T_{flare}) is above 500 °C, but not within 1000-1200 °C - 90%, if the temperature in the exhaust gas of the flare (T_{flare}) is above 500 °C and it is within 1000-1200 °C This way, only when exhaust gas temperature is 1000-1200 °C, flare efficiency is 90%; when it is 500-1000 °C or 1200 °C above, efficiency is 50%; when it is below or equal to 500 °C, <i>efficiency is 0.</i> This is assessed as conservative and no-harm, because: - exhaust gas temperature is always lower than combustion temperature; when exhaust gas temperature is above 1000 °C, combustion temperature is also ensured as above 1000 °C; - since combustion temperature is automatically controlled by PLC unit, the 1000-1200 °C combustion temperature range can be regarded as always met. Also, flare PLC/instruments had no malfunctions or errors during working time. It is worthwhile to note that as confirmed by the manufacturer, the flare is complying with various stringent European standards and additionally its construction resp. its auxiliary equipment guarantees minimised emissions. The manufacturer emphasizes the typical combustion efficiency of 99.9 % ^{/53/}. Therefore, the verification team is of the opinion that
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	the tool approach is not feasible for this project's particular situation and the approach followed by the Project Participant is practical and results in conservative estimation of emission reduction. Therefore, the deviation is accepted.
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4.2 Accuracy of GHG Emission Reduction or Removal Calculations

GHG emissions reductions for the project and the baseline scenario were correctly calculated using the formulae described in the applied methodology for this project. GLC's Assessment Team has reviewed the emission reduction (ER) spreadsheets ^{/5/} and checked all the formulae and found they are correct and in line with the monitoring plan of the registered CDM PDD and the applied monitoring methodology.

The emission reduction due to capturing of methane, its utilisation in onsite gas engines and its combustion in an onsite enclosed high temperature flare was estimated by measuring the required parameters using appropriate monitoring equipment and using the formulae described in the methodology ACM0001 (version 11) and the CDM PDD except the deviation described in the above section. All the monitored parameters are described above in section 4.1. All ex-ante parameters which are used in the calculation of emission reduction are presented in section 3.1 of the MR transparently. It is confirmed that all the ex-ante parameters have been correctly used in the emission reduction calculation.

The following Table 5 shows the ex-ante parameters used in the ER calculation which are in compliance with registered CDM PDD, except for new waste MCF. It is worthwhile to note that MCF for new waste is adopted as 0.8 instead of 1.0 indicated in PDD, which is assessed as reasonable and thus is accepted, based on physical inspection of the landfill, which revealed that the project may not satisfy the requirement of MCF 1.0. For details please refer to CAR 22.

Table 5: Ex-ante parameters used in the ER calculation

Parameter	Description	Value	Unit	Source
MD _{Hist,y}	The amount of methane destroyed historically for the previous year before the start of project activity.	0	-	/6/
-	Regulatory requirement relating to landfill gas	n/a	n/a	
GWP _{CH4}	Global Warning Potential of methane	21	tCO ₂ e/tCH ₄	/7/
ρ _{CH4,n}	Normal density of methane (At standard temperature and pressure: 0 degree Celsius and 1.013 bar)	0.7168	kgCH ₄ /Nm ³ CH ₄	/7/

w_{OM}	Weighting for operation margin emission factor	0.5	-	/11/
w_{BM}	Weighting for build margin emission factor	0.5	-	/11/
ϕ	Model correction factor to account for model uncertainty	0.75	-	/11/
DOC_f	Fraction of degradable organic carbon that can decompose	0.5	-	/11/
DOC_j	Fraction of degradable organic carbon of waste type j	43	% for Wood and wood product	/11/
		40	% for Pulp, paper and cardboard	
		15	% for Food, food waste, and beverages and tobacco	
		24	% for Textile	
		20	% for Garden, yard, and park waste	
		0	% for Glass, plastic, metal and inerts	
k_j	Decay rate of waste type j	0.035	1/yr for Wood and wood product	/11/
		0.07	1/yr for Pulp, paper and cardboard	
		0.4	1/yr for Food, food waste, and beverages and tobacco	

		0.07	1/yr for Textile	
		0.17	1/yr for Garden, yard, and park waste	
		0	1/yr for Glass, plastic, metal and inerts	
MCF	Methane Correction Factor	0.8	-	/11/
OX	Oxidation Factor (reflecting the amount of methane that is oxidized by the covering material)	0.1	-	/11/
F	Volume fraction of methane in landfill gas	50	%	/11/
p _{j,x}	Composition of waste type j in year x, where x is the year where waste GLC's A _{T,x} is being deposited.	18.50	% for Wood and wood product	/15/
		1.57	% for Pulp, paper and cardboard	
		33.75	% for Food, food waste, and beverages and tobacco	
		1.57	% for Textile	
		3.75	% for Garden, yard, and park waste	
		40.87	% for Glass, plastic, metal and inerts	

The detailed emission reduction calculation is presented in the ER-spreadsheet ^{/5/}. All the formulae and the calculation procedure were checked by GLC's Assessment Team. It was observed that the emission reduction was determined in a conservative manner. The emission reduction being claimed during the period from 2011-02-26 to 2012-07-31 is 66.7% lower than the estimated emission reductions in the registered CDM-PDD, as given in the Table 6 below.

Table 6: Emission Reductions

Period	Estimated VCUs as per CDM PDD	Achieved VCUs as per Monitoring Report
Emission Reductions	847,484 Year 2011: 483,731 (=580,477 * 10 / 12) Year 2012: 363,753 (=623,576 * 7 / 12)	281,951
% Deviation (+/-)	-	(-) 66.7 %

The reduction in achieved emission reduction is considered appropriate due to various conservative approaches in the emission reduction calculation, variation in process parameters and uncertainties applying models to predict methane generation. However, no justification is required since the achieved emission reduction is lower than the estimated value in the CDM PDD.

4.3 Quality of Evidence to Determine GHG Emission Reductions or Removals

During the verification site visit, all relevant documents were checked to assess the correctness and quality of data submitted by the project participants, which are used to determine emission reductions.

Total amount of landfill gas captured, total amount of landfill gas fed to power plant, total amount of landfill gas fed to flare unit, volumetric fraction of CH₄ in landfill gas and flare temperature were verified with data provided by the project's data acquisition system. GLC's Assessment Team performed a "data authenticity check" in which all available raw data ^{/14/} has been again downloaded from the project's data acquisition system. Such checking aimed to ensure that only authentic/not-modified monitoring records data were used for the emission reduction calculation (thus ensuring that the measurement records made available in the MS-Excel format "raw data" input files and measurement records reported in the monthly emission reduction spreadsheets were not intentionally or unintentionally edited/modified during the generation of these files). The downloaded raw data in encrypted format ^{/14/} was handed over to GLC's Assessment Team in order to perform the data authenticity check. This check was done in 2 steps. First step was to convert the data available in encrypted format ^{/14/} to readable data in MS-Excel format. In a second step, the so converted data has been compared against the data presented as "raw data" in the ER-spreadsheets ^{/5/}.

As a result of the checks, by comparing files previously generated by the project participants against the files generated by the auditors, GLC's Assessment Team was able to confirm that the generated monthly data is identical to the data provided in the monthly emission reduction calculation spreadsheets previously generated by the project participants as part of the monitoring/reporting process. While no deviations or differences were found when comparing the files, and by assuming that the downloaded data format is a reliable, un-editable and authentic monitoring data (by taking for grant that data earlier recorded in the data acquisition unit cannot be edited or modified), the performed data authenticity check has thus successfully and sufficiently confirmed that only authentic and not-modified monitored

measurement data (from the project's data acquisition system) were previously used by the project participants for the calculation of emission reductions reported in the Monitoring Report.

The amount of electricity imported by the Project from the grid and the amount of electricity exported by the Project to the grid were verified with the monthly invoices provided by the power grid operator ^{/49/}.

All records needed for monitoring are archived in line with the requirements of the registered monitoring plan. No significant, lack of evidence and missing data were detected during onsite verification. Hence, the verification team confirms that the monitoring system ensures required quality to ensure the quality of the monitored data. All internal data are subjected to QA/QC measures, which are continuously developed and improved ^{/54/}.

All the meters (landfillgas flow meters, landfillgas pressure meters, landfillgas temperature sensors, gas analyser and electricity meters) were calibrated at appropriate calibration frequency (described in section 4.1) to ensure the quality of the data. In cases calibration was delayed, conservative approaches have been applied. Hence GLC's Assessment Team confirms that the quality of data is reasonably good and the emission reductions have been calculated conservatively.

It was verified through onsite interviews that the plant's team involved in the monitoring of project activity is well experienced. Hence, the verification team concludes that competent staff is employed by the project proponent to carry out the relevant tasks with sufficient accuracy. Furthermore, it was learnt from the onsite interviews with the operating team that a training was provided by the manufacturer of the flaring station ^{/43/}, this training included the monitoring devices for LFG quality, LFG quantity, Flaring parameters as well as data acquisition.

GLC is of the opinion that this method of estimation is accurate and results in conservative estimation of emission reduction and is in line with the applicable VCS requirements.

4.4 Management and Operational System

GLC's Assessment Team was able to verify that authorities and responsibilities for monitoring and reporting of all data related to the emission reductions were clearly defined for the monitoring period from 2011-02-26 to 2012-07-31. Moreover, all project related equipment was confirmed to be operated by qualified and trained staff ^{/43/}. The frequency of monitoring measurement, as well as applicable calibrations and reporting details were conducted as outlined in the monitoring plan available in the registered CDM PDD.

The allocation of the responsibilities is followed as described in the registered VCS PD. Routines for the archiving of data are defined and documented. During the site visit all the required plant records were verified and found that the data is consistent with the provided in the VCS-Monitoring Report ^{/4/} and ER sheets ^{/5/}.

The monitoring plan described in section 3.3 of the MR is confirmed to be correct. All the parameters of the monitoring plan are monitored using appropriate metering system. The parameters related to landfill gas flow, landfill gas methane content, flare's exhaust gas temperature, electricity generation and electricity consumption are measured continuously. Reporting and recording is done on a minute-by-minute base (to the electronic data acquisition system), except for electricity generation and electricity consumption, where it is done on a monthly base.

Available raw data is downloaded daily on site and sent to the Project Participant's head office, where it is stored and further processed. Data is then available and stored on site and in the head office as a backup. It is therefore confirmed that the data monitoring and data storage represents good monitoring practice.

GLC's Assessment Team has interviewed the plant personnel who are involved in the monitoring of the parameters that are used to determine the emission reductions of the project activity. It is confirmed based on the interviews that the plant's team is competent enough to monitor the parameters as described in the monitoring plan.

As discussed above, GLC's Assessment Team concludes that management and operational system of the project is implemented and operated well. Thus it ensures the quality of data which is required in calculating the emission reductions. Description of the monitoring plan including organisation structure and data management system provided in section 3.3 of the MR is confirmed to be correct.

5 VERIFICATION CONCLUSION

Germanischer Lloyd Certification GmbH (GLC) has performed the 1st verification of the project: “Bantargebang Landfill Gas Management & Power Generation” with regard to the applicable relevant requirements for VCS project activity.

The project activity encompasses the collection of LFG generated at the Bantargebang landfill, its destruction (by combustion in a high temperature enclosed flare) and its utilization as gaseous fuel for electricity generation in an electricity generation facility. All electricity generated at the electricity generation facility is exported through the electricity grid, thus displacing equivalent amount of electricity that would otherwise be generated by fossil-fuel sources (power plants) and new additions in the absence of the project activity. For periods when the electricity generation facility is not (fully) operational or for periods when more LFG is available than the electricity generation facility can convert into power, one enclosed high temperature flare is available for safe combustion of the LFG with the goal of avoiding emissions of methane (CH₄) into the atmosphere.

This verification covers the period from 2011-02-26 to 2012-07-31 (including both days). It is noteworthy that due to commissioning, not all instruments and devices have been in place and fully operational on the starting date of this monitoring period (2011-02-26). On this date, the electricity generation facility was able to export electricity to the grid. This was monitored from the early beginning by the use of electricity meters. The remaining instruments related to landfill gas flow, landfill gas methane content and flare’s exhaust gas temperature have been commissioned in June 2011. Thus, for the months of February, March, April and May 2011, emission reductions from capture and destruction of methane contained in landfill gas are not claimed. For these months, only the emission reductions for CO₂-displacement (through electricity exported to the grid) are claimed and verified.

PT Navigat Organic Energy Indonesia is responsible for the collection of data in accordance with the validated monitoring plan and the reporting of GHG emissions reductions from the project.

It is GLC’s responsibility to express an independent verification statement on the reported GHG emission reduction from the project. GLC does not express any opinion on the selected baseline scenario or on the validated and registered CDM PDD.

The scope of this verification covers the determination of voluntary greenhouse gas emission reduction generated by the above mentioned project. GLC conducted the verification based on the registered CDM PDD ^{/15/}, validation report ^{/6/}, monitoring report ^{/4/}, ER sheets ^{/5/}, the approved consolidated monitoring methodology ACM0001 “Consolidated baseline and monitoring methodology for landfill gas project activities” in its version 11 and supporting documents made available to the verifiers by the project participants. The verification included:

- i) Checking whether the design of the project is implemented and installed as planned and described in the registered CDM PDD;
- ii) Checking whether the provisions of the monitoring methodology and the monitoring plan in the CDM PDD were consistently and appropriately applied;
- iii) The collection of evidence supporting the reported data;
- iv) Checking whether the installed monitoring equipment is calibrated appropriately.

As a result of the verification, GLC confirms that

- all operations of the project are implemented and installed as planned and described in the registered CDM PDD,
- the monitoring system is in place and functional and is in line with the applied monitoring methodology and the monitoring plan,
- the installed equipment essential for estimating emission reductions works reliable and is calibrated appropriately,
- GHG emission reductions are calculated without significant misstatements in a conservative and appropriate manner.

The verification of reported emission reductions is based on the information made available to GLC and the engagement conditions detailed in this report. Hence, GLC cannot be held liable by any party for decisions made or not made based on this report. Also all the documents checked during on-site visit and verification process will be kept confidential and will not be disclosed without the Project Participant's consent as required by VCSA.

In GLC's opinion, the GHG emissions reduction for the "Bantargebang Landfill Gas Management & Power Generation" as reported in the Monitoring Report issued on "29 December 2012, version 3.1" are calculated without material misstatements in a conservative and appropriate manner. The GHG emission reductions were correctly calculated on the basis of the approved consolidated monitoring methodology ACM0001 "Consolidated baseline and monitoring methodology for landfill gas project activities" in its version 11 and the monitoring plan contained in the validated CDM PDD for the project.

Germanischer Lloyd Certification GmbH herewith confirms that the project has achieved emission reduction during the reporting period from 2011-02-26 to 2012-07-31 (including both days) as follows:

GHG Emission Reductions or Removals	Unit	2011	2012	Total during the 1 st verification period
Baseline Emissions	tCO ₂ e	125,836	156,210	282,046
Project Emissions	tCO ₂ e	63	32	95
Leakage	tCO ₂ e	0	0	0
Net GHG emission reductions or removals	tCO₂e	125,773	156,178	281,951

It is confirmed that the project activity has achieved total emission reduction of **281,951 tCO₂e** during the monitoring period from 2011-02-26 to 2012-07-31 (including both days).

Hamburg, 21-01-2013

Germanischer Lloyd
Certification

ANNEX 1: ABBREVIATIONS

CAR	Corrective Action Request
CDM	Clean Development Mechanism
CDM-EB or EB	CDM Executive Board
CH ₄	Methane
CL	Clarification Request
CO ₂	Carbon dioxide
CO ₂ e	Carbon dioxide equivalent
DOE	Designated Operational Entity
EIA	Environmental Impact Assessment
ER	Emission Reduction
ER sheet	Emission Reduction calculation spread sheet
FAR	Forward Action Request
GHG	Greenhouse gas(es)
GLC	Germanischer Lloyd Certification GmbH
LoA	Letter of Approval
MP	Monitoring Plan
MR	Monitoring Report
NA	Not Applicable
PD	Project Design Document
PP	Project Proponent or Project Participant
PPA	Power Purchase Agreement
QA/QC	Quality Assurance / Quality Control
t CO ₂ e	Ton of Carbon Dioxide equivalent
UNFCCC	United Nations Framework Convention on Climate Change
VCS	Verified Carbon Standard
VCSA	VCS Association
VCS PD	VCS Project Design Document
VCU	Verified Carbon Unit

VVS	CDM Validation & Verification Standard
VVM	VCS Validation and Verification Manual

ANNEX 2: REFERENCES

Reference	Author: Title, version, date of issue
/1/	VCS Project Description document for project : -“Bantargebang Landfill Gas Management & Power Generation” (version 1.0, dated 2012-10-19) -“Bantargebang Landfill Gas Management & Power Generation” (version 2.0, dated 2012-12-03)
/2/	Verified Carbon Standard (VCS): VCS Version 3 Requirements Documents: - VCS Standard, v3.3, issued: 2012-10-04 - VCS Program Guide, v3.4, issued: 2012-10-04
/3/	CDM-EB: Clean Development Mechanism Validation and Verification Standard (Version 02.0). EB65, Annex 4.
/4/	PT Navigat Organic Energy Indonesia: -VCS Monitoring Report of Bantargebang Landfill Gas Management & Power Generation. Version 1.0, issued 2012-09-15 -VCS Monitoring Report of Bantargebang Landfill Gas Management & Power Generation. Version 3.2, issued 2013-01-15
/5/	PT Navigat Organic Energy Indonesia: Emission Reduction Spreadsheets. Version 3. File names: - VCSMR1.ER 01-2011 JUN CH4 Abatement.xlsm - VCSMR1.ER 02-2011 JUL CH4 Abatement.xlsm - VCSMR1.ER 03-2011 AUG CH4 Abatement.xlsm - VCSMR1.ER 04-2011 SEP CH4 Abatement.xlsm - VCSMR1.ER 05-2011 OCT CH4 Abatement.xlsm - VCSMR1.ER 06-2011 NOV CH4 Abatement.xlsm - VCSMR1.ER 07-2011 DEC CH4 Abatement.xlsm - VCSMR1.ER 08-2012 JAN CH4 Abatement.xlsm - VCSMR1.ER 09-2012 FEB CH4 Abatement.xlsm - VCSMR1.ER 10-2012 MAR CH4 Abatement.xlsm

Reference	Author: Title, version, date of issue
	- VCSMR1.ER 11-2012 APR CH4 Abatement.xlsm - VCSMR1.ER 12-2012 MEI CH4 Abatement.xlsm - VCSMR1.ER 13-2012 JUN CH4 Abatement.xlsm - VCSMR1.ER 14-2012 JUL CH4 Abatement.xlsm - VCSMR1.ER 15 Consolidated ER Version 3.0.0.xlsm
/6/	TÜV NORD CERT GmbH: CDM validation report for the project activity "Bantargebang Landfill Gas Management & Power Generation". Report No: 8000374274- 09/272, dated: 2012-06-22
/7/	CDM-EB: "Consolidated baseline and monitoring methodology for landfill gas project activities", ACM0001, version 11.
/8/	CDM-EB: "Tool for the demonstration and assessment of additionality" Version 05.2, EB39 Annex 10
/9/	CDM-EB: "Tool to determine project emission from flaring gas containing methane", version 1.0, EB28 Annex 13
/10/	CDM-EB: "Emissions from solid waste disposal sites", Version 6.0.1, EB66 Annex 46
/11/	CDM-EB: "Tool to calculate the emission factor for an electricity system", Version 2.2.1, EB63 Annex 19
/12/	CDM-EB: "Tool to calculate project or leakage CO ₂ emissions from fossil fuel combustion", Version 02, EB41 Annex 11
/13/	Germanischer Lloyd Certification GmbH: CDM GHG Services Manual (incl. procedures and forms)
/14/	PT Navigat Organic Energy Indonesia: Raw data from Memograph. Read out in presence of GLC's Assessment Team. Transferred to GLC's Assessment Team for Data Authenticity Check. File Names: - rec_data DB003A04267 0014 2011-03-03 10-59-29.dat - rec_data DB003A04267 0015 2011-06-21 07-38-00.dat

Reference	Author: Title, version, date of issue
	- rec_data DB003A04267 0016 2011-06-21 07-56-00.dat - rec_data DB003A04267 0017 2011-06-21 08-08-00.dat - rec_data DB003A04267 0018 2011-06-21 08-09-00.dat - rec_data DB003A04267 0019 2011-06-21 08-10-00.dat - rec_data DB003A04267 0020 2011-06-22 07-44-00.dat - rec_data DB003A04267 0021 2011-06-22 15-12-00.dat - rec_data DB003A04267 0022 2011-06-23 10-31-00.dat - rec_data DB003A04267 0023 2011-06-23 10-32-00.dat - rec_data DB003A04267 0024 2011-06-23 16-20-00.dat - rec_data DB003A04267 0025 2011-06-23 16-21-00.dat - rec_data DB003A04267 0026 2011-06-24 14-50-00.dat - rec_data DB003A04267 0027 2011-06-24 18-06-00.dat - rec_data DB003A04267 0027 2011-08-22 01-03-00.dat - rec_data DB003A04267 0027 2011-10-20 01-07-00.dat - rec_data DB003A04267 0027 2011-10-30 03-00-00.dat - rec_data DB003A04267 0027 2011-10-30_02-00-00.dat - rec_data DB003A04267 0027 2012-01-01 12-44-00.dat - rec_data DB003A04267 0027 2012-03-04 18-49-00.dat - rec_data DB003A04267 0027 2012-05-07 15-16-00.dat - rec_data DB003A04267 0028 2012-06-04 18-48-00.dat - rec_data DB003A04267 0029 2012-06-04 18-49-00.dat - rec_data DB003A04267 0030 2012-06-05 15-06-00.dat - rec_data DB003A04267 0031 2012-06-05 15-09-00.dat - rec_data DB003A04267 0032 2012-06-15 17-12-00.dat - rec_data DB003A04267 0033 2012-06-15 17-14-00.dat - rec_data DB003A04267 0034 2012-06-18 13-43-00.dat - rec_data DB003A04267 0035 2012-06-18 13-50-00.dat - rec_data DB003A04267 0035 2012-08-17 20-31-00.dat
/15/	PT Navigat Organic Energy Indonesia: Project Design Document for CDM project activity "Bantargebang Landfill Gas Management & Power Generation", version 02.0, dated 2012-06-21. Available online: http://cdm.unfccc.int/Projects/DB/RWTUV1340626014.25/view
/16/	IPCC: 2006 IPCC Guidelines for National Greenhouse Gas Inventories: work book
/17/	Signing of contract between the Government of DKI Jakarta with the joint operation of PT Navigat Organic Energy Indonesia and PT Godang Tua Jaya for the improvement of facilities for management and operation of integrated waste processing plant of Bantargebang. Dated 2008-12-05

Reference	Author: Title, version, date of issue
/18/	Acceptance Reports, signed by PT Navigat Organic Energy Indonesia on 2010-06-30: Equipment acceptance for gas engine with the engine number 5638011 (internal ID: GE-01) Equipment acceptance for gas engine with the engine number 5638021 (internal ID: GE-02)
/19/	Agreement between PT Navigat Organic Energy Indonesia and grid operator PT PLN (PERSERO) concerning the designated Commercial Operating Date of the Project as set to 2011-02-25.
/20/	Acceptance Reports, signed by PT Navigat Organic Energy Indonesia on 2011-03-02: Equipment acceptance for gas engine with the engine number 6643151 (internal ID: GE-03) Equipment acceptance for gas engine with the engine number 6643261 (internal ID: GE-04)
/21/	Hofstetter Umwelttechnik AG: Commissioning of flare unit and degassing station HOFGAS LPM 2x4000/2500 CDM. Dated 2011-06-25
/22/	Acceptance Reports, signed by PT Navigat Organic Energy Indonesia on 2011-07-16: Equipment acceptance for gas engine with the engine number 2208748 (internal ID: GE-09) Equipment acceptance for gas engine with the engine number 2208747 (internal ID: GE-10)
/23/	Acceptance Reports, signed by PT Navigat Organic Energy Indonesia on 2011-07-27: Equipment acceptance for gas engine with the engine number 2208776 (internal ID: GE-07) Equipment acceptance for gas engine with the engine number 2208780 (internal ID: GE-08)
/24/	Acceptance Reports, signed by PT Navigat Organic Energy Indonesia on 2012-02-13: Equipment acceptance for gas engine with the engine type JGS 320 GS-LL (internal ID: GE-11) Equipment acceptance for gas engine with the engine type JGS 320 GS-LL (internal ID: GE-12)
/25/	Endress + Hauser: Technical Information Deltabar S PMD 70. Document ID: TI382P/00/EN/06.09 No. 71095431
/26/	Endress + Hauser Instruments International AG, Kaegenstrasse 2, 4153 Reinach / Switzerland: Statement regarding the maximum permissible error (MPE). Issued 2012-12-04.
/27/	Endress + Hauser: Technical Information Mnigrad M TR 10. Document ID: TI256T/02/en

Reference	Author: Title, version, date of issue
	71073145
/28/	Endress + Hauser: Technical Information Cerabar M PMC51. Document ID: TI436P/00/EN/02.10. No. 71109889
/29/	Calibrations for Flow Meter Deltabar S PMD 70 with serial number E105920109D: Final Inspection Report. Issued on 2011-02-01 by Endress + Hauser GmbH + Co.KG, Germany.
/30/	Calibrations for Flow Meter Deltabar S PMD 70 with serial number E105910109D: Final Inspection Report. Issued on 2011-02-01 by Endress + Hauser GmbH + Co.KG, Germany. Calibration Certificate no. DEBEEAVR, calibration date 2011-12-09. Issued by Endress + Hauser Messtechnik GmbH + Co.KG, Germany.
/31/	Calibrations for temperature sensor Omnigrad M TR10 with serial number: E1046A14152 Factory Calibration N°E10F8714180, issued 2011-01-25 by Endress + Hauser Sicestherm S.r.l., Italy Calibration Certificate no. DEBDFEVR, calibration date 2011-12-21. Issued by Endress + Hauser Messtechnik GmbH + Co.KG, Germany.
/32/	Calibrations for pressure sensor Cerabar M PMC 51 with serial number E101AD01128: Factory Calibration certificate, issued on 2011-01-25 by Endress + Hauser GmbH + Co.KG, Germany. Calibration Certificate no. DEBEBHVR, calibration date 2011-12-21. Issued by Endress + Hauser Messtechnik GmbH + Co.KG, Germany.
/33/	Calibrations for Flow Meter Deltabar S PMD 70 with serial number E1058F0109D: Final Inspection Report. Issued on 2011-02-01 by Endress + Hauser GmbH + Co.KG, Germany.

Reference	Author: Title, version, date of issue
/34/	Calibrations for Flow Meter Deltabar S PMD 70 with serial number E1058E0109D: Final Inspection Report. Issued on 2011-02-01 by Endress + Hauser GmbH + Co.KG, Germany. Calibration Certificate no. DEBEDHVR, calibration date 2011-12-14. Issued by Endress + Hauser Messtechnik GmbH + Co.KG, Germany.
/35/	Calibrations for temperature sensor Omnigrad M TR10 with serial number E1046B14152: Factory Calibration N°E10F8814180, issued 2011-01-25 by Endress + Hauser Sicestherm S.r.l., Italy Calibration Certificate no. DEBDEFVR, calibration date 2011-12-21. Issued by Endress + Hauser Messtechnik GmbH + Co.KG, Germany
/36/	Calibrations for pressure sensor Cerabar M PMC 51 with serial number E101AE01128: Factory Calibration certificate, issued on 2011-01-25 by Endress + Hauser GmbH + Co.KG, Germany. Calibration Certificate no. DEBECAVR, calibration date 2011-12-21. Issued by Endress + Hauser Messtechnik GmbH + Co.KG, Germany.
/37/	Calibrations for Flow Meter Deltabar S PMD 70 with serial number E105900109D: Final Inspection Report. Issued on 2011-02-01 by Endress + Hauser GmbH + Co.KG, Germany. Calibration Certificate no. DEBEECVR, calibration date 2011-12-09. Issued by Endress + Hauser Messtechnik GmbH + Co.KG, Germany.
/38/	Calibrations for temperature sensor Omnigrad M TR10 with serial number E1046914152: Factory Calibration N°E10F8614180, issued 2011-01-25 by Endress + Hauser Sicestherm S.r.l., Italy Calibration Certificate no. DEBEADVVR, calibration date 2011-12-21. Issued by Endress + Hauser Messtechnik GmbH + Co.KG, Germany

Reference	Author: Title, version, date of issue
/39/	Calibrations for pressure sensor Cerabar M PMC 51 with serial number E101AC01128: Factory Calibration certificate, issued on 2011-01-25 by Endress + Hauser GmbH + Co.KG, Germany. Calibration Certificate no. DEBECCVR, calibration date 2011-12-14. Issued by Endress + Hauser Messtechnik GmbH + Co.KG, Germany.
/40/	NUK NENNING UND KRUMM GmbH/Germany: Technical Information Gas Analyzing System NGA5.
/41/	NUK NENNING UND KRUMM GmbH/Germany: OPERATING INSTRUCTIONS Stationary Gas Analysing System for landfill gas NGA 5.
/42/	Factory calibration certificate for methane analyser serial number A15161, calibrated on 2010-11-09, issued by NUK NENNING UND KRUMM GmbH/Germany
/43/	Training Certificates for operational staff, issued on 2011-06-25 by Hofstetter Umwelttechnik AG.
/44/	Calibrations of gas analyser NUK, performed by operational staff: 2012-04-12, 2012-04-18, 2012-04-25, 2012-05-02, 2012-05-14, 2012-05-21, 2012-05-28, 2012-06-04, 2012-06-12, 2012-06-15, 2012-06-19, 2012-06-25, 2012-06-29, 2012-07-02, 2012-07-05, 2012-07-09, 2012-07-13, 2012-07-16, 2012-07-20, 2012-07-24, 2012-07-27, 2012-07-31, 2012-08-06.
/45/	Standard Gases used for calibration: PT. STARGAZ Services Indonesia. Certificate of Analysis no. 21.1012. Cylinder no. 00522. Gas composition: Carbon dioxide: 39.9 %, Methane: Balance. Stability 24 months. Dated 2012-03-02. PT. STARGAZ Services Indonesia. Certificate of Analysis no. 88.1012. Cylinder no. 09460. Gas composition: Carbon dioxide: 35.04 %, Methane: 38.40%, Nitrogen: Balance. Stability 24 months. Dated 2012-05-23. PT. STARGAZ Services Indonesia. Certificate of Analysis no. 89.1012. Cylinder no. 00415. Gas composition: Oxygen: 20.80 %, Nitrogen: Balance. Stability 24 months. Dated 2012-05-23.

Reference	Author: Title, version, date of issue
/46/	PT.PLN (PERSERO) Commissioning of electricity meter manufactured by Actaris, serial number: 53081963. Dated 2010-09-23 (Internal ID EM-02).
/47/	PT.PLN (PERSERO) Commissioning of electricity meter manufactured by EDMI, serial number: 211556670. Dated 2012-01-10 (Internal ID EM-02).
/48/	PT.PLN (PERSERO) Commissioning of electricity meter manufactured by EDMI, serial number: 212063056. Dated 2012-05-23 (Internal ID EM-04).
/49/	Invoices from PT. PLN (PERSERO) to PT Navigat Organic Energy Indonesia covering the following periods: 2011-02-26 to 2011-04-01 2011-04-01 to 2011-05-01 2011-05-01 to 2011-06-01 2011-06-01 to 2011-07-01 2011-07-01 to 2011-08-01 2011-08-01 to 2011-09-01 2011-09-01 to 2011-10-01 2011-10-01 to 2011-11-01 2011-11-01 to 2011-12-01 2011-12-01 to 2012-01-12 2012-01-12 to 2012-01-02 2012-01-02 to 2012-02-01 2012-02-01 to 2012-03-01 2012-03-01 to 2012-04-01 2012-04-01 to 2012-05-01

Reference	Author: Title, version, date of issue
	2012-05-01 to 2012-05-23 2012-05-23 to 2012-06-01 2012-06-01 to 2012-07-01 2012-07-01 to 2012-08-01
/50/	JUMO GmbH & Co. KG: Construction and application of thermocouples. Data Sheet 90.1000. Document ID: 02.08/00073412
/51/	Hofstetter Umwelttechnik AG: Commissioning of thermocouple. Dated 2011-06-25.
/52/	Hofstetter Umwelttechnik AG: Operating instruction for degassing unit HOFGAS LPM 2x4000/2500 CDM. Edition 5.11
/53/	Hofstetter Umwelttechnik AG: Statement regarding Methane destruction efficiency of Hofstetter high temperature flares. Issued 2010-12-23.
/54/	PT Navigat Organic Energy Indonesia: QA/QC Documents (examples): Work Instruction Daily Weekly Data Evaluation. No. : NOEI/DCC/WI-OM/01, Ver. : 2.0.0, Date : 2012-07-31 Daily Production Report for Bantargebang Landfill Gas Management & Power Generation, dated 2012-07-31. Corrective preventive action request (CPAR) (as example of internal CAR). Dated 2012-07-02
/55/	National Council for Climate Change (= Designated National Authority - DNA for the Republic of Indonesia): Letter of Approval. Issued 2011-01-27. Available online: http://cdm.unfccc.int/filestorage/e/1/821CMNUA7RLGQ9OK3WV6YD4SZXI0JP.pdf/6486%20HC A.pdf?t=U2R8bWY2anJlfDBmnzrRfWnQyxJ_mqOtPI1
/56/	Power Purchase Agreement between PT Navigat Organic Energy Indonesia and PT PLN (Persero), signed 2011-02-10.

Reference	Author: Title, version, date of issue
/57/	PT Navigat Organic Energy Indonesia: grid emission factor calculation spreadsheets: JAMALI grid, year 2009 JAMALI grid, year 2010
/58/	Ministry of Energy and Mineral Resources: Emission Factor of Java-Bali Interconnected System 2008-2009.xls Emission Factor of Java-Bali Interconnected System 2010.xls
/59/	Waste Tonnage certificates (monthly), 2011 and 2012
/60/	Calibration certificate for weighbridge, Serial Number CC 049461032: LETTER OF CALIBRATION TEST RESULT, No : 510.3/8764/MT-2, Uncertainty ± 20 kg ($k=2$), issued by KARAWANG METROLOGY CENTER, 2012-07-03, valid until 2013-07-02.
/61/	VCS Association: Validation and Verification Manual, Ver 3.0, 2012-10-04

ANNEX 3: RESOLUTION OF CORRECTIVE ACTION AND CLARIFICATION REQUESTS (FINDINGS' LIST)

For Validation:

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
CAR 1 (2012-10-25) The description of project crediting period in 1.6 is not in compliance with 3.9.3. of VCS standard Version 3.2.	03 December 2012 (1st round): The project credit period has been updated to comply with clause 3.9.3 of the VCS Standard. The new credit period is 26th of February 2011 to 31st of July 2033, both days inclusive.	2012-12-04 (1st round): Upon checking revised VCS PD it is confirmed that the current description is in accordance with VCS standard. CAR is closed.	Ok/Closed

For Verification:

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
CAR 1 (2012-10-19) * The "VCS Monitoring Report Template" has not been applied completely. Some editorial mistakes and format changes can be found. Additionally, the title page requires information on the following: "Document Prepared By (individual or entity)". Furthermore, the instruction for the title page stipulates: "All boxes must appear in the final document." This has not been correctly applied.	02 November 2012 (1st round): Format of Monitoring Report has been amended in several aspects (fonts, etc.) to comply with the VCS requirement. Including the title page to address the DOE's concern in this CAR.	2012-11-14 (1st round): The revised VCS Monitoring Report (version 2.0) has been reviewed by GLC's Assessment Team. It can be confirmed that the above mentioned version has been amended in terms of editorial mistakes and application of the "VCS Monitoring Report Template" in its version 3.1. In contrary to the above, the revised VCS Monitoring Report (version 2.0) shows a few inconsistencies. The "Report Version: 1.2" as indicated in the footer from pages 1 to 40 is not consistent to the "version 2.0" as indicated in the box of the title page and in the rest of the document. Additionally, the pagination of all pages is not correct. Finally, the monitoring report title has not been presented in the format as requested by the "VCS Monitoring Report Template" in its version 3.1. Hence, CAR 1 is still open.	Ok/Closed

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
	30 November 2012 (2nd round): In the latest version, (a) The MR title has now been corrected to exclude the monitoring report period, etc. (b) The pagination has been corrected, including the report version on each page of the document.	2012-12-05 (2nd round): OK. GLC's Assessment Team has revised the application of the VCS Monitoring Report Template" (version 3.1) in the revised VCS Monitoring Report (version 3.0). It can be confirmed that the document now is using the correct fonts and sizes as per instructions by the template. Additionally, the footer section has been revised and is now showing the correct report version and the correct pagination. Hence, CAR 1 is closed.	
CAR 2 (2012-10-19) * Details, provided in section 1.2 of the VCS MR, require the following: "Indicate ... and whether the project is a grouped project." This has not been applied.	02 November 2012 (1st round): An explicit statement to this effect has been added in Section 1.2 of current VCSMR.	2012-11-14 (1st round): OK, GLC's Assessment Team can confirm that the section "1.2 Sectoral Scope and Project Type" of the revised VCS Monitoring Report (version 2.0) now indicates the following: "This Project is not part of a grouped project." Hence, CAR 2 is closed.	Ok/Closed

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
CAR 3 (2012-10-19) * The evidence "VCSMR1.05" is used in section 1.5 of the VCSMR and showing a Power Purchase Agreement (PPA). However, the provided evidence cannot be deemed sufficient as both the original and translated version present only the first page of a 33 pages contract without signature or stamps.	02 November 2012 (1st round): We would like to underline that the emphasis on the original statement (VCSMR version 1.0) points to the COD, which is a significant milestone under the said PPA, but NOT the PPA itself. As this statement seems to raise confusion, the reference to the PPA has been deleted in the current VCSMR.	2012-11-14 (1st round): OK. Section 1.5 of the revised VCS Monitoring Report (version 2.0) has been reviewed by GLC's Assessment Team. The definition of the "Project Start Date" can be deemed correct as it is set to the day following the agreed Commercial Operation Date (COD, 2011-02-25). Project Start date is 2011-02-26, as this day can be deemed as first full day under commercial conditions after a phase of commissioning and testing. Finally, the removal of the reference "VCSMR1.05" can be accepted as this eases now the comprehension of the way how the Project Start date was defined. Hence, CAR 3 is closed.	Ok/Closed
CAR 4 (2012-10-19) * While the VCSMR is in a table in section 2.1 describing relevant dates for project implementation, the evidences for 15 Oct 2010 (VCSMR1.04, in both original language and English) are stating details of a "diesel engine" what	02 November 2012 (1st round): Kindly note that this document were issued by an institution outside the Project Developer's control. This error is part of institutional challenges associated with absence of other similar installation lodging similar permit in the past. We have notified the issuing office on the matter	2012-11-14 (1st round): OK. Project Participant has explained the situation already during the on-site visit. GLC's Assessment Team has been visiting the project site and can confirm that all engines on site are fuelled with Landfill Gas (LFG), which is generated and collected on Bantargebang Landfill Site and converted into electrical power by the Project Activity.	Ok/Closed

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
cannot be deemed as correct as this would be contradictory to the project description. Additionally, evidences are not showing the correct manufacturer of that engine nor the right engine type/model.	several times and requesting a revision. However, we have not been able, until today to get the correct version. Nevertheless, we have provided the DOE with engine specifications / maker as well as visit to the site to confirm and assure that the relevant engines are all landfill module gas engines.	Additionally, GLC's Assessment Team gained inside the working conditions and behaviours of local authorities. Considering the conditions seen there, the statement of Project Participant can be accepted. Hence, CAR 4 is closed.	
CAR 5 (2012-10-19) * While the VCSMR uses in its section 5.1 the parameter “kj” for the ex-post calculation of theoretical methane generation, this parameter and the resulting values have not been presented in the section 3.1.	02 November 2012 (1st round): This error has been corrected. Parameter kj is now included in Section 3.1 of the latest VCSMR	2012-11-14 (1st round): OK. GLC's Assessment Team has reviewed the revised VCS Monitoring Report (version 2.0) with special focus on sections 3.1 and 5.1. The document in its section 3.1 “Data and Parameters Available at Validation” now provides details on the parameter “kj”, defined as “Decay rate of waste type j”. The presented values are the default values as presented by the following tool of the CDM: “Emissions from solid waste disposal sites” in its version 6.0.1 (as per EB 66, Annex 46). Hence, CAR 5 is closed.	Ok/Closed

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
CAR 6 (2012-10-19) * The VCSMR presents in its section 3.2.1 details of the equipment to measure the following parameters: - LFGtotal,t - LFGelectricity,t - LFGflare,t The provided information about the used devices and their accuracy cannot be deemed correct as for the same instruments several sources (as presented either in the VCSMR, in calibration certificates or in the Technical Information of the manufacturer) are stating different accuracies. Thus, the validity of initial calibration certificates is not clear. Additionally, the accuracy of some of the presented devices depends of the set span, this has not been taken into	02 November 2012 (1st round): This CAR were issued prior to the site visit. During validation meeting, we have provided the DOE with full access of technical specifications for our flow metering system (the DeltabarS , Omnigrad TR10, Deltatop, Cerabar M, Energy Manager (RMC621), and Memograph M) Essentially, the DeltabarS, Omnigrad TR10 and Cerabar M are interconnected by the Deltatop unit, which then send signal to the Energy Manager which calculates the volumetric flow-rate from all input points, and subsequently transmit this signal to the Memograph M for data storing and presentation. This relationship has now been clarified in the VCSMR for the respective parameters. Please see CAR 19 with regard to how the term “accuracy” are being defined by flow meter manufacturer, and the meaning of term given in the calibration certificates – from the manufacturer perspective.	2012-11-14 (1st round): Please see GLC's Assessment to CAR 19! The presentation of the following parameters has been reviewed in the revised VCS Monitoring Report (version 2.0) : - LFGtotal,t - LFGelectricity,t - LFGflare,t The actual presentation of accuracies cannot be deemed correct as this is inconsistent with Project Participant's answer (b) to CAR 19. Thus, the Maximum Permissible Error has not been applied in the presentation of the above mentioned parameters. Hence, CAR 6 is still open.	Ok/Closed

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
account.	30 November 2012 (2nd round): Additional note has been added into Section 3.2.1.1-3 of VCSMR1 (Monitoring Equipment) information concerning the application of accuracy and Maximum Permissible Error.	2012-12-05 (2nd round): OK. GLC's Assessment Team reviewed the presentation of the following parameters in the revised VCS Monitoring Report (version 3.0) in its section 3.2.1 3.2 (Data and Parameters Monitored): - LFG_{total,t} - LFG_{electricity,t} - LFG_{flare,t} The above mentioned parameters now include a note explaining the differences between the term "accuracy" as indicated by the manufacturer when testing a device under laboratory conditions and the application of a "Maximum Permissible Error". The presentation of these parameters can now be deemed correct and consistent to other statements by Project Participant. Hence, CAR 6 is closed.	
CAR 7 (2012-10-19) * For the majority of the monitoring equipment (presented in section 3.2.1) neither manufacturer's	02 November 2012 (1st round): These information has been provided to the validation team during the validation meetings, and the important one has been referenced and included	2012-11-14 (1st round): OK. The following recommendations and Good Industrial Practices have been applied: - All devices to determine LFG_{total,t}, LFG_{electricity,t} and LFG_{flare,t}: yearly	Ok/Closed

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
recommendations for calibration needs nor all calibration certificates have been provided to the Assessment Team.	as part of the document submission of this first rounds. (See VCSMR1.32 – 40)	- Device to determine wCH4: weekly - Devices to determine ELim,y and ELex,y: Not in ownership of Project Participant. - Device to determine Tflare,h: yearly replacement Hence, CAR 7 is closed.	
CAR 8 (2012-10-19) * While as per registered PDD, the amount of electricity imported by the project from the grid (parameter ELim,y) and the amount of electricity exported by the project to the grid (parameter ELex,y) are to be reported in MWh, the VCSMR and the related spreadsheets present the parameter in kWh.	02 November 2012 (1st round): This has been corrected in Section 3.2.1.5 (ELim,y), 3.2.1.6 (ELex,y) and 4.1.2.2 (the presented Total MWh, Table 3) of the VCSMR.	2012-11-14 (1st round): Sections 3.2.1.5 (Electricity Imported) and 3.2.1.6 (Electricity Exported) of the revised VCS Monitoring Report (version 2.0) present now the values in both kWh and MWh, what can be deemed correct. Furthermore, Section 4.1.2.2 (Consolidated Net Amount of Renewable Electricity Exported and Emission Displaced) provides as well values in kWh and MWh. Thus, the presentation of values for the parameters ELim,y and ELex,y in the revised VCS Monitoring Report (version 2.0) can be deemed correct. Finally, a review of the Emission Reduction spreadsheet ("VCSMR1.ER 15 Consolidated ER Version 2.0.0.xlsm", worksheet "EL_LFG") shows, that the presentation on this level is only done in kWh.	Ok/Closed

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
		Hence, CAR 8 is still open.	
	30 November 2012 (2nd round): In this submission, VCSMR1.ER15, sheets EL_LFG has been corrected to present the final (aggregated) values of electricity exported and imported in MWh.	2012-12-05 (2nd round): OK. GLC's Assessment Team reviewed the Emission Reduction spreadsheet ("VCSMR1.ER 15 Consolidated ER Version 3.0.0.xlsm", worksheet "EL_LFG") and can confirm that this spreadsheets presents the final values for the parameters $EL_{im,y}$ and $EL_{ex,y}$ are now reported in MWh, as required by the PDD. Hence, CAR 8 is closed.	
CAR 9 (2012-10-19) * While the VCSMR states a value of 32,960 kWh for the amount of electricity exported by the project to the grid (parameter $E_{Lex,y}$), the related spreadsheet "Consolidated Emission Reduction Version 1.0.xlsx" states a value of 60,059,979 kWh. This issue represents an inconsistency in the	02 November 2012 (1st round): This error has been corrected in the current VCSMR.	2012-11-14 (1st round): OK. The revised VCS Monitoring Report (version 2.0) presents now in its section 3.2.1.6 the parameter $E_{Lex,y}$ with a value of 60,086,159 kWh or 60,086.159 MWh. This is consistent to the value as presented in "VCSMR1.ER 15 Consolidated ER Version 2.0.0.xlsm". Hence, CAR 9 is closed.	Ok/Closed

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
monitoring.			
CAR 10 (2012-10-19) * As per Monitoring Plan of the registered PDD, the following parameters should be reported: *EFCO_{2,i} *FC_{i,j,y} A presentation of these parameters is missing in the section 3.2 of the VCSMR. Furthermore, clarification needs to be given whether there are no additional units/equipments/machineries involved due to the project which in absence of the project will not be used and no additional fuel consumed.	02 November 2012 (1st round): Parameter EFCO₂ has been added in Section 3.2.2.3, whereas Parameter FC_{ijy} has been added in Section 3.2.2.9.	2012-11-14 (1st round): OK. While reviewing the revised VCS Monitoring Report (version 2.0), GLC's Assessment Team can confirm the following: - Section 3.2.2.3 (Emission factor of fossil fuel used) now provides the parameter EFCO_{2,i} for both (1) Grid Emission Factor Calculation and (2) On-site Fossil Fuel Consumption. - Section 3.2.2.9 (Fuel Consumption of Heavy Duty Machinery) is presenting the parameter FC_{i,j,y} as "Amount of fuel consumed by process type j, where process j is the type of additional units/equipment/machineries that are employed due to project activity which in absence of project will not be used". - Section 4.2.2 (Project Emission from On-Site Fuel Usages) explains why this Emission source has been accounted as zero (0) for the monitored period. Additionally, GLC's Assessment Team has not seen any type of waste compactor while visiting the project site.	Ok/Closed

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
		Hence, CAR 10 is closed.	
CAR 11 (2012-10-19) * Both methodology ACM001 and the registered PDD stipulate for calculation of MDproject,y resp. MDproject,p resp. MDproject,t: "The sum of the quantities fed to the flare(s), to the power plant(s), ... must be compared annually with the total quantity of methane generated. The lowest value of the two must be adopted as MDproject,y." This requirement has been taken into account in equation 1 in the VCSMR, but cannot be reproduced in the monthly spreadsheets (sheet "Calculated Data").	02 November 2012 (1st round): Thank you for this comment. This error has now been corrected in each of the methane abatement spreadsheet (VCSMR1.ER 1-14), Sheets ("Calculated Data").	2012-11-14 (1st round): OK. While reviewing the monthly methane abatement spreadsheets (VCSMR1.ER 1-14), Sheets ("Calculated Data"), it can be seen that the formula to calculate MDProject,p compares the value of LFGtotal,t with the sum of values of LFGelectricity,t and LFGflare,t. The lower value resulting from this comparison is used then to calculate MDProject,p. Hence, CAR 11 is closed.	Ok/Closed

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
CAR 12 (2012-10-19) * The conversion from the parameters MDproject,t and PEflare,t (as presented in [kg CH4] in the sheet "Calculated Data" in the monthly spreadsheets) into "Sum of MD_Project" resp. "Sum of PE_flare" is not traceable as the sheet "Consolidated Results" refers to the Pivot Function of the software what cannot be deemed as traceable.	02 November 2012 (1st round): The pivot function as well as the sheets consolidated results have been removed from the monthly methane abatement spreadsheet.	2012-11-14 (1st round): OK. While reviewing the monthly methane abatement spreadsheets (VCSMR1.ER 1-14), it can be seen that these files provide in total 3 worksheets: 1) Raw Data 2) Calculated Data 3) Downtime Log For calculation of MDproject,t and PEFlare,t sheet (2) "Calculated data" is used. All calculations in this worksheet are traceable and accessible. Data can be reproduced and is comprehensively presented. Hence, CAR 12 is closed.	Ok/Closed
CAR 13 (2012-10-19) * The spreadsheet covering the period of April 2012 (VCSMR1.ER 11-2012 APR Methane Abatement Activity.xlsx), May 2012 (VCSMR1.ER 12-2012 MEI Methane Abatement Activity.xlsx) and June 2012 (VCSMR1.ER 13-2012 JUN Methane Abatement Activity.xlsx) are showing in the sheet "Raw Data" values	02 November 2012 (1st round): This incompleteness was due to incorrect cell copying method. This redundant information has been cleared off from all methane abatement spreadsheets.	2012-11-14 (1st round): OK. GLC's Assessment Team has reviewed the following revised methane abatement spreadsheets: - VCSMR1.ER 11-2012 APR CH4 Abatement.xlsm - VCSMR1.ER 12-2012 MEI CH4 Abatement.xlsm - VCSMR1.ER 13-2012 JUN CH4 Abatement.xlsm Other values than values related to the presented month have not been found anymore in the sheet "Raw data".	Ok/Closed

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
for "00.01.1900 00:00" which cannot be deemed as correct.		Hence, CAR 13 is closed.	
CAR 14 (2012-10-19) * Table 1 (Measurement Results for Calculation of Methane Destroyed by the Project) and Table 5 (Project Emission from Flaring Landfill Gas), presented both in the VCSMR, are showing inconsistencies while being compared with the monthly spreadsheets.	02 November 2012 (1st round): The values contained in both table has now been updated to reflect the current emissions contained in the latest spreadsheets.	2012-11-14 (1st round): GLC's Assessment Team reviewed the revised VCS Monitoring Report (version 2.0) in order to assess updates made by Project Participant. - Table 1 in version 1.0 and Table 1 of the version 2.0 of the VCS Monitoring report have the same designation: "Measurement Results for Calculation of Methane Destroyed by the Project". The presented values of version 2.0 have been compared with the values presented in "VCSMR1.ER 15 Consolidated ER Version 2.0.0.xlsm", worksheet: "MD_Project". It can be stated that the values have been updated but the values presented for the parameter MDproject,p for June 2011 are still presented inconsistent in both report and spreadsheet. - Table 5 in version 1.0 and Table 4 of the version 2.0 of the VCS Monitoring report have the same designation: "Project Emission from Flaring Landfill Gas". The presented values of version 2.0 have been compared with the values presented in "VCSMR1.ER 15 Consolidated ER Version 2.0.0.xlsm", worksheet: "PE_Flaring". It	Ok/Closed

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
		can be stated that the values have been updated and that the presentation in both report and spreadsheets is consistent. Hence, CAR 14 is still open.	
	30 November 2012 (2nd round): In this submission, MDproject, p in Table 1 has been corrected in accordance to the Consolidated Spreadsheet Version 3.0.0. Furthermore, a cell reference error was found in the consolidated spreadsheet, and this is corrected in this version. (See log change text for detail of area updated)	2012-12-05 (2nd round): OK. GLC's Assessment Team reviewed the Table 1 of the revised VCS Monitoring Report (version 3.0) and the related spreadsheets. It can be confirmed that the values for MD_{project,p} have been revised. The presentation of these values is now consistent to the values as calculated in the Emission Reduction spreadsheet ("VCSMR1.ER 15 Consolidated ER Version 2.0.0.xlsm", worksheet "MD_Project"). As further explained by Project Participant in his answer from 2012-11-03 (2nd round), a cell error occurred while linking the monthly spreadsheets to the consolidated spreadsheet. The column in the consolidated spreadsheet showing "MD_{project} in CO₂" was previously linked to the cell presenting "Net methane abated" in the Monthly spreadsheets ("worksheet Calculated Data"), instead of being linked to the Cell "Methane captured, used & destroyed". GLC's	

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
		Assessment Team can confirm that this error has been corrected. Hence, CAR 14 is closed.	
CAR 15 (2012-10-19) * While comparing the Table 3 (Consolidated Net Amount of Renewable Energy Sold to Grid and CO2 Displaced) as per VCSMR with the presented spreadsheet (VCSMR1.ER 15 Consolidated Emission Reduction Version 1.0.xlsx; sheet "EL-LFG"), some inconsistencies appeared. Additionally, the presented evidences (VCSMR1.31) cannot be deemed sufficient in some cases.	02 November 2012 (1st round): The values in Table 3 has now been updated to reflect the most current version of the spreadsheet, and hence address the inconsistencies issue. This CAR were issued prior to the site visit, and all evidences were already presented to the auditor during the validation meeting.	2012-11-14 (1st round): Table 3 in version 1.0 and Table 3 of the version 2.0 of the VCS Monitoring report have the same designation: "Consolidated Net Amount of Renewable Energy Sold to Grid and CO2 Displaced". The presented values of version 2.0 have been compared with the values presented in "VCSMR1.ER 15 Consolidated ER Version 2.0.0.xlsm", worksheet: "EL_LFG". It can be stated that the values have been updated and that the presentation in both report and spreadsheets is consistent. Hence, CAR 15 is closed.	Ok/Closed
CAR 16 (2012-10-19) * While comparing the presented calibration protocols for the gas analyser (VCSMR1.29) with the related	02 November 2012 (1st round): (a) Time issue. The calibration reports were written by the engineer who was performing the calibration. This piece of information (despite the	2012-11-14 (1st round): OK. (a) Project Participant's answer can be accepted. While the staff on site is signing with actual date	Ok/Closed

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
raw data, several issues appear: - The time indicated on the calibration protocol does not fit into the presented data and thus cannot be reproduced. - For the time of calibration, data is varying according to calibration procedure. It is not stated how this has been taken into account while calculating the emission reductions.	field title) should be interpreted as the "report time" rather than the "calibration time" The precise time of the calibration would be those recorded / indicated by the Memograph. All spreadsheet now indicates the period of which the gas sensors were undergoing calibration ("in-calibration"). (b) The period of which the gas analyzers are being calibrated is now clearly indicated in the spreadsheet. During this period, no emission reduction from methane abatement is claimed (MDproject = 0)	and time, the exact time of calibration can be reproduced by Project Participant's data logging system. The gas analyzer has a 2-position-key switch. Position 1 is indicated as "Measuring", while position 2 shows "Calibrate". The position of the key is registered as digital value in the data log. Position 1 gives the signal "o.k.", position 2 gives "Cal". By using this information, calibration periods can be identified. As raw data coming from the data logging system is not editable (and must be converted into a readable spreadsheet by using a special software), the information of calibration events can be traced. (b) This approach of Project Participant has been assessed by GLC's Assessment Team. As the real value of methane (CH₄) for the calibration-period is not registered, Project Participant prefers not to claim any emission reduction. This is conservative. The above described way to distinguish times with and without calibration has been formalized by Project Participant and whenever the status "calibration" is registered, the value for MDproject,p is indicated as zero, so no emission reduction is claimed. Hence, CAR 16 is closed.	

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CAR 17 (2012-10-19) * The evidence "VCSMR1.29 GA 00. QIAL CH4 Instrument History.pdf" is showing several calibration dates but not all of the related protocols have been presented to the Assessment team. Furthermore, the corrective actions to be taken if tolerance found is higher than 2 % are not described.	02 November 2012 (1st round): This CAR were issued prior to site visit. The complete calibration reports have been presented to the validator during the validation meeting. Where tolerance were found to be higher than 2%, the Central Coordination office will issue a written request for corrective action to the site manager and request to have investigation to be carried out and documented. An example of such occasion is given in VCSMR1.45, issue on 2nd of July 2012.	2012-11-14 (1st round): OK. Project Participant is right, the calibration reports listed as reference in the document "VCSMR1.29 GA 00. QIAL CH4 Instrument History.pdf" have been presented on-site. Additionally, the corrective actions to be realized if tolerance of the gas analyzer is found is higher than 2 % are now explained and described in a presented document "Corrective preventive action request (CPAR)" (file name: VCSMR1.45 CPAR #001.pdf). Hence, CAR 17 is closed.	Ok/Closed
CAR 18 (2012-10-19) * As described for the parameter w_{CH_4}, the gas analyser was not calibrated regularly but intermittently in the period between start of the Monitoring period (26-02-2011) and 12 April 2012. No evidence of these intermittent calibrations has been presented to the assessment team. Furthermore, no	02 November 2012 (1st round): The gas analyser were first calibrated during commissioning. Since, then calibration were not performed until 12 April 2012. The first calibration, indicated a significant negative drift, embodying a conservative readings as elaborated below. The analyser reads a methane concentration of 55.9% for a standard gas with known concentration of 60.1%. As such, prior to this calibration, the	2012-11-14 (1st round): GLC's Assessment Team reviewed section "3.2.1.4 Methane in Landfill Gas" of the revised VCS Monitoring Report (version 2.0). The field "Any comment" contains the statement, that prior "to 12 April 2012 the instrument were calibrated only intermittently". It is not clear if these intermittent calibrations have been documented in a report or if they have been done without any reporting.	Ok/Closed

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statement of the manufacturer regarding the prescribed calibration frequency has been provided.	analyser has been under reporting the methane content significantly. Subsequently, the methane concentration recorded is conservative throughout this period. The statement for the calibration frequency as determined by the manufacturer were given in VCSMR1.36B page 19.	The evidence VCSMR1.36B (VCSMR1.36B GAE NUK NGA5 Operation Manual.pdf) provides a document titled: "OPERATING INSTRUCTIONS Stationary Gas Analysing System for landfill gas", issued by NENNING UND KRUMM GmbH, Germany. (Edition 02.11/Ne). On page 19, Section "11. Servicing instructions" can be found. In this section, a weekly calibration of the analyzers is recommended. This document represents the manufacturer's recommendations. Hence, CAR 18 is still open.	
	30 November 2012 (2nd round): In this submission, the relevant statement in Section 3.2.1.4 has been clarified to state that calibration has been made only once during instrument commissioning without instrument drift being indicated in the report.	2012-12-05 (2nd round): OK. A review of section 3.1.2.4 shows that the gas analyzer was only calibrated during the overall commissioning of the degassing unit, and that the next reported calibration has been carried out on 2012-04-12. As the intermittent calibrations cannot be evidenced, they are not considered any longer. Hence, CAR 18 is closed.	

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CAR 19 (2012-10-19) Some of the presented calibration protocols of the instruments used for monitoring purposes are showing a higher deviation than the maximum permissible error set by the manufacturer. This has not been considered in calculation of the Emission Reductions.	02 November 2012 (1st round): This CAR relates to the term “accuracy” in the flow instruments Technical Specification (VCSMR1.32A) and “Maximum Permissible Error” as used in the calibration reports issued by Endress+Hauser for instruments tagged as FIR/TIR/PIR (VCSMR1.33.G/I/L, VCSMR1.34 G/K/N, and VCMR1.35 D/H/K) In order to respond to this CAR, we requested clarification from Endress+Hauser and were explained the following: (Please note that signed statement to this effect is still being sought from Endress+Hauser) (a) The term “accuracy” used in the technical specification relates to the instrument measurement accuracy when tested using laboratory / test gas under laboratory conditions. Whereas these terms are useful to compare the quality of one instrument to another (during purchasing decision for example), the accuracy can not be replicated under “operating conditions” for non-laboratory gas. As such, the accuracy expectation given in the technical specification can not be used as an	2012-11-14 (1st round): a) OK. This can be accepted. Thus, for the term “accuracy” the same unit is tested several times, measured values are compared with design values. Finally, a statement can be given how accurate the unit resp. the device works under test (i.e. laboratory) conditions. b) OK. As Project Participant did not request any higher tolerance of reading, manufacturer’s default value of 0.5% for the Maximum Permissible Error is used. This can be accepted. c) This statement can be accepted. All devices used for flow determination (i.e. flow meter, pressure sensor, thermometer) have been inspected in December 2011 (see as evidences VCSMR1.33.G/I/L, VCSMR1.34 G/K/N, and VCMR1.35 D/H/K). All meters passed the test which means all devices were found to be below manufacturer’s default value of 0.5% for the Maximum Permissible Error. Thus, calibration was neither necessary nor conducted, but the revised VCS Monitoring Report (version 2.0) often	Ok/Closed

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
	operational-performance benchmark of the instrument itself. (b) The manufacturer also clarify that the term "Maximum Permissible Error or MPE" refers to the tolerance of reading that is deemed "acceptable" by the end-user, NOT by the manufacturer. For instrument with commercial trading implications, MPE is generally agreed and stipulated in the commercial contracts. Where the end-user does not provide this information, Endress+Hauser sets an MPE value as per generally accepted measurement tolerance limit, in our case +/-0.5%. (c) During calibration, Endress+Hauser will only re-calibrate the instrument when the reading were found to be above the MPE. Where inaccuracy were found to be below the MPE, calibration is deemed not necessary, and only the reading is reported. This protocol is in compliance with the requirement of ISO/IEC17025 as stated in the "Calibration Report" issued by Endress+Hauser.	states in sections "QA/QC procedures to be applied" that the "instrument were calibrated by Endress+Hauser complying with procedure outlines by ISO/IEC17025 standard were found to be within acceptable Maximum Permissible Error of 0.50%". This is inconsistent to the statement (c) of Project Participant's answer. Additionally, the works carried out by Endress and Hauser have been on site as per provided evidences. This is presented wrongly in some cases. The evidences provided by Project Participant in regard of flow measuring equipment (i.e. flow meter, pressure sensor, thermometer; documents VCSMR1.33.G//L, VCSMR1.34 G/K/N, and VCMR1.35 D/H/K) show that these devices have been checked and found ok in December 2011 (on 9th, 14th and 21st). As all of these used devices passed the factory test in late January 2011 resp. early February 2011, the applied calibration period of less than 12 months is acceptable and represents Good Industrial Practice. Additionally, the documents issued by the manufacturer after having performed the on-site-checks have a validity of 12 months.	

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	In conclusion, the Projects flow-related instrumentations were not operating outside the manufacturer's guideline and/or acceptable norms as given by the manufacturer (in the calibration report) and not against the technical specifications. Consequently, no implications needs to be further considered in the emission reduction calculation.	Please provide the signed statement from the manufacturer. Hence, CAR 19 is still open.	
	30 November 2012 (2nd round): [Pending to statement from E+H]	2012-12-05 (2nd round): OK. The revised VCS Monitoring Report (version 3.0) now states in sections "QA/QC procedures to be applied" for all devices used to monitor the parameters - LFG_{total,t} - LFG_{electricity,t} - LFG_{flare,t} the following: <i>"The instrument were serviced & inspected by Endress+Hauser complying with procedure outlines by ISO/IEC17025 standard, and were found to be within acceptable Maximum Permissible Error of 0.50% of max</i>	

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
		measurement range". These statements are now consistent with Project Participant's answer from 2012-11-02 (1st round) to CAR 19 and to the written statement of the manufacturer (assessment see below). Additionally, statements in regard of factory calibration have been revised, where not necessary (e.g. calibrations on 2011-12-14 were not done in the factory but on site). Shortly after submission of this findings list, a statement from Endress + Hauser Instruments International AG, Switzerland, dated 2012-12-04, was provided to GLC's Assessment Team for Assessment (Project Participant's evidence number: VCSMR1.46). In this statement the manufacturer confirms the statements already explained by Project Participant in his answer from 2012-11-02 (1st round) to this CAR 19, bullet points a), b) and c). Hence, CAR 19 is closed.	
CAR 20 (2012-10-19) Operational hours are not reported either in MR or in ER sheet.	02 November 2012 (1st round): In order to report "Operational Hour" of Power Plant, a downtime log sheet are created on each of the	2012-11-14 (1st round): GLC's Assessment Team has chosen the methane abatement spreadsheet for May 2012 (VCSMR1.ER 12-	Ok/Closed

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
	methane abatement activity spreadsheet (VCSMR1.ER 1-14), as Sheets ("Downtime Log"). These sheets list ALL incidents where the power plant is shown to have withdrawn less than 1Nm³ of landfill gas. Those incidents where these has occurred more than 59 minutes are classified as "downtime" and highlighted with purple background. The aggregate total of these occurrences are calculated as total downtime hour, and deducted against the duration of reporting to arrive to the "operation hour" of the power plant in that respective month. This information has now been included in the Consolidated Emission Reduction Spreadsheet (VCSMR1.ER 15), and the aggregate total is presented as "value monitored" in parameter OH_PP in Section 3.2.2.	2012 MEI CH₄ Abatement.xlsm) to assess the presented approach. The following points are not clear: - The application of the benchmark "less than 1 Nm³ of LFG" cannot be followed. As per "Downtime Log" of the above mentioned spreadsheet, the period from 01 Mai 2012 21:40:00 to 01 Mai 2012 21:45:00 is presented as downtime. While comparing this period with the "Raw data" it can be seen, that in each of this 5 minutes the flow to the gensets has been 2 Nm³/h. This is inconsistent with the presented benchmark as per Project Participant's response. - The determination of the benchmark "less than 1 Nm³ of LFG" has not been explained. - As per "Downtime Log" of the above mentioned spreadsheet, the "Total duration of time of which the power plant receiving <=1Nm³ of gas for more than 59 minutes has been considered for downtime: - o While the benchmark is defined in Project Participant's answer as "less than 1 Nm³ of LFG", the above cited statement is defining it as "equal to or less than 1 Nm³ of LFG". This is not consistent. - o It is not clear why only events longer than 59 minutes are taken into account while summarizing downtime. Hence, CAR 20 is still open.	

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	30 November 2012 (2nd round): <u>Clarification of the value of the applied benchmark</u> We shall clarify that the Power Plant (in it's entirety, not viewed as individual engine) is considered "not-operating" when at any given <i>minute</i> it does not withdraw any gas from the main supply. Thus the determination of downtime being 1Nm3 is reviewed on minute basis. Thus, the applied benchmark is 1Nm3/min, <u>not</u> 1Nm3/hour The value in the "raw data" however presents flow-rate at any given minute as Nm3/hr. Thus, in order to achieve the equivalent Nm3/min, this value needs to be divided by the 60minutes of an hour. For example, in the period stated by the auditor, the flow-rate to the power house is indicated to be 2Nm3/hour in Sheet("Raw Data"). Thus, in that particular minute, gas is flowing at a rate of 0.03Nm3/min – as such the Power Plant is considered not operating. We would like to re-confirm that the applied cut-off	2012-12-06 (2nd round): OK. The clarification as provided by Project Participant can be deemed sufficient. When considering that the recorded value of LFG flow (i.e. 2 Nm³/h) must be divided by 60 (as the value is displayed in Nm³/h but the recording is doing minutewise), the resulting flow cannot be used to operate any gas engine. Thus, the whole power plant can be considered to be out of operation. Additionally, the approach to consider a value of ≤1 Nm³/min as benchmark can be followed. As soon as the values are going to be higher than ≤1 Nm³/min (or its equivalent of ≤60 Nm³/h), it is evident that the solenoid valve in the pipeline feeding the power plant has given the order to open (for start, gas is then sucked to start any engine) or the valve is not yet closed (as any engine is about to stop). Thus in that cases the plant is considered to be in operation.	

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	range is <u>less than 1 Nm³/min</u>. For avoidance of doubt, the power plant is considered operating when the withdrawal rate is 1Nm³/min or more. Whether or not we set the benchmark at 0.01Nm³ per minute or 1.5Nm³ per minute is however a highly subjective discussion. The intention to put a number is simply to provide clarity of the boundary of the mathematical problem, which is a more robust practice than setting a numerical 0 Nm³ per minute. This is because higher value indicates that the power plant already enters into transitional mode toward operation. <u>Clarification of why only values above 59 minutes is included in the calculation</u> Further difficulties is posed with the requirement to report "operational hour" (instead of "operational minute"), that is the number of "full 60minutes" of which the Power Plant is operating. As can be seen in the downtime log, there are many incidents where the Power Plant is down only for a period of less than 30 minutes. The hour of which this incident occur, can not thus be regarded as downtime hour. In that particular hour, the Power	Furthermore, the explanation by Project Participant regarding the term operational hours can be accepted. As the "operational hours" should be reported, the applied approach is deemed acceptable. Hence, CAR 20 is closed.	

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	Plant is considered operating. As such, only the number of which the Power Plant is down by more than 59 minutes (ie 60 minutes or above) continuously is added into the calculation of number of downtime hour, which in the latest version of spreadsheet has been rounded up to the nearest hour. The rounding (introduced this in this submission) results in the correction of operation hour from originally 9,637hr to 9,614hr throughout the reporting period. This information is corrected in Section 3.2.2.8 of the VCSMR. In this particular exercise, it is known of what is considered as “conservative” or “non-conservative” calculation. This is because the intention of the requirement to report this particular parameter is not provided by neither the methodology nor the PDD, and is absence in the equation to calculate emission reduction. Thus, the value reported in this monitoring report represents <u>the best indication, given the available information</u> – has no influence to the amount of emission reduction to be certificated.		

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CAR 21 (2012-10-19) In "EF JAMALI 10 version 1 0 0" and/or "EF JAMALI 09 version 1 0 0": - lower limit of Density of Fuel Oil (MFO & HFO) is different from validated PDD; - on "References & Assumptions" sheet, there are some typo errors in plant names (eg. No. 78 in 2009 file and No. 143 in 2010 file) - for group A2 plants, the natural gas consumption in MMBTU is calculated using conversion factor 1.054 and then Natural Gas Consumed in GJ is calculated back using slightly different conversion factor 1.05435. - NCV diesel instead of NCV fuel oil is applied in calculation of Fuel Oil Consumed in GJ. - In the sheet "2009 group m	02 November 2012 (1st round): (a) Lower Limit Density for MFO & HFO. The source data use for MFO & HFO (VCSMR1.27 EF 03) does not quote for lower limit for fuel oils. However, it is agreed that it is prudent and conservative to assume a lower limit density to be the same value as per upper limit of diesel oils. The same approach has been taken for the registered PDD, and will be applied here. (b) Incorrect name for Plant 78 (2009) and Plant 143 (2010). Plant 78 and Plant 143 should be named "PLTGU Priok ST 2.0", instead of "PLTGU Priok ST 1.0". Correction has been made to these effects on both files. Thank you. (c) Inconsistent Conversion Factor In order to assure consistency in conversion factor used, all factors are taken out from the calculation sheets, and linked to a conversion factor table located at Sheet ("References & Assumption"). (d) Incorrect NCV for Fuel Oil. This error has	2012-11-16 (1st round): Issues are addressed except (b) and (f). (a) in "EF JAMALI 10 version 1 0 0" and "EF JAMALI 09 version 1 0 0", Density of Fuel Oil (MFO & HFO) applies value of 0.990 t/m³, the upper limit value in PERTAMINA's publication. However, registered PDD applies 0.955 t/m³ as fuel oil density and the ex-ante "6486 EF Calculation.xlsx" further indicates 0.955 as median value of lower limit 0.920 and upper limit 0.990 t/m³. This is how this issue was raised. In updated "VCSMR1.JAM 17 EF JAMALI 09 version 2.0.0" and "VCSMR1.JAM 18 EF JAMALI 10 version 2.0.0", 0.955 as median value of 0.920 and 0.990 is applied as fuel oil density, which is consistent with registered PDD. Issue is addressed. (b) -in "EF JAMALI 09 version 1 0 0", plant 78's name is indicated as "PLTGU Priok ST 1.0" which is the same as plant 77's. It is to be noted that in the corresponding EF file obtained from the ministry, plant 77 and 78 also have	Ok/Closed

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power plant", several plants from "2009 Original" which have both power generation and fuel consumption data and do not belong to LCMR are not included in the list of Group A1, eg. No. 180 - In the sheet "2010 group m power plant", plant No. 184 which does not have fuel consumption data and does not belong to PLTA/PLTP is not included in the list of Group A2. Filtering criteria for group A1 and group A2 needs to be clarified. - In the sheet "2010 group m power plant", plant No. 168's emission is blank.	been corrected in the current version of spreadsheet. Thank you. (e) Unintentional exclusion of some power plants in Group A1 for Year 2009. This has now been corrected in the current version of spreadsheet. Thank you. (f) Exclusion of PLTM Curug (Plant 184) from Group M power plant. PLTM Curug does not have any fuel consumption because it is a micro hydro power plant ("Pusat Listrik Tenaga Micro Hydro"). PLTM Curug itself is a series/sets of run-off river micro hydro power plants, not a single unit. The abbreviation of PLTM / PLTP / PLTA and their exclusions have now been clearly indicated in sheet ("References & Assumption") (g) Unintentional exclusion of Plant 168 in 2010 Group M Power Plant. This has been corrected in the spreadsheet. Thank you.	the same name. Upon checking the file of version 2 0 0, it is confirmed that plant 78 is revised to "PLTGU Priok ST 2.0" in "References & Assumptions" sheet; however it is not revised in all other sheet. Explanation/clarification is necessary regarding: how the real name of plant 78 is found and why it is not revised in all places. -in "EF JAMALI 10 version 1 0 0", "References & Assumptions" sheet, plant 143's name is indicated as "PLTGU Priok ST 1.0" which is the same as plant 142's. In all other sheets of this file as well as in the corresponding EF file obtained from the ministry, plant 143's name is indicated as "PLTGU Priok ST 2.0". Upon checking "References & Assumptions" sheet of the file of version 2 0 0, it is confirmed that plant 143 is revised to "PLTGU Priok ST 2.0". This is assessed as correct. (c) in EF files of version 2 0 0, it is verified that the inconsistency factor has been addressed. A parameter BTU_J is defined as as 1,054.35 J/BTU. It is included in conversion factor table of "References & Assumptions" and the relevant equations are linked to it. (d) in EF files of version 2 0 0, it is verified that the NCV of fuel oil is now correctly applied. (e) in 2009 EF file of version 2 0 0, it is verified that the	

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		incorrectly excluded power plants, eg. 180 are now included. (f) It is verified that the abbreviations of plant types are now added in "References & Assumptions" of EF files version 2.0.0. However, data source is not indicated; and unlike other plant types, PLTM, indicated as "Pusat Listrik Tenaga Mikro Hydro" or "Micro Hydro Power Generation", can not be found in plant type list of the EF file obtained from the ministry. Thus the source of PLTM's definition still needs to be indicated. (g) in 2010 EF file of version 2.0.0, it is verified that plant 168's emission is now correctly counted in. This CAR is still open.	
	30 November 2012 (2nd round) (a) Explanation of Plant 78 as PLTGU ST 2.0 (instead of PTGU ST 1.0). The original data for year 2009 were presumably has typological error, which is later corrected for year 2010. It shall be noted that on both record the plants were identified to have been commissioned in 1994, and as such is likely to be referring to the same plant.	2012-12-10 (2nd round): OK. (a) Upon checking EF file for 2010, it is found that plant 142 is indicated as Priok ST 1.0 while plant 143 is indicated as Priok ST 2.0 with COD 1994 and capacity 200 MW. Since plant 77 in EF file for 2009 is Priok ST 1.0, it is assessed as reasonable that the plant 78 naming Priok ST 1.0 is a type error, given that its indicated COD is also 1994 and MW is also indicated as 200 MW	

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	(b) The definition of "PLTM". We can not obtained further formal information from what has been explained & discussed with the auditor during the validation meeting – including a confirmation from an official of the Ministry via a black-berry messenger. The following shall be noted about the PLTM being discussed (PLTM Curug, Plant number 184): (a) This Plant consumes no fossil fuel and thus a renewable energy by definitions. (b) The Plant generates 14,171MWh representing only to 0.011% of the entire grid generation. If this value is zeroed (ie omitted entirely), it has no impact to the end result emission factor, and emission reduction to be certificated. It is thus reasonable to deduce that the absence of formal designation on Plant 184 has no impact to the emission reduction reported, beyond the allowable materiality threshold for a project of this scale.	(b) The assessment team noted the difficulty in clarifying definition of project type PLTM. Since plant 184 Curug is indicated as without fuel consumption in EF sheet from the ministry, it is assessed reasonable to assume a renewable energy nature and thus the missioning of PLTM's definition is assessed as immaterial.	

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CAR 22 (2012-10-19) For calculation of $MG_{Pr,y}$ in ER sheet: -it should be explained why MCF value for new waste is different from that value in validated PDD. -kj of pulp, paper and cardboard is applied as 0.105 (0.07*1.5) instead of 0.07 in validated PDD without given explanation.	02 November 2012 (1st round): (a) The MCF value has been corrected to include only pre-Project value, instead of taking the PDD approach of expecting future MCF value. The underlying reasons why we are taking a different approach is clarified in the "any comment" section of the relevant parameter. (b) The earlier applied k-factor was incorrect and this has been corrected to 0.07 as per PDD. Thank you.	2012-11-16 (1st round): Issues are addressed except (c): (a) justification in updated MR regarding adoption of MCF value 0.8 for new waste is assessed as reasonable and thus is accepted, based on physical inspection of the landfill, which revealed that the project may not satisfy the requirement of MCF 1.0. (c) it is verified that in updated ER sheet, kj of pulp, paper and cardboard is now applied as 0.07, which is in compliance with registered PDD. However, the parameter "B_PAPER" in ER sheet is a typed-in value and does not get updated along with the change of kj for paper, thus the calculated result of $MG_{Pr,y}$ is not affected by the change of kj paper from 0.105 to 0.07. This is incorrect and needs revision. This CAR is still open	Ok/Closed
	30 November 2012 (2nd round) In this submission, the parameter B_PAPER has now been updated as a formula with kj as a dependent. This update results in the change of total theoretical methane from 658,669tCO₂ to 652,784tCO₂. This is updated in Section 3.2.2.6 and Section 5.1	2012-12-10 (2nd round): OK. Upon checking version 3 0 0 of ER sheet, it is confirmed that B_PAPER is revised correctly.	

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CAR 23 (2012-10-19) Electricity export and import data for July 2012 in ER sheet actually covers the period from 10am 1st July 2012 to 10am 1st August 2012.	02 November 2012 (1st round): The lapsing data has now been excluded and this is clearly indicated in Consolidated ER Spreadsheet Sheets ("EL_LFG").	2012-11-14 (1st round): GLC's Assessment Team reviewed the ER spreadsheet, folder "EL_LFG". While the period starting on 1.Jul.12 10:00 AM and ending on 1.Aug.12 10:00 AM has been completely presented (as JUL 2012 A), the period starting on 31.Jul.12 11:59 PM and ending on 1.Aug.12 10:00 AM has been deducted from the presented results. Data have been read out on Project Participant's electricity meters for the first 10 hours of 2012-08-01. The read out data is compiled in the same spreadsheet (folder "1 august 2012"). The result of this 10 hours period is presented as "JUL 2012 B" and is deducted from the whole amount of exported electricity. Hence, CAR 23 is closed.	Ok/Closed
CAR 24 (2012-10-19) *Incoming waste data for July 2012 is missing in ER sheet. *The submitted waste report for	02 November 2012 (1st round): (a) The date for July 2012 has been added into Sheet ("AT,X") (b) The document for February 2011 has been	2012-11-14 (1st round): OK. (a) in updated ER file, it is verified that the incoming waste data for July 2012 has been added in A_T,x and the	Ok/Closed

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
February 2011 is incomplete and does not contain incoming waste data for the month *The submitted waste report for February 2009 is not clear and the incoming waste data for the month cannot be read.	resubmitted (see VCSMR1.28 WB 26). (c) The document for February 2009 as VCSMR1.28 WB 02 is also submitted, although it is outside reporting period.	value, 157492.02 tonnes, is consistent with waste record. (b) VCSMR1.28 WB 26 is checked and it is verified that according to this document, incoming waste in Feb. 2011 as indicated as 152249.28 tonnes in ER sheet, is correct. (C) Although year 2009 is not in this monitoring period, waste dumped in year 2009 is involved in calculation of $MG_{PR,y}$ of this monitoring period. VCSMR1.28 WB 02 is checked and it is verified that according to this document, incoming waste in Feb. 2009 as indicated as 140580.44 tonnes in ER sheet, is correct. This CAR is closed.	
CAR 25 (2012-10-19) On MR page 13, data unit of F is %v/v while value applied is 0.5. Value does not correspond to the unit.	02 November 2012 (1st round): This value has now been corrected in the current version of VCSMR1.	2012-11-14 (1st round): GLC's Assessment Team can confirm, that the value has been corrected. In revised VCS Monitoring Report (version 2.0), Section 3.1., the value for Parameter "F" is now presented as 50%. Hence, CAR 25 is closed.	Ok/Closed

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CAR 26 (2012-10-19) Geographical coordinates of the project in MR is inconsistent with that in validated PDD.	02 November 2012 (1st round): At the time of validation, it was not clear where the power house would be located within the 120hectare landfill. As such the PDD use the general geographical coordinate of the landfill. In this monitoring report, the exact coordinate of the power house can now be accurately pin-pointed. As consequence, the geographical coordinate presented in the VCSMR1 is more accurate than that stated in the PDD.	2012-11-14 (1st round): OK. The explanation of Project Participant concerning the unclear location of the power house can be accepted. Hence, CAR 26 is closed.	Ok/Closed
CAR 27 (2012-10-19) For transparency, in months where electricity meter replacements or wiring system change for metering electricity happened, figures on each jointly signed transaction statement should be reported instead of only given the calculated result.	02 November 2012 (1st round): This has now been corrected in both the consolidated emission reduction spreadsheet Sheet ("EL_LFG"), and VCSMR1 (Table 3)	2012-11-14 (1st round): A review Table 3 of the revised VCS Monitoring Report (version 2.0) and the ER spreadsheet (sheet "EL_LFG") shows, that the following months are not reported as complete but in parts (A-B-C resp. A-B): JAN 2012 A JAN 2012 B JAN 2012 C MAY 2012 A MAY 2012 B JUL 2012 A	Ok/Closed

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
		JUL 2012 B* While only for "JUL 2012 B*", a note indicates the reason for splitup ("adjustment to represent lapsed period between billing period and monitoring period"; see CAR 23 for further details), the other split ups are not sufficiently presented. Hence, CAR 27 is still open.	
	30 November 2012 (2nd round) In this submission, Table 3 (Section 4.1.2) is now complimented with footnote explaining the underlying reasoning for the split-up.	2012-12-06 (2nd round): OK. A review Table 3 of the revised VCS Monitoring Report (version 3.0) shows several footnotes containing explanation why the respective months have been split up. Hence, CAR 27 is closed.	
CAR 28 (2012-10-19) For flow meter for LFGelectricity;t the serial number on local calibration certificate indicated is inconsistent with information from on-site physical inspection.	02 November 2012 (1st round): This is a typo by the calibrator. NOEI is seeking Endress+Hauser to correct the document.	2012-11-14 (1st round): Please provide corrected document as soon it is available. Hence, CAR 28 is still open.	Ok/Closed

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
	30 November 2012 (2nd round). Endress+Hauser issued revision and the revised document has been added as supporting document number VCSMR1.34G2.	2012-12-06 (2nd round): OK, GLC's Assessment Team reviewed the revised document (internal evidence number: VCSMR11.34G2). GLC's Assessment Team can confirm that the serial number now stated in the document is consistent to the conditions met on site (SN E1058E0109D). Hence, CAR 28 is closed.	
CAR 29 (2012-10-19) During on-site visit, it was found that the serial number of electricity meter of the grid company (position tag EM-04) has been wrongly indicated on MR.	02 November 2012 (1st round): This error / inconsistency has been corrected in the VCSMR1.	2012-11-14 (1st round): GLC's Assessment Team found out during on site visit, that the electricity meter of the grid company (position tag EM-04) has the following serial number: 21 20 63 056 As the presented serial number of the above mentioned device has been wrongly indicated on version 1.0., the revised VCS Monitoring Report (version 2.0) now presents the correct number, consistent with the number	Ok/Closed

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		verified on site by GLC's Assessment Team. Changes in the report have been done for the parameters ELim,y and ELex,y. Hence, CAR 29 is closed.	
CAR 30 (2012-10-19) There are several typos in QA/QC description for Tflare in MR regarding the commissioning and supposed replacement dates of the thermocouple.	02 November 2012 (1st round): The date of which the unit must be replaced has been corrected to 1 year since date of commissioning which is 25th of June 2012. Thank you.	2012-11-14 (1st round): While reviewing QA/QC description for Tflare in revised VCS Monitoring Report (version 2.0), GLC's Assessment Team states the following: - While the thermocouple has been commissioned on 2011-06-25, it should be replaced no later than 2012-06-24 (one year later). - The actual replacement date 30th August 2011 is not clear. Hence, CAR 30 is still open.	Ok/Closed
	30 November 2012 (2nd round) Agreed. The supposed date of replacement and actual replacement date has been updated on the relevant section accordingly.	2012-12-06 (2nd round): OK. While reviewing the QA/QC description for parameter T_{flare} in the revised VCS Monitoring Report (version 3.0), the following changes have been done: The replacement date has been correctly set to 2012-06-	

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		24 The actual replacement date has been corrected to 2012-08-30, what is now logic compared with project's implementation and resulting replacement dates. Hence, CAR 30 is closed.	
CAR 31 (2012-10-19) During on-site verification, several inconsistencies (e.g. for 3 months) were found between the converted raw data and the raw data presented in the monthly ER files.	02 November 2012 (1st round): The error was due to the limitation of the number of data points that can be handled by the processing software. To reach a monthly aggregation, data must be collated several times, and this is prone to human error. As a response to this, in the subsequent design of data storage and reporting, data is downloaded and collated using excel automation on daily basis. The error has now been corrected. Thank you.	2012-11-14 (1st round): OK. While on site, Project Participant downloaded all available raw data from project's data log hardware in presence of GLC's Assessment Team. Data has been handed over to the GLC's Assessment Team. In a second step, data was processed by Auditors by using the specialized software to reproduce the presented monthly data. As indicated by the finding, some inconsistencies have been found, namely for the months of June 2011, August 2011 and Feb 2012. The following spreadsheets have been recently checked by DOE and found consistent to available raw data: - VCSMR1.ER 01-2011 JUN CH4 Abatement.xlsm - VCSMR1.ER 03-2011 AUG CH4 Abatement.xlsm - VCSMR1.ER 09-2012 FEB CH4 Abatement.xlsm The explanation of Project Participant can be accepted as	Ok/Closed

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		it is as well GLC's Assessment Team's experience that the software is not able to reproduce the whole data set needed for one month. Hence, CAR 31 is closed.	
CAR 32 (2012-10-19) The Calibration of the raw gas analyzer used to monitor the parameter wCH4 was delayed. Conservative approach should be applied for measurement results according to UNFCCC guideline.	02 November 2012 (1st round): The first calibration report indicates that throughout the period of which calibration has been delayed, the analyser reports significant under reporting of methane content. This is by all means are conservative to the emission reduction claimed and thus no further correction is necessary.	2012-11-14 (1st round): The first available calibration report during project operation dates 2012-04-12 (VCSMR1.29 GA 01. 2012APR12 QIAL weekly calibration.pdf). The methane analyzer is showing a concentration of 55.9 % of methane while using a test gas with a certified gas quality of 60.1 %. Thus, the error identified in the delayed calibration error is beyond the maximum permissible error of the measuring equipment (2 %). It is not clear how this fact has been considered for the period without any documented calibration for the methane analyzer (from 2011-02-26 to 2012-04-12). Hence, CAR 32 is still open.	Ok/Closed

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	30 November 2012 (2nd round) In the first calibration report, the gas analyzer indicates a methane concentration of 55.9% for a gas with a known concentration of 60.1% (in the standard gas). This implies that the gas analyzers under-report the amount of methane in landfill gas by 4.2%, which is beyond the maximum permissible error of +/-2%. As consequence to this, the calculated baseline emission from methane captured from landfill gas is already most conservative.	2012-12-06 (2nd round): OK. When comparing the value measured (55.9 %) while a certified standard gas was supplied to the methane analyzer with a concentration of 60.1 %, it can be seen that the actual indication is with -4.2 % lower (in absolute numbers) and even with -7.0 % lower, if calculated to a relative number. This means that the analyzer was constantly reporting a lower value than the real value was. Thus, this error is greater than the maximum permissible error of ± 2.0 %. As this error is showing lower values, this error does not lead to higher emission reductions. Thus, the application of the measured values is already conservative, as a deduction of 7 % is already applied due to the occurred error. Hence, CAR 32 is closed.	
CAR 33 (2012-10-19) It should be clarified whether combustion temperature of flare is considered as flare operational	02 November 2012 (1st round): The flare combustion temperature is a part of the "flare operational temperature". (See VCSMR1.39B) The flare manufacturer, HOFSTETTER has clarified	2012-11-14 (1st round): Project Participant is requested to give more details on his statements: - VCSMR1.39B is a document of more than 220 pages. Please indicate the reference inside this	Ok/Closed

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parameter, and if so, how it is monitored and how the monitoring result is considered in flare efficiency determination. Clarification should also be given regarding how flare's operational range of gas flow rate is considered in flare efficiency determination.	that the correct range of flare operational temperature should be between 800-1200degC, instead of the earlier thought 1,000 to 1,200degC. This has now been corrected in all spreadsheet and the VCSMR1. The combustion temperature is measured by S-type thermocouple, and auto-monitored and controlled by a Programmable Logic Controller (PLC) to always operate to reach an optimum (dynamic) set point between 1,000 to 1,200 degC, by varying air intake as soon as sufficient pressure (flow) is detected. This PLC's set point are controlled & password protected by the manufacturer and can not be changed by the project operator. Thus, monitoring only constitutes whether or not the PLC is functioning as intended. If the PLC is well functioning then the exhaust gas temperature (T_{flare}) will reach the higher end of the limit without exceeding 1,200degC, and simple observation of the parameter T_{flare} reading can deduce the state of the PLC. The PDD and the methodology determined the flare efficiency from exhaust gas temperature, not by the combustion temperature.	document that "flare combustion temperature is a part of the 'flare operational temperature'". - Please provide an evidence to the statement of the manufacturer "that the correct range of flare operational temperature should be between 800-1200degC". This statement is in contradiction with the design of the flare which indicates a combustion temperature of 1,000 up to 1,200 °C (VCSMR1.39B, page 19). - While VCSMR1.35B is showing a maximal designed flow of 2,625 Nm³/h, it is not clear for what flowmeter (serial number) this calculation was provided. Additionally, the maximal designed flow of 2,625 Nm³/h is contradictory to the designed flow of the flare booster station as per VCSMR1.39B, page 19. Hence, CAR 33 is still open.	

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	As with the gas flowrate not to exceed 2,500Nm3/hr, this has been physically limited by the extent of the capacity of the flow meter used (BGB.FIR.81.3) which is designed for maximum allowable flow of 2,625Nm3/hr (See VCSMR1.35B)		
	30 November 2012 (2 nd round); (a) Kindly refer to page 60 of VCSMR1.39B ("Temperature Regulation"). This section describes the pivotal function of combustion temperature in controlling the flare operation. Combustion temperature is measured and compared to the temperature set point of the PLC to ensure that the flare reached the desired combustion temperature. (b) The statement from the manufacturer were misinterpreted. The correct "optimum operating temperature" is 1,000 – 1,200degC. This upper and lower limits are all corrected in the spreadsheet, and Section 3.2.1.8 of the VCSMR1 (c) VCSMR1.35B is relevant to the pitot tube (DC unit) attached to the flowmeter. The coupling can only provided by means of visual inspection, and as such a photographic evidence is provided;	2012-12-06 (2nd round): OK. a) Page 60 of VCSMR1.39B (document: Flare specifications, section "Temperature Regulation") states that a "proper combustion does not necessarily need a constant temperature of e.g. 1200°C. The temperature depends on the - Gas quality (30...50... % methane) - Gas quantity (min...max flow) - Combustion air quantity". It becomes evident that to ensure a proper combustion the flare combustion temperature must be considered as "flare operational parameter". b) While reviewing section 3.2.1.8 of the revised VCS Monitoring Report (version 3.0), it can be seen that the "combustion temperature" has been set to 1.000...1.200 °C. This statement is	

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	The maximum flow of the flow-meter depends on the pipe (standard) diameter of which the DC unit is fitted. As consequence there is slight lapse between the flow-meter capacity (2,625Nm³/hr) and the capacity of the flare unit (2,500Nm³/hr). As with any system, the smaller capacity becomes the limiting capacity, and hence the real capacity of the entire system.	consistent with the design of the flare which indicates a combustion temperature of 1,000 up to 1,200 °C (VCSMR1.39B, page 19). c) While comparing the documents VCSMR1.35B and VCSMR1.35L it becomes evident, that the indicated flow of 2,625 Nm³/h is the maximum flow capacity which the combination of inner pipe diameter and installed flow measuring devices are designed for. The normal range of this flow measuring device is indicated as between 500...2,500 Nm³/h. It is understood that the maximum capacity must be slightly higher to compensate operational events (such as sudden release of condensate inside the LFG-piping system may increase the flowrate for a short time). Thus, the limit of 2,500 Nm³/h can be accepted. Hence, CAR 33 is closed.	
CAR 34 (2012-10-19) On MR page 9, date of document "VCSMR1.12" is incorrectly indicated as 22 July 2016.	02 November 2012 (1st round): This information has now been corrected in the current version of the VCSMR1. Thank you.	2012-11-14 (1st round): OK. Revised VCS Monitoring Report (version 2.0) now correctly provides "22 Jul 2011" as date for the evidence "VCSMR1.12". This is in accordance with the presented documents under the above mentioned number.	Ok/Closed

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		Hence, CAR 34 is closed.	
CAR 35 (2012-10-19) The names of the flow rate parameters slightly deviate from the names in PDD.	02 November 2012 (1st round): The name changes reflects the time interval of how the data is recorded. The PDD stated that volume is to be reported every hour, whereas the Project uses a much higher data recording resolution (per minute). These differentiation improves the accuracy of reporting rather than the other way around, as such does not constitute a deviation. The name changes are now explained in the "any comment" section of the respective flow parameter.	2012-11-14 (1st round): OK. Project Participant's answer can be accepted. As explained by Project Participant, the higher data recording frequency leads to higher data accuracy and thus is more conservative in terms of calculation of emission reductions. Hence, CAR 35 is closed.	Ok/Closed
CAR 36 Monitoring results for MD_project on the final ER sheet can not be traced back to the monthly files or consolidated ER file.	02 November 2012 (1st round): The Consolidated ER Sheet (15) is taking information from 14 monthly abatement activity sheets. The linkage has now been clarified, and should be traceable to it's entry.	2012-11-14 (1st round): OK. The revised monthly spreadsheets have been compared with the consolidated spreadsheet (folder MD_Project). The data presented in the consolidated spreadsheet can now be traced back to the monthly files and are consistent. Hence, CAR 36 is closed.	Ok/Closed

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CAR 37 (2012-10-19) While the VCSMR states as "Instrument II" to determine the amount of electricity exported by the project to the grid (parameter ELex,y) and the amount of electricity imported by the project from the grid (parameter ELim,y) the following model: "EDMI MKGE-N680", it has been verified by the assessment team that the model is EDMi Mk6E-N680 instead. This is an inconsistency between the VCSMR and the situation on site.	02 November 2012 (1st round): The model has now been corrected in the current version of VCSMR1. Thank you.	2012-11-14 (1st round): OK. Corrections of the model have been done in sections 3.2.1.5 "Electricity Imported" and 3.2.1.6 "Electricity Exported". Hence, CAR 37 is closed.	Ok/Closed
CAR 38 (2012-10-19) MR section 2.2 "Deviations from the Monitoring Plan " should be correctly filled in ,with information given regarding deviations from MP.	02 November 2012 (1st round): Section 2.2 now explains deviation from monitoring plan with respect to the way flare efficiency is determined / assigned, and it's justification / impact to emission reduction calculation.	2012-11-14 (1st round): Section 2.2 (Deviations from the Monitoring Plan) of the revised VCS Monitoring Report (version 2.0) contains the deviations from the Monitoring Plan for the presented monitoring period. It is explained that Project Participant opted to deviate from the hourly determination of flare	Ok/Closed

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		efficiency (as per CDM-Methodological "Tool to determine project emissions from flaring gases containing methane", EB 28 Annex 13). Instead of the hourly determination, Project Participant applies determination of flare efficiency minute wise (as allowed by the CDM Methodological tool "Project emissions from flaring" (Version 02.0.0, EB 68, Annex 15). Further explanations are given in Section 4.2.1 of the revised VCS Monitoring Report (version 2.0). In general, it is GLC's Assessment Team's opinion, that: - A minute wise application of flare efficiency's determination is acceptable given the circumstances of the project (flare not running long time as gas engines are combusting the main part of collected LFG). - If Project Participant is applying stipulations of the CDM Methodological tool "Project emissions from flaring" (Version 02.0.0, EB 68, Annex 15), these stipulations must be fulfilled completely. Thus, the presented formula (equation 3 of the revised VCS Monitoring Report, version 2.0) does not apply the complete set of stipulations of "Option A: Default value". Project Participant is only calculating as follows: "Methane destruction efficiency of the flare unit (flare efficiency) at time t is set exclusively based on temperature of flare exhaust gas at respective time t(T(flare,t)". Furthermore, it is not clear, while applying the version 2 of the mentioned tool, where	

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		the measurement of TEG,m as per tool are taken. Finally, it is not clear why Project Participant describes only this deviation from monitoring plan as other deviations also occurred. Hence, CAR 38 is still open.	
	30 November 2012 (2nd round) (a) Deviation related to determination of flare efficiency. Project Proponent wish to clarify that the intention is not to apply Version 2.0 of the Tool. The ER calculation applies default value efficiency as per Version 1.0, with exception the resolution of time assignment (per minute instead of per hour). The reference to Version 2.0 is only to illustrate that the same time resolution is also adopted in this later version indicating that the approach is not entirely dissented. This is now clarified in Section 2.2.1 of VCSMR (b) Other Deviation. A summary of non-	2012-12-06 (2nd round): OK. GLC's Assessment Team reviewed the section 2.2 "Deviations from the Monitoring Plan" of the revised VCS Monitoring Report (version 3.0): a) It is now clearly stated, that Project Participant does not to apply the version 2.0 of the CDM "Methodological Tool: Project Emissions from Flaring". It is understood that this tool is only used to illustrate that the approach to determine flare efficiency on a minute-by-minute-base is accepted while further developing the above mentioned tool. Anyway, as per registered PDD, the Methodological "Tool to determine project emissions from flaring gases containing methane" in its version 1 has to be applied. The application of this tool not being in question, the deviation to determine Flare efficiency on a	

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	conformance to the Monitoring Plan has been added as Section 2.2.2	minute-by-minute approach can be accepted given the following 2 reasons: 1) The flare operational time on the project activity is very limited, so that a period of consecutive 60 minutes (what is needed to apply correctly the Methodological "Tool to determine project emissions from flaring gases containing methane" v1.0) is rare. 2) Due to developments based on the experience in application of the Methodological "Tool to determine project emissions from flaring gases containing methane" v1.0, the CDM-Executive Board decided in July 2012, that a minute-by-minute determination of the flare efficiency is acceptable. So, given the circumstances of the project and recent improvements in CDM Methodological tools, it is the opinion of GLC's Assessment Team that the chosen approach for determination of flare efficiency is conservative and acceptable. b) For the sake of completeness and transparency, Project Participant listed in section 2.2.2 of the revised VCS Monitoring Report (version 3.0)	

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		other items, which cannot be considered as deviations but as non-compliances (delayed calibrations of methane analyzer, delayed replacement of thermocouple of the flare). Hence, CAR 38 is closed.	
CL 1 (2012-10-19) * It is not clear, how often the presented project crediting period under the CDM is renewable.	02 November 2012 (1st round): The CDM Credit Period is renewable twice every seven years. Explicit statement to this effect is now made in Section 1.6 of the VCSMR1.	2012-11-14 (1st round): OK. In Section 1.6 "Project Crediting Period" the following is stated: "Under CDM, the Project credit period commenced in 1st of August 2012 and can be renewed twice every 7 years". Hence, CL 1 is closed.	Ok/Closed
CL 2 (2012-10-19) * While the VCSMR in section 2.1 is in a table describing relevant dates for project implementation, the evidence for 30 Jun 2010 (VCSMR1.02) is not showing in detail what engine type it is about. Additionally, the translation does	02 November 2012 (1st round): Additional engine acceptance report for GE-01 and GE-02 has now been added into the list of supporting documents (VCSMR1.02A and 1.02B respectively). Both documents contains the detail of the engine.	2012-11-14 (1st round): OK. The presented evidences VCSMR1.02A and 1.02B are indicated as "form of equipment acceptance report" and show the following: - Engine Manufacturer - Engine Types - Engine Serial Numbers - Generator Serial Numbers	Ok/Closed

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
not clearly state the number of engines.		- Number of engines Hence, CL 2 is closed.	
CL 3 (2012-10-19) * It is not clear why in evidences VCSMR1.12 the engine size is not the same as per manufacturer statements in evidences VCSMR1.10 resp. 1.11.	02 November 2012 (1st round): This document is issued by the relevant authority to enable the PPA to come into effect. The document records the alternator serial number instead of the engines. NOEI is not in position to affect the output of the document other than requesting clarification of why this is the case.	2012-11-14 (1st round): OK. Project Participant has explained the situation already during the on-site visit. GLC's Assessment Team has been visiting the project site and can confirm that the mentioned engines as per VCSMR1.10 resp. 1.11. are installed on site, are fuelled with Landfill Gas (LFG), which is generated and collected on Bantargebang Landfill Site and converted into electrical power by the Project Activity. Additionally, GLC's Assessment Team gained inside the working conditions and behaviours of local authorities. Considering the conditions seen there, the statement of Project Participant can be accepted. Hence, CL 3 is closed.	Ok/Closed

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
CL 4 (2012-10-19) * The "VCS Monitoring Report Template" stipulates for section 2.1 the following: • Describe the implementation status of the project activity(s). • Provide information regarding the operation of the project activity(s) during this monitoring period, including any information on events that may impact the GHG emission reductions or removals and monitoring. " While the presented table is marked as "major milestones relevant for implementation", it is not clear if this table should be seen as information regarding the operation. Additionally, it is not clear why milestones are not completely set in chronological order.	02 November 2012 (1st round): Each of the milestones presented in the table are affecting the implementation status and thus the operation of the project, and subsequently it's ability to claim emission reduction. For example, (a) the acceptance of each new gas engines & its certification by the authority (whichever date the later) determines increased operation capacity of the Project, and thus it's ability to displaced grid electricity. (b) the completeness of commissioning of the degassing station means that the project commences the counting on it's methane abatement ability, thus supporting the information outlined in Section 4 (Table 1) where methane abatement commenced only in June 2012. Accordingly, the table has now been re-titled to: "Major milestones relevant to the implementation and thus operation of the emission reduction project", to emphasize this effect. Each milestone has now been rearranged in chronological order to the date of the document.	2012-11-14 (1st round): GLC's Assessment Team reviewed the revised VCS Monitoring Report (version 2.0) with a special focus on its section 2.1 (Implementation Status of the Project Activity). While the implementation status of the Project Activity has been clearly and sufficiently described, operational information is still not sufficiently presented in regard of "any information on events that may impact the GHG emission reductions or removals and monitoring". It can be confirmed that the table in section 2.1 is now presenting "Major milestones relevant to the implementation and thus operation of the emission reduction project" and that the milestones have been rearranged to chronological order. Hence, CL 4 is still open.	Ok/Closed

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
	30 November 2012 (2nd round) Additional paragraph describing operational events that may impact GHG emission reduction and removal are added, making reference to the relevant section of the VCSMR1.	2012-12-06 (2nd round): OK. GLC's Assessment Team reviewed the revised VCS Monitoring Report (version 3.0) with a special focus on its section 2.1. It can be confirmed that now an additional paragraph is now providing "information on events that may impact the GHG emission reductions or removals and monitoring". Thus, the requirements of the "VCS Monitoring Report Template" are applied. Hence, CL 4 is closed.	
CL 5 (2012-10-19) * The VCSMR presents in its section 3.1 the "Data Unit / Parameter: ρ (CH₄,n)." It is not clear to which Methodology the stated value refers to and which conditions are used as reference. Additionally, it is not clear if this parameter is only used for calculation of baseline emissions or as well for calculation of project emissions.	02 November 2012 (1st round): The normal density of methane is calculated as a fixed value in Equation 14 of PDD. This parameter are used by both baseline and project emission calculations, and this is now explicitly stated.	2012-11-14 (1st round): OK. As per PDD (equation 14 and section B.6.2), the parameter ρ (CH₄,n) is defined as per Approved consolidated baseline methodology ACM0001 "Consolidated baseline and monitoring methodology for landfill gas project activities" in its version 11. Reference Temperature is set to 237.15 K, reference pressure is 101,325 Pa. Furthermore, Section 3.1. of the revised VCS Monitoring Report (version 2.0) states under "Purpose of data" for the above mentioned parameter: "Used for calculation of	Ok/Closed

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
		methane capture & destruction, affecting both the baseline & project emission". Hence, CL 5 is closed.	
CL 6 (2012-10-19) * The function and the devices used as "energy computer (RMC621)" and as a "memograph (RSG40)", presented in section 3, is not described in a clear and transparent manner.	02 November 2012 (1st round): The data logger used is Memograph M (RSG 40) supplied by Endress+Hauser. This unit function as data storage, display, and manager, taking all input from two flow and energy manager units RMC621, also supplied by Endress+Hauser. The RMC621 units are taking data transmitted from every instruments in the degassing station, analyse them where necessary and dispatch them to the data logger. This relationship is now clarified in Section 3.2.1 (Online Parameter).	2012-11-14 (1st round): OK. The clarification presented in Section 3.2.1. of the revised VCS Monitoring Report (version 2.0) can be deemed sufficient. The functions of the Memograph M (RSG 40) as data logging and manging system and of the Flow and Energy Manager (RMC621) as flow and energy computer to convert the flow into normalized conditions have been explained clearly. Hence, CL 6 is closed.	Ok/Closed
CL 7 (2012-10-19) * The VCSMR states in the QA/QC procedure for Tag-ID BGB.FIR.0006 for the parameter LFGelectricity,t that this flowmeter has been replaced with	02 November 2012 (1st round): We wish to reconfirm that BGB.FIR.0006 is replaced by BGB.FIR.0007 (and not BGB.FIR.0003). This error has been corrected in the current version of VCSMR1. Thank you.	2012-11-14 (1st round): OK. The replacement procedure can be accepted. The QA/QC procedure for Tag-ID BGB.FIR.0006 for the parameter LFGelectricity,t now shows the following: "05 October 2011: The instrument were decommissioned,	Ok/Closed

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
BGB.FIR.0003. It is not clear why this has been done as BGB.FIR.0003 is in use for LFGtotal,t.		and to be replaced with BGB.FIR.0007 with larger capacity.” Hence, CL 7 is closed.	
CL 8 (2012-10-19) * While the VCSMR states in its chapter 3.3.3 a reference to the same chapter, it is not clear what exactly is meant with this reference.	02 November 2012 (1st round): This reference suppose to refer to Section 3.2.1 (“Details of Online Parameter”). This error has been corrected in the current version of VCSMR1. Thank you.	2012-11-14 (1st round): OK. Reference inside Section 3.3.3 (Data Acquisition Process) of the revised VCS Monitoring Report (version 2.0) is now done to chapter 3.2.1 “Details of “On-line Parameter”). Hence, CL 8 is closed.	Ok/Closed
CL 9 (2012-10-19) * The source for the equation 3 to determine the methane destruction efficiency is not clearly presented.	02 November 2012 (1st round): Equation 3 of VCSMR1 is constructed from combining Equation 6 & 16 of the PDD. Statement to this effect is now made explicit in the current version of VCSMR1.	2012-11-14 (1st round): OK. The revised VCS Monitoring Report (version 2.0) provides a statement, that “fugitive methane is calculated using formula constructed from combining Equation 6 and Equation 16 of the PDD”. Hence, CL 9 is closed.	Ok/Closed

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CL 10 (2012-10-19) * In regard of the methane abatement spreadsheets (e.g. VCSMR1.ER 01-2011 JUN Methane Abatement Activity.xlsx), the difference between the sheet "Original Data" and "Raw Data" is not clearly traceable. Furthermore, it is not clear why in both sheets a formula has been inserted (row B, "Time Value"). Additionally, it is not traceable why in the mentioned sheets "Original Data" negative values are set into brackets. Finally, it is not clear why some methane abatement spreadsheets present 4 sheets ("Original Data", "Raw Data", "Calculated Data", "Consolidated Results"), while others are presenting only 3 ("Raw Data", "Calculated Data", "Consolidated Results").	02 November 2012 (1st round): (a) We wish to clarify that Sheet ("Calculated Data") always taking data from the Sheet ("Raw Data"), at any given month. However, in certain months, the original output of the software can not be used directly as basis for calculation, as many contains text input that can-not be computed. Consequently, in some months, the exact output is provided as sheet ("Original Data"), which later cleared from data noise into Sheet("Raw Data") which subsequently used as basis input for the Sheet ("Calculated Data"). (b) The column "Time Value" has been removed from all sheets as they are redundant.	2012-11-14 (1st round): As described in CAR 31, GLC's Assessment Team checked the available raw data. So, DOE can confirm that in some monthly data sets content a lot of text (e.g "Invalid value!"). Considering this fact, the use of an additional data to clear the text is appropriate and acceptable. Additionally, with Project Participant's explanation it becomes clear why some monthly spreadsheets present 3 sheets, while other are presenting 4 sheets It can be confirmed by GLC's Assessment Team that the column "Time Value " has been removed from the presented spreadsheets. However, it is still not clear why data as per sheet "Original Data" negative values are set into brackets. Hence, CL 10 is still open.	Ok/Closed

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	30 November 2012 (2nd round); In the latest version of spreadsheet, all number formatting has been updated to present negative number with minus (-) signed, instead of within bracket.	2012-12-06 (2nd round): OK. GLC's Assessment Team checked the revised monthly spreadsheets. It can be confirmed, that customized formatting has been removed, numbers are now presented in standard format without using any special format. Hence, CL 10 is closed.	
CL 11 (2012-10-19) * In some months, raw data are not fully available, i.e. less than 44,640 resp. 43,200 values for a 31 resp. 30 days period. It is not clear why data are missing.	02 November 2012 (1st round): When the power plant lost its grid connectivity, it loses its entire power supply, including those used for the data-logger. As such, the number of data sets are lower than the maximum number possible for every given months. This lower number of data sets, however, has no impact to the emission reduction claimed as absence of data means that emission reduction can-not be claimed in the first place.	2012-11-14 (1st round): OK. As described in CAR 31, GLC's Assessment Team checked the available raw data. So, DOE can confirm that some monthly data sets do not provide the maximum possible quantity of data being registered every minute. The missing time is justified by the lost of power connection. These events can be traced in the events log of the specialized software as "Power on" and "Power off". As Project Participant emphasizes, no emission reductions are claimed for times logged data cannot be presented.	Ok/Closed

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
		Hence, CL 11 is closed.	
CL 12 (2012-10-19) * Table 1 of the VCSMR, presents "Measurement Results for Calculation of Methane Destroyed by the Project". It is not clear if the provided values are used for overview only or if they are used for further calculation.	02 November 2012 (1st round): The value represented in each of the column in VCSMR1(Table 1) represents each and every parameters that must be monitored, as outlined in the PDD/VCSMR1. A clarification statement has been inserted (prior to Table 1) to assert that the provided value displays the aggregated periodical amount of the parameters, but MD_Project can not be reproduced from the data displayed as the actual computation occurs at much lower resolution.	2012-11-14 (1st round): OK. The statement in Section 4.1.1 (Methane captured and destroyed by the Project) of the revised VCS Monitoring Report (version 2.0) is as follows: The "value of methane destroyed cannot be reproduced using the data displayed in table 1". This statement makes clear that Table 1 is displaying values as overview and summary. Hence, CL 12 is closed.	Ok/Closed
CL 13 (2012-10-19) * The sheet "Calculated Data" is providing a "summary of reading". The formula behind the summary is taking in some spreadsheets more values into account than values available. This is not traceable.	02 November 2012 (1st round): This has been corrected in the current version of methane abatement spreadsheet for each respective months. Thank you.	2012-11-14 (1st round): OK. While checking the monthly spreadsheets, it can be seen that exactly the values presented for the respective month are taken into consideration for calculation of the "summary of reading". Hence, CL 13 is closed.	Ok/Closed

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CL 14 (2012-10-19) * The file "VCSMR1.ER 15 Consolidated Emission Reduction Version 1.0.xlsx" is presenting in its sheet "MD_Project" in column H the "No of Days". While the applied formula is counting the quantity of days between "Period Start" and "Period End", it is not clear why this has been done in this way.	02 November 2012 (1st round): The column counting number of days has been removed from Sheet ("MD_Project") to prevent further confusion. This was originally used to adjust the period applicable for MG_PR, this adjustment is now performed in sheet ("MG_PR") using number of month, instead of exact number of days.	2012-11-14 (1st round): It is verified that the column counting number of days has been removed from "MD_Project". Since number of days is not involved in calculation of this parameter this is assessed as acceptable. Also it is verified that for calculation of MG_PR, number of months is used for incomplete years in the monitoring period, and the calculation is correct. This CAR is closed. [note: this is a CL not a CAR]	Ok/Closed
CL 15 (2012-10-19) In "EF JAMALI 10 version 1.0.0" and/or "EF JAMALI 09 version 1.0.0": - English translation of headers should be added in sheet "2010 original" - Abbreviations should be given for key terminations, eg. for type of power generation units, type of fuel, etc.	02 November 2012 (1st round): (a) For EF JAMALI 10, English translation to the heading were inserted as "comment" in spreadsheet "2010 Original", so not to change the original heading / structure. (b) The classifications of power plants group are now clarified in the spreadsheet ("References & Assumptions") (c) Web-hyperlink to Reference 2/4 are provided in the spreadsheet. However, there	2012-11-14 (1st round): (a) it is verified that English translation has been added as comments to the heading. (b) it is verified that the classification of power plants group is also added. However, clarification is still needed for PLTM. Refer to CAR 21. (c) it is verified that some website links are added. (d) it is verified that the data source of the formula is	Ok/Closed

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- website links of Ref. No 2&3&4, should be provided if applicable; - data source of the formula for HHV-LHV conversion should be given; - for consistency and transparency, instead of typing in values in formulas, all conversion factors should be listed out and any the formula applying them should link to the relevant cells; - explanation should be given regarding why choice of coal field numbers for HHV-LHV conversion calculation is different from the choice in validated PDD.	are no web-link to Reference 3. (d) Formula conversion now refer to the "Coal Conversion Fact" published by World Coal Institute.	added. This CL is closed.	
CL 16 (2012-10-19) For electricity export/import, it is indicated that the electricity meters tagged as "EM-01" and "EM-03" were not available for the entire monitoring period. This slightly deviates from	02 November 2012 (1st round): EM-01 and EM-03 will have the same Class as EM-02 and EM-04 and thus the two sets of (aggregated) readings are not expected to have significant measurement deviations. EM-01 and EM-03 must meet standards set and agreed in the PPA, and	2012-11-14 (1st round): GLC's Assessment Team inspected while being on site the whole set of electricity meters available. The following situation could be verified: Electricity Meter EM-01:	Ok/Closed

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what's stated on PDD page 43. Clarification should be provided regarding the influence to monitoring/emission reduction accuracy.	validated by PLN's Calibration Laboratory. However, the difference of operatorship means that NOEI will be fully in control for maintaining services and calibration of the unit. A statement confirming that the absence of EM-01/03 has little impact to emission reduction has now been inserted in the "any comment" section of the relevant parameters.	Ownership: Project Participant Manufacturer: Actaris Model: SL7000 Type: SL761A071 2010 Accuracy class: 0.2s Serial Number: 36050236 Electricity Meter EM-02: Ownership: PLN (grid operator) Manufacturer: EDMI Model: Mk6E-N680 Accuracy class: 0.2s Serial Number: 211556670 Electricity Meter EM-03: Ownership: Project Participant Manufacturer: Itron Model: SL7000 Type: SL761A071 2012 Accuracy class: 0.2s	

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
		Serial Number: 53131005 Electricity Meter EM-04: Ownership: PLN (grid operator) Manufacturer: EDM Model: Mk6E-N680 Accuracy class: 0.2s Serial Number: 212063056 It could be verified on site as it is stated in the revised VCS Monitoring Report (version 2.0) that during the recent monitoring period, the meters EM-02 and EM-04 are the primary data source for determination of parameters parameter ELim,y and parameter ELex,y.. This is supported by the use of the evidences "VCSMR1.31" as they state the reading of the meters EM-02 and EM-04. Thus is it not clear why as source of data in the revised VCS Monitoring Report (version 2.0) for parameter ELim,y and parameter ELex,y is stated: "Grid interface transaction meters at Position Tag EM-01 and EM-03". These electricity meters have neither been available the whole monitoring period nor are they	

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
		specified by type, model or serial number in the revised VCS Monitoring Report (version 2.0). Hence, CL 16 is still open.	
	30 November 2012 (2nd round) The relevant statements in Section 3.2.1.5 and Section 3.2.1.6 of VCSMR1 have been corrected accordingly.	2012-12-06 (2nd round): OK. Section 3.2.1.5 and Section 3.2.1.6 of the revised VCS Monitoring Report (version 3.0) have been assessed by GLC's Assessment Team. It is now clearly indicated, that for monitoring of parameters $EL_{im,y}$ and $EL_{ex,y}$ the "Grid interface transaction meters at Position Tag EM-02 and EM-04" have been used. Additionally, Project Participant provided the following statement: "During this monitoring period, EM-01 and EM-03 were not available for the entire reporting period. As consequence, data from its equivalence meters EM-02 (equivalent to EM-01) and EM-04 (equivalent to EM-03) were used as primary data source during this reporting period." Thus, the meters EM-01 and EM-03 are mentioned in this report for the sake of transparency and completeness. Hence, CL 16 is closed.	

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
CL 17 (2012-10-19) The VCSMR states in its section 3.3.1 on page 30 that "what is described in this report...were not fully effective during this reported period." It should be clarified what were and what were not fully effective.	02 November 2012 (1st round): The only part that were not fully effective were the implementation of daily data acquisition & quality control procedure (Section 3.3.4). Rather than describing what is not yet effective, we have clarified the starting time for the activity. These activities has been tested / simulated since May 2012, but not yet fully approved / deployed until June 2012.	2012-11-14 (1st round): As per VCS Monitoring Report Template (v3.1), section 3.3 should provide a "Description of the Monitoring Plan". Thus, the recently covered monitoring period should be considered while describing the Monitoring Plan. It is not clear, why plans for the future are described and not a review of the past period has been provided. Hence, CL 17 is still open.	Ok/Closed
	30 November 2012 (2nd round) Additional paragraph describing monitoring (reporting) implementation prior to June 2012 is added into Section 3.3.4 of VCSMR1.	2012-12-06 (2nd round): OK. GLC's Assessment Team reviewed section 3.3 of the revised VCS Monitoring Report (version 3.0). As explained by Project Participant and normal to project operations, reporting issues are continually improved. The Project Participant describes now in a clear and transparent way that project's reporting was improved at the end of the monitoring period (June 2012). In June 2012, the reporting was reorganized in accordance with	

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		Project Participant's own quality requirements. Additionally, it could be verified by GLC's Assessment Team during the onsite visit, that all data and values presented and used for emission reduction calculations are free of tampering and that the monitoring procedures installed on the site allow clear checks of the correctness of data used. Hence, CL 17 is closed.	
CL 18 (2012-10-19) On MR page 5, it is indicated that the project credit period is 26th of February 2011 to 31st of July 2019, with twice renewals at most. Since the project has been registered as CDM project, clarification needs to be provided against its compliance with relevant VCS guidelines.	02 November 2012 (1st round): Section 1.6 (Project Crediting Period) has been revised. CDM Registration is now clearly stated, and a full reference to VCS Standard 3.9.3. The Project Credit Period is now 26 February 2011 to 31 July 2033.	2012-11-14 (1st round): OK. The revised VCS Monitoring Report (version 2.0) now states a crediting period from 2011-02-26 to 2033-07-31. The VCS standard in its version 3.2 provides the guidelines for determination of the project crediting period. Section 3.9.3 of the VCS Standard (v3.2) gives an example of the eligibility for projects under VCS and one other Greenhouse-Gas-Programs, such as CDM. Thus, a CDM project with a seven year twice renewable project crediting period is not eligible for VCU issuance beyond the end of those 21 years. This means for the Project Activity, that the end of the 21 years is marks as well the end of the VCS crediting period, in cases that the proper	Ok/Closed

Description of Finding (CAR, CL, FAR) <i>Describe the finding in a transparent manner i.e. state clearly what is required and why; address the context (e.g. section)</i>	Project Participants Response <i>This section shall be filled by the PP. The finding shall be addressed with suitable arguments and evidence</i>	GLC's Assessment <i>The assessment shall include how the finding is closed i.e. how it is found that the response is assessed to be appropriate and meeting the specific requirement of the finding. In case the response is not satisfactory, additional response and DOE assessments (#2, #3, etc.) shall be sought.</i>	Final Conclusion (OK or OPEN)
		VCS crediting period ends later, which is the case here (start date: 2011-02-26, twice renewable 10 years period, end date: 2041-02-25). As CDM period for this Project Activity starts on 2012-08-01, it ends latest after 21 year, thus the CDM end date is determined as 2033-07-31. Hence, CL 18 is closed.	
FAR 1(2012-10-19) QA/QC improvements based on the issues described in this finding list should be checked during next verification. Especially, regarding CAR 16& CAR 17, whether time has been recorded more accurately and what actions have to be considered if deviation is found higher.		n/a/	

* based on desk review by 2012-09-28 prior to on-site verification without auditors' full access to all necessary documents. These documents have been made available during the on-site visit.